

# 5th Applied Active Inference Symposium (Part 3, Nov 14, 2025) ~ LIVE

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Hello everyone. Welcome to the fifth applied active inference symposium day 3, November 14, 2025. My name is Zorera Manad. I'm honored to introduce the first presenters of today, Steph McCertie and Andreas Kuloscicos. Uh, presenting money and the free energy principle. Okay, the stage is yours, please. >> Thank you. Uh, thank you to the active inference institute for hosting this symposium and having us give this talk. This is money and the free energy principle. My name is Steph McCertie. I'm the head of blockchain research and education at Wolam research and my collaborator is Andrew Kulicowskis, PhD in mathematics who runs maths for wisdom. In this presentation, uh, we thought it would be a good idea for me to share a little bit about my story and how I arrived at the active inference institute. And I'm going to share a connection that I see in um that I believe is is true in um free energy principle and money. and then I'm going to give it to Andrew to speak a little bit more about the math of energy, entropy, and free energy and pull back the curtain a little bit. So my story begins around 2014 when I read these two books back to back and they kind of set the stage for the next decade of my two approaches of research. Um, one direction and was cosmic evolution by astrophysicist from Harvard, Eric Chaisan, which kind of said something along the lines of um how much free energy is flowing through a physical system normalized by mass and time. And it points to this evolution across the universe in a direction of complexity. And in the other direction on the other end of the spectrum was this book called why information grows by Cesar Hidalgo statistical physicist from MIT. And that kind of gave me another way of seeing a similar story which was how do you measure and determine order and structure in systems and that kind of painted a picture of information growth or entropy minimization. So, this colored a spectrum or two two roads for me and I'm going to just fill in some personal events over the past 10 years that helped color this in. So in 2015 I found Bitcoin and while I didn't understand it completely technically at the time I became fascinated by the consensus mechanism called proof of work which to me really was the connection between energy

computation and value. And it was interesting because while reaching this distributed consensus in a new way, we could organize uh economies or economics or money and in different ways and and that gave me just a sense that we could be looking at either ends of the spectrum. In 2017, I had a a another question that I really wanted to answer, which was uh about solar energy. So, I went to work for Tesla in 2017 to understand a little bit more about the proliferation of solar energy and solar panels. it was obvious that in nature, the sun provides ample free energy. And I expected the same to be true for our economy that more of our energy production would come from the sun. Um Tesla had an interesting approach which said there are constraints and limitations on the current electrical grid infrastructure. Uh to get around that one needs high free energy density of batteries. So that colored in another kind of two points on my spectrum here. Then in 2020 I learned about Wolfram physics and this was Steven Wolfram's fundamental phys physics project and Steven Wolfram has an approach to physics which is deeply computational and it's computation with a capital C. It's how do you look at simple rules and how do that how does that lead to emergent complexity and that filled in a couple more gaps in my understanding of the major pillars of physics general relativity quantum mechanics and statistical mechanics or he calls uh the second law of thermodynamics. So I was seeing this happen. I joined Wolfram. I was tasked to lead the research team at Wolfram Blockchain Labs and there I studied many blockchains, many consensus protocols and many cryptocurrencies and I learned about a new uh blockchain in 2024 called Ki network which really captured my attention for two reasons. It has a novel consensus protocol called proof of entropy minima which is a variant of Bitcoin's proof of work and it has a two token architecture which is important because money has several important characteristics and if you have more than one token you can separate those characteristics and be more uh true to what those characteristics of might be. So all of my attention uh over the past year and a half has been on understanding that more and that led to um coming to the active inference institute a couple of months ago where I met Andrew. We're both investigating the math of active inference and we headed off. Um, Andrew began working for me in helping me to understand the two token architecture of Qui Network and the math of active inference. And we've had a a wonderful back and forth over the past couple of months investigating these things. And so it feels like I've this is my story and I've been orbiting around um the free energy principle. I've been looking at it from a physical energy perspective and an informational computational perspective and all those things kind of have brought me to to want to learn more. So I'm here as an investigator still on a path and

journey trying to to to learn more. But um I wanted to say one thing about technically about Y network and about the consensus mechanism because it might show a little bit more about um what we mean when we say proof of entropy minima. So this paper in 2023 by Carl Creeder and Shakara Shashri um put out the the way that this is calculated. So proof of Shannon entropy minimization is what we're speaking about here. And what we really want to talk about um at least in the blockchain world is when Bitcoin is doing its mining, this is what people refer to. It's running this secure hash algorithm with a 256bit digest. And this algorithm SHA 256 has an output of 256 ones and zeros. And so the total possible state space of any particular hash is 2 to the 256. Now what can you do with that? Um well if you put a simple rule or a simple constraint on that state space for example if you look at the number of consecutive leading zeros in the output of a SHA 256 hash beyond some threshold you can start to find interesting properties and you can use those properties in a distributed computer. So, here's an output, 256 bits. I've colored in the first nine leading zeros, and that's the that's what we're measuring. And we say in this space, 256 bits, the first nine in this example are zeros. And that lets us say something like we are going to quantify that number of bits and remove them or accumulate them. We're going to accumulate them from one block to the next perpetually. And in that way, what we're doing is we're accumulating uncertainty and removing it. And that's the sense in which we're talking about proof of Shannon entropy minimization. That's a lot of words, but what does it look like visually? Here is the output of 5,000 SHA 256 outputs in integer form. And as you can tell, there's not much signal here. But when you uh impose a constraint like this simple rule consecutive leading zeros, you can start to see that we get this uh kind of hierarchy of events uh that some events are rarer than others. And you can use that fact to sequence events in a hierarchy. And that can be used as setting a certain time interval in a distributed computer. Uh because there's there's not one global clock in this distributed computer. And so being able to get into the nuance of establishing a frequency or a time interval in several levels really gives us this hierarchical ordering which means we can establish when uh one output is greater than another. And that can lead to establishing a dominant blockchain and a subordinate blockchain. And that's important because we can distribute the algorithm in any geography and get the same statistical results off of it. So we're lifting some statistical properties out of this distribution of work and um pi network takes this to its uh as far as I can tell takes it to its limit and just really extends and expresses that idea to um to scale to it to a large degree. And so the rule is simple. It is the change in entropy is the number of states removed from one block to the next. But it's a small

difference that makes all the difference. And I just want to say one last thing. The other important fact about P network is that it is a two token architecture. And this is important because in the characteristics of money like store value, medium of exchange and unit of account, if all of those are lumped into a single token, it makes that token uh carry a lot of burden. And when you separate the characteristics of money, you get around something called Gresham's paradox, which simply states that if you've designed a global decentralized monetary system to to make a superior form of money, let's say in this case it's Bitcoin, could be gold, the participants are incentivized to hoard that money. And that's in direct tension with another characteristic of of a monetary system, which is if you're hoarding, you're not exchanging. And so it diminishes that value. So it's important that we kind of pull apart these characteristics. And in this network, because we're measuring entropy, we are playing with the idea of calling the  $\chi$  token an entropy measure. and we're calling  $\psi$  an free energy measure. So there's something here to play with. Um and um it's still in pursuit. But with that said, I want to pause there and let Andrew uh take the reigns here and speak about the math of energy, free energy, and entropy. Andrew, >> thank you. uh Steph uh and thank you to the active inference institute. Uh we Steph and I met through the active inference institute. So I recommend if you want to meet people um that's the place to go and I am always interested in meeting people because I lead a investigatory community math for wisdom. So um check that out. Uh and so thank you Steph for all of your uh lifelong insights and intuitions. It's a wealth uh that uh brought you to Chi and Quai and brought you as an investigator to try to make sense, you know, does this work? Could it work? Should it work? What would that look like? What's the structure? And that brought you to active inference because as you said, there's these two currencies. One that seems related to entropy and scarcity, the quai, another that's related to energy, the cost of electricity. um and that's chi and what's the natural relationship between that in an economy. So this uh slide uh shows the summary of what I want to talk about which is um this equation relating energy equals free energy plus entropy. And so of course free energy is active inference's um valuable asset or and so we're um that's why we're here and so this formula is very good and I've always wanted to understand it better. So uh my theme is that um and this was said by Carl Fristen uh quoting um Richard Fineman at keynote address that uh uh if I create it I can understand it which kind of means like a lot of times I'm not understanding things I'm not able to create them. So some things I will tell you I know and some things I'll tell you I don't know. Um so the summary is that to explain this mathematical equation um and then to uh explain in terms of uh cryptocurrency the distinction between

belief in an economic system and belief in belief in the economic system. Those are the themes. And then uh I have a PhD in math but specifically algebraic combinatorics which is the math for counting things. But really what it means is if you look at an algebraic equation, can you say what it's actually trying to tell us? You know, what is actually it it what what does it mean really? You know, so the way I write these equations, uh I write them out like this. And if you'll see the entropy term has a  $q$  of  $x$  and a  $\log$  of  $1$  over  $q$  of  $x$ , but the energy term has a  $q$  of  $x$  and a  $\log$  of  $1 / p$  of  $x$  comm  $y$ . These are probability distributions. And the main theme of my um comments is that um here they're different but if you made this equal to  $q$  of  $x$  then you would get entropy. So if you this kind of leads to the theme that entropy is a sort of degenerate energy and so now let me talk about uh energy. I mean first I'll talk about entropy and then I'll talk about energy. So let's rethink these things. And uh you may have seen this formula for entropy. I've always been curious like why does it have a minus sign? Becomes very hard on my mind. Uh so uh it turns out that negative of a  $\log$  is just  $\log$  of  $1 / q$  of  $x$ . So first of all, you'll always see me writing this because I'm allergic to negative signs unless they really have a purpose. Okay, so this is my personal allergy. Um another thing is that and Steph saw that this is important.  $Q$  of  $X$  appears in two places. That's like one actor in two roles. Okay. So it's important to be not confused. You know that they're playing two very different roles. One will be knowledge gain and the other one will be the knowledge lens. So you can see there's a distinction here between an observer and an observe. The knowledge lens is uh all about the expected value and that's coming from an observer looking at everything through a lens of different probabilities they're assigning or sometimes they're called different weights. But what are they looking at? Well, they're taking the expected value of this term which is the knowledge gained. Okay, why is that the knowledge gained? Well, this is a way of writing and so let's remember that this is a probability distribution. So that means that these values you know  $x$  can be cat or whatever the probability of cat will be a number maybe like 30%. Uh but or you know  $o$  comma 3 uh or  $o_3$  if you're American I got living in 0. So uh  $q$  of  $x$  is going to be greater than um equal to zero and less than or equal to one. Now if you're very astute uh um uh you'll note that uh if this is zero this is going to be a big problem because one over zero is against the law. So um but here this could be zero. So all of a sudden we're seeing that these two roles start to imply different things that in the case of what I call the observed um you're not allowed to have zero probability. I mean nature wants it to be there or just don't talk about it. Okay. But from the point of the observer, you're in control. You can carve things up. You can say, "Hey, nothing happened here. I'm observing." Okay? So, there's a huge qualitative

difference and we pick up on this by caring about these things. So, let's go back to say this is a positive number and it's less than uh one. Okay? Less than or equal to. If it was one that means you're certain then it would be one over one. That's one. Log of one is zero. So when you're certain the entropy is zero, there's no I mean not the entropy but this part is zero and the knowledge gained is zero. So that's that makes sense. No knowledge gained if you're certain. Um but otherwise like let's say this is  $1/2$ . One over  $1/2$  is going to be two. And you can see that whatever this fraction is but if you flip it over it's going to be bigger than one. And when you take a log of something that's bigger than one is going to be bigger than zero. Okay. So these are positive numbers. So negative log of  $q$  of  $x$  is actually a positive number. Okay, that's the that's one point. And so let's look more specifically and we're going to be working in log base 2 because it's these probabilistic choices, you know, this or that. That's comfortable, but we could be working in other logs. But if you look at  $1$  over  $1/8$ . So  $1/8$  is a probability. You flip this over, it becomes  $\log_2$  of 8. 8 is  $2 * 2 * 2$ . We write it as 2 to the 3. And then um log base that is just pulling down that exponent. It's three. Okay. So, and that means that you have three choices. So, like  $1/8$  would be the probability of flipping a coin three times and getting something specific like tails and then tails and then head. Well, that's basically three bits of information. That's three choices, binary choices. So, that's three units of knowledge gained. Um and so when we rethink of entropy, we see that all these numbers are positive. Everything here is positive. Uh and then oh let's look at this knowledge lens and you just talk about that. So you could your knowledge lens could be how you're looking at the world. Let's say you're looking planetarywise and let's say it's very crude but you're going to say look it's one quarter land and three quarters water. Okay. So if you want to talk about the ruggedness or smoothness of the planet, you'd say, well,  $3/4$  is very smooth and then one quarter is ruggedness. So it puts things in perspective. But it could be, let's say, one quarter youth,  $3/4$  adults. Okay? Then you want to look at the loudness of music that's preferred at a festival. You could balance it in like that. Or it could be like the relationship between me and everybody else. So it could be like one quarter me in terms of how we're feeling and three/4ers everybody else. More likely it's more like three quarters me, one quarter the rest of you. Um so that's how it's balanced. So these are examples. You can see how then that would you know then you can actually calculate a case. Uh so let's say it's let's put together these knowledge lenses. So let's say it's one quarter. Oh to do entropy. You see, these are both the same thing, but they're in different situations. So  $1/4$  and you flip over the  $1$  over 4, that'd be four.  $1/4 * \log_2$  of 4 plus, let's say  $3/4 \log_2$ , you flip over the  $3/4$ s, you get  $4/3$ . So if we calculate it out, uh 4 is  $2 * 2$ . So log

$2$  of  $4$  is  $2$ .  $1/4 * 2$ . Uh  $\log_2 4$  over  $3$ , that's basically roughly  $40\%$ , which is  $2$  over  $5$ . I just made it I like fractions so I wrote it like that times three quarters. Okay. So the knowledge lens is the first fraction and then the the knowledge gained the surprising. So in the case of the one quarter like it's more surprising you know you get two in terms of the land part but the water part's less surprising. That's two-fifths of surprise on on on that end. So and then three quarters is how often you have that and you put it all together one quarter  $* 2$  that would be like  $1/2$ . I can write  $10$   $20$ th  $3/4 * 2$ - fifths that'd be  $620$ ths. You can add it up and you can see like who contributed more entropy. Actually, it was the land that contributed more entropy because if it was only one quarter, but it contributed a lot of surprise. Okay, so the land contributed more. The rarer one contributed more entropy. These building blocks the the water there's three quarters water but it's just not as surprising. So it's not collecting as much entropy. And so this is the expected value of knowledge gained. The knowledge gained from the land is two. Like wow, something happened. Knowledge gained from the sea, like if an asteroid hits into the sea, it's oh well, not such a big deal. Not not not such big news. So that's two-fifths. Um so but the chance of it happening, this asteroid falling on land or water, well  $3/4$  chance it's going to fall there, one quarter there. So you're kind of adding it up. And then maybe something I'll be talking about it's interesting this relationship uh  $10$  to six right  $10$  over  $20$  to  $6$  over  $20$  we'll be looking at that okay how that changes so why don't we do that why don't we go to the next slide because we're so into entropy we really want to understand it because you'll see I really I actually don't understand it so and I think it's a secret probably like no one really understands it and we'll we'll try to show that why but it's uh entropy is ambiguity though I think that part I kind of understand because and entropy means that um um well from a physicist point of view and I have a bachelor's degree in physics it means that you have more possible configurations that will give you the same result okay so it becomes like a combinatorial thing so it's very it's very that part that part's ambiguity and doubling entropy is something conceivable kind of straightforward let's do that and so we'll look at a specific case let's say there's  $n$  equal prob probabilities each with  $1/n$  and  $n * 1/n$  equals  $100\%$  is  $1$ . Okay, so the formula for entropy becomes this. Um you've you're summing over all the possibilities and each of these building blocks has the probability in front which is  $1/n$  times  $\log_2$  of  $1$  over the probability which is  $1$  over  $1$  over  $n$ . And you'll flip this over you'll get  $n$ . Okay, so let's look at it concretely. And we did this one. So eight times, but see now there's eight of them because we've got, you know, that was just a a separate building block, but now we're going to have all eight building blocks. And what you'll notice is that the  $8 * 1/8$  cancel. So when everything's very uniform,

there's nothing really contributed by the knowledge lens to entropy. It's kind of saying like we're we roll out, we're not, you know, we're not affecting the knowledge gain because we're just very neutral. We're so neutral, we're not going to really affect knowledge gain. So it's all left to the  $\log_2$  of 8. But you'll see that okay  $\log_2$  of 8 we saw it was three. If we do it with 64 which is 8 squared 64 you'll get again 1 over 64. That part cancels. That's just all neutral. It's 64 equal probabilities each with 1 over 64 chance. Okay. You'll look at the  $\log_2$  of 64. This is the knowledge gained. That's 2 to the 6 because it's  $8 * 8$  um 8 squared 2 or it's  $2^6$  cubed so that's 2 to the 6 so it's going to be six and so you can see it doubled okay so the lessons is is that in this particular case when everything's very neutral there's a the knowledge lens is you know has equal probabilities um then the only contribution is from the s what they what what we call surprise I call knowledge gained um and that in order to double it, you really have to square it. Okay, so that's a big deal. Like you have to square it if you want to double the entropy. If you want to double the knowledge gained and so basically like if you threw a coin three times, you got to throw a coin three more times. Okay? And then so that'll be squaring. So like the entropy if if you did a thousand computations or choices, let's say, right? And you want to double the entropy, you got to go up to a million. Okay. Or if you want to go up from a million, you got to go up to not a billion, but a trillion, right? Like, so doubling entropy is not something you would want to ever do really, you know, especially not in that type of Shaw 56 game, you know, where it's all uniform. Okay, so doubling's clear. Now, what about having though? Okay, because not everything's uniform. And so if you go backwards, okay, you can go backwards from 64 to 8, 8 to, you know, uh, well, let's say we go to four to two and whatever. So you get to two. How do you go have that? Right? Like where do you go? Right? So that's this diagram. If you only have two choices, one will have probability um  $x$  and the other one will probably have  $1 - x$ . So like if  $x$  is 10%, then the other one would be 90%. Right? And so you add this up. This will be  $x \log_2 1/x$ . This will be  $1 - x \log_2 1/(1 - x)$ . Right? Well, you can play around with that. And I drew this blue graph to show that because I don't know how to solve that and I don't think anybody knows how to solve this type of thing. Um, unfortunately, but we can graph it with my Excel spreadsheet. That's my computer of choice. And so then, um, or actually I use Open Office Calc. I can't afford Excel. So 0 to 100%. Um, and then uh, the entropy goes up to one. Okay. So  $12 \log_2 2 + 12 \log_2 2$  uh  $\log_2 2$  of two is 1.  $1/2 + 1/2$  is one. Do you can believe me on that if that was too fast for you? But the question is okay this will go down. So that's the peak. So kind of makes sense. Maximum ambiguity when both probabilities are the same.



Now we could get half the entropy. I'm sorry maximum yeah entropy maximum ambity. We could get half the entropy if we were willing to skew it. Okay. And it turns out that happens at  $1/n$ . So it's  $1/n$  this way,  $8/9$  that way for some magical unknown reason. But that  $1:8$  ratio and this is approximate, you know, it's these are real numbers. So but I try I like fractions. So  $1/n$  to  $8/9$ ths. Uh that is when this goes by two. And so I made this whole thing like if you want to go half again it'd be to get  $1/4$  it'd be  $124$ th to  $23/24$ ths. you know  $1/8$   $157$  uh you know  $158$  to  $5758$  and you can see the sequence going down and so that is a fascinating sequence um and these integers are not known I mean they're not in the encyclopedia of integer sequences okay so that means that this is treading new ground right but you could see this one attention because like in order to have the ambiguity that means that you somehow are able to skew things to get half Okay. And if we saw um and if you want to say well who's who's got the ambiguity part right like is it in this A or is it in the B part right so if A is the one ninth part or is it in the B part? Well it's um  $2.3$  of this will be  $2.3$  to  $1$ . Okay. But when you go to  $124$ th it's going to be  $3.2$  to  $1$  and then  $4.1$  to one and five  $4$ . Now this is very nice because you know instead of these huge numbers growing like this is growing slow first of all. So when you want to like we were doubling you know like when we were going up to double the entropy we're just going out like crazy. But now all of a sudden we're just getting a slightly more certainty by saying we're going to have a slightly smaller percentage and then a slightly smaller percentage and then we're going to feel like the knowledge gained by having is going to be halved. Let's see. I got to be careful. the expected knowledge gain let's say the expected knowledge gain we have so very curious but this almost looks like the integers but these aren't integers and of course you know these are rounded uh so but these aren't integers so there's a whole world I call these invisible numbers okay and so that's what's coming up so I went too long on this but now you feel like you don't know entropy and I don't know entropy we're we're even okay so let's talk about energy then so energy um I also don't know about energy um but I I know that there's Boltzman's distribution and it relates probability to energy and energy states. And so I think the crucial thing I kind of figured out reading this is that at equilibrium. Okay, so if you've got a system at equilibrium and it turns out this is a theme first step, but like that's useless. Okay, so in the useless case, which is kind of like the default case, like the death of the universe case, it's just a a uniform temperature and you just can't get a hold of it. just things bouncing around sharing that energy and they're just okay. So in that case um you can tell how all the energy states are going to distribute themselves and then you can look at the sum. So each energy state would be this separate thing. You can look at the sum

that's called the canonical partition function. I'm so awesome. And so then if you take that percentage that's a probability. I am going to do a little trick here. This is called  $Z$  a lot of times, but I'm going to just put in this uh curly  $E$  to say really you can just not ignore this  $Z$  if you just put in this curly  $E$ , you know, because it'll be  $E$  to the it's just another  $E$  factor basically. Just convert this number. It's a number and you just convert it to that kind of number and you so the point of that is that it's like a ground state energy. Okay. So will you know so  $e$  to the  $i$  minus epsilon is just recalibrating the energy to a ground state. You're just ch moving it up or down you know to so that you don't worry about. So that's the first thing we'll do. Now let's look at what that means then in terms of flipping it around like because we want to know the energy. So if we flip it around we go okay let's take the log that get rid of the  $e$  right. So now the log is equal to this difference of energies over uh Boltzman's constant times the temperature. Well negative we've already seen how to handle that. It's the log of  $1/p$  to the  $i$   $p$  sub  $i$  and that'll be that. So wow we got our knowledge gained. We see that it's a difference in energy compared to the base energy and divided by temperature. So I'll be writing like energy divided by temperature etc. and it makes things more messy but it's basically true but actually you can this is just a multiplicative constant you can get rid of it if you introduce this number  $B$  and these just numbers you know so you can introduce it like this um and you put  $E$  to this Boltzman constants times the temperature it's all flipped over that can be the base for your logarithm and so if you just say look I'm not going to work in uh natural logarithms I'm going to work with this log with this case okay then the whole thing goes away okay so and it's really not temperature it's one over temperature and it's you know got this constant in it so one over temperature is in a certain sense more natural coldness is more natural than warmness because coldness when things are cold and there that means they're moving around very slowly uh they're much more receptive to energy they're actually more adaptive like you give it a little bit of energy and it will give a lot of flexibility a lot more increase in ambiguity which is to increase a lot of increase in possibilities if you just give it a little energy. So like in the human body, everything's slowed down. That's the point. Everything's controlled, right? So that you give it a little bump and things could happen in different directions. That's really the the derivative of um of uh entropy versus energy. That's the way you really want to talk about. So you really want to talk about like one over temperature, but you can think of that as the interface. Um, and and that'll that's maybe I'll say more about that later, but it's really just the interface like you're playing with the speed of things. So, you know, if you want things to be just play with the speed of time, so to speak,

that'll change the temperature to hot or cold or whatever. Okay, that's just a maybe not so important. But see, basically, if we do that, then we can take we don't just have then the information gain, the knowledge gained, but we have the expected value. So let's take the same probability and let's take the expected value and that'll be the expected value of energy. Okay. So when we say energy sometimes we mean a single state but we some like look over all the states. Now if you look over all the states with the same probability well then you're getting entropy. So expected energy is entropy. Okay. But that's because we're at equilibrium. Okay. So that's a crucial thing about Boltzman's distribution is that it's not really it's it's like when you have this special case the default case where like um uh we're going to look through the knowledge lens of our probably like we're in our own messy room and it's just I I don't know what to say like it's as messy as it's ever going to be like there's no one there's no mother to make us clean it up let's say there's nobody to we're just ourselves right our natural selves kind of drift to this state where like um it's just like that so but in in disequilibrium. You see, we're going to make these different. We're going to say no, our mother came or our father came to be more maybe to blame our fathers, not our mothers, let's say. And then they said, oh no, you can't, you know, what are you doing? Right? So, that's the expected energy. I expect you to clean it up. That's right. So, this tension. Okay. So, now let's get to what we're talking about because we're so smart and so ready. Um, we want to explain this equation down here, which I introduced us with. It turns out it's super easy to derive. Okay, but I think conceptually boris because we're going to just start here comparing these two probability distributions and this line is says it all. It's just so trivial. It's so trivial you won't understand it. So I'm going to give a little story to help you understand how trivial this is. The story is that it's a dialogue between two people. I think Steph says that's maybe the easiest way to understand like me and Steph, right? And so uh but I like for me it's just more visceral two minds the unconscious and the conscious. Okay. And so a lot of times in active influence we see the X and the Y. The Y is the evidence. And I think of that as the sensory mind. Okay. And a lot of times we write it for P. We don't write it for Q because P is the probability distribution that's tapping into that sensory evidence. Okay. But there's also inferred causes, right? Like so maybe the evidence is that there was a shadow but the inferred cause is that I'm walking under a tree right so um so the unconscious notices everything right but the conscious may not care about the shadow but cares about the tree because I don't want to walk into the tree right I don't mind walking into the shadow big deal so uh so then I have this dialogue between these two so the unconscious mind is like 100 billion neurons giving me all these answers you know every single pixel

Right? Everyone's got an answer. They're all pandemonium as they call it. But I'm trying to compress that and this is all I know. I know all these things. But I'm trying to say no. It's smarter to think in terms of what I don't know. Tell me Andrew like what you don't know about entropy. Right? Get to the point. What you don't know about energy. Right? So I boil that down to 100,000 black boxes, cortical columns, um questions, words. Every word is something I don't know. Like so if I say horse, I'm really saying what is a horse? And then the other mind tells me, oh here's a picture of a horse, here's a fact about a horse, here's a horse, you know, etc. So what we have here, and this is like an autoencoder, and this is from the '90s when I was at school at UCSD. Um, you compress all, and this is something a computer could do on its own even back in the 90s. It could say, look, the computer doesn't have to be trained. It's just going to try to randomly, you know, by trial by error compress things. It's going to compress them and decompress them. and it's going to see how it happens. So, it's going to compress this information into this language that's more sparse and and pure and it's going to decompress it, de decode it, encode and decode. Did it get the same thing back? If it did, that's valid. Was that a better compression than what you had before? Okay, so that's a typical game. And so the idea is that I have a chasm an abyss between the mind that knows which is basically representing the whole world which is basically like you know my unconscious just tapped into my body tapped into the world on the other side I have a mind divorced from the world you see how I like I'm abstract thinker whatever I just have no clue right but by being a tricky question thinker you know embracing my not knowledge we work it out we got the same information two forms now so imagine that dialogue and now let's say okay I've got the sensory information. I can use my conscious mind to slip it in here and break it up into two factors. Okay,  $Q$  compared to  $P$  and then just one over  $Q$ , you know, so I can multiply by one. I can multiply by  $Q$  on top and  $Q$  on the bottom. I can do that, but it mentally breaks this up into two different things. And that is it. Okay, that's all it is. It's just a slick using of the  $Q$  to break things up. I can take the log, it makes it additive. So instead of multiplying, I'm summing, which can be nice. I do my thing where I flip everything upside down, right? So  $Q$  over  $P$  is the same as  $1$  over  $P$  over  $Q$  because I want to understand what I'm reading, right? I want to be able to create it. So if I write it this way, I say, oh, that's knowledge gain. That's knowledge gain. That's knowledge gain. But there are different knowledge gains. This is  $Q$  of  $X$ . This is  $P$  of  $X$ , comma  $Y$ . This is how they compare. And now I say, now I'm going to add my knowledge lens to the whole thing. So now with the energy and this is obviously disequilibrium you know the the observed world the sensory world is the knowledge gained

but the knowledge lens how the observer is looking at it is the Q of X it's the conscious is observing the unconscious right and when you do that though when you play this game um and when you break things up like that you're going to get two parts and that's the interesting part like you're going to get a useful term term and a useless term you're going to get a disequilibrium term where you can actually do work because there's this tension I can clean up my room and there's going to be the lazy day where I don't okay so there's lazy and is you know you can't have and clean up his room if you don't allow for lazy and that's the law of the universe right so so that's basically what this is saying and so you can do the same type of thing here uh maybe just to go back so the free energy is saying okay my father wants me to clean my room How does that look like to me? Okay, that's the free energy. How does it look like to me? I don't see I I don't see the notion of cleanliness. That's like where did that come from? And here's the here's the total picture of it. Okay, where it's not about me, it's about reality. Cleanliness is a real thing. Okay, but here it's like how I look at cleanliness. And then here's just me. Okay, so now we're going to say you know you can take this equation which is more common you know with the free princ you can break it up into divergence and knowledge gained or also known as surprisal um I'm no longer surprised I'll just call it knowledge gained so um but it's just this formula here  $P(X, Y)$  you know we're looking at two things the evidence and the knowledge um inferred uh state you know so be a state or an action whatever It's if you're not familiar with this notation when you when you have the probability of two things it's just the probability of the one let's say the evidence plus given the evidence what's the probability of the other okay the the cause. So it's probability of the shadow and then given the shadow what's probability of the tree. Okay you just break that up now. And then if you just do that you flip everything over and then you've got the free energy breakup. And there's other breakups I think it would just work the same. If you noticed, lazy and didn't use basis theorem. Basis theorem would say, well,  $P(X \text{ given } Y) * P(Y)$  equals  $P(Y \text{ over given } X \text{ time } P(X, \text{ right?})$  And then you can flip over and you can say, well, then you can say, you know, we want to do priors and we want to do it, but we can't. It's all intractable and so we're going to flip it all over. So, we use basis theorem, etc., etc. There's a whole story. Lazy Andress said, well, lazy Andress and the Andrus, the Fineman Andress said, look, if I can't create, I don't understand it. I was trying to create all that and I just got this and lazy said why do you need all that right that's the problem with me so let's now apply this to cryptocurrency uh we worked hard time trying to make sense of coin chi and etc and what's the point and the c the cool thing is is that um it's um question of like does it make sense to

have two currencies and you can see just like energy has kinetic energy and potential energy where kinetic energy is kind of useless it's the energy of moving things around bumping into each other. Potential energy says there's like a unifying thing that everyone can respond to. It's not local. It's a global thing like my father. So um uh just like in that way you can have two currencies and in particular the particular problem with Bitcoin because there's only one currency it's priced in the dollar. So if the goal of this in Steve uh Steph's heart is to uh say hey can we have a civilization a world where um it has a economic currency that's not um um that's that that that's not depending on fiat currency. It's not this current system. Can we have an independent system that's not dependent on nation states and powers etc. Right? Well then it's got to be able to price itself. So it does kind of need two currencies. Right? It shouldn't be priced in the dollar or whatever. Um, and so there's a need, but on the other hand, like in order to get people to want to support this, they really want to win, you know, by mining and things like that or investing or whatever, they need a currency that allows them to speculate, but we also need a currency for everyday things. Well, this comes at a cost. Okay, so if you're not familiar, like Bitcoin mining, uh, I think LA recently, it's like \$20 billion a year. Okay, so \$20 billion a year just up in smoke in electricity costs. It's huge. Now what are you getting for that? You're getting a \$2 trillion economy. Okay, so that becomes an interesting ratio. Okay, like but per year like so 1% per year you have to devote just to go up and smoke but then you get this \$2 trillion economy which is not a joke. So that ratio became very interesting like what does that mean? And we compared that uh to the price in dollars. And this is long because you as you can see it it starts at \$1, \$10, \$100, \$1,000, 10,000, and at the very top it's \$100,000, which is like the current price for Bitcoin, changing as I speak. So if you do a log chart, you can see like there's ups and there's downs, but like that up is a huge up and that down is a huge down, right? I mean, it went from a thousand down to maybe 200, right? it went from 10, you know, 15,000 down to maybe, you know, 6,000 or something, you know. So, it's going up like that. What you'll notice, uh, if you look at that ratio, mining cost market cap, it swings, you know, so it goes down to 2%, it goes up to, uh, 7% and then down to 2% and 6% and so on. It's swinging up and down. Now a very curious thing is that when this mining cost is high comparing to market cap I mean which means lots of people are spending a lot of money mining right compared to what it's actually supporting okay that's when the price is down so there's a curious thing like when the price is and you kind of make sense you know we saw this whole the the the econ the economy of bitcoin being this that there has to be an even rate of creating these bitcoins if and it's a contest it's these lottery tickets like Steph was showing

you When there's a lot of people wanting to get the lottery ticket, the lottery ticket has to be hard to get and becomes increasingly, you know, exponentially harder. And when there's not a lot of people, they got to make it easy so that people would mine it because they need a steady rate of these bitcoins being created one every 10 minutes or so. Okay, maybe one every 5 seconds. So, uh, that type of thing. So, uh, what's the connection? Okay, and I really don't know. I don't know. But um it's about the cost of doing business I suspect. So if you want a civilization then you know you look at the Egyptian primitives pyramids let's say what is that all about? It may just simply be the cost of doing business. Like if you want to have a well functioning civilization you say look we're going to build pyramids right there's got to be no reason for that. It's just simply that that's how we organize ourselves. Okay. And if that's 10% but that can elevate everyone's standard of living why not right? our um society is built on fiat money, the Federal Reserve system. There's these obligations to have exponential growth in uh currency in in in in economic value. Like it's, you know, we got to grow by 2 or 3% a year. Uh which is obviously unsustainable, but we're forced to do that. And if you can't do that, then you have to go uh find a ghetto where you can uh have victims and then you can sign them up for telephone plans that they never ordered and things like that. So you have to have uh what some presidents call some kind of countries, let's say. You just have to have all kinds of horrible things in order to have this moral cast system, you know, and to have that, you know, where everyone can look down on somebody and not be so unhappy about themselves that I'm a telemarketer, but I don't use drugs, you know, I use drugs, but I buy them with money, you know, etc. So horrible thing. That's part of the cost of a civilization, you see. Now, what if we didn't have to have, you know, wars and uh, you know, 5% military expenses, whatever. What if we just could make our own money, right? Okay. So, we burn it up in smoke. Okay. So, maybe that's not the best system, but maybe it's better than what we have, right? I think that's an important thing to say. So, this cost of doing business um, and then it what it becomes, this is kind of like one of my concluding thoughts or maybe my few thoughts about cryptocurrency, but is this distinction between belief in the system and belief in belief in the system. So the sensory system believes in the world there's something there's an answer I got it but see these miners they believe more there's no economy there like you know in she inquire when bitcoin started out there's no no there's nothing being used but people believe that they should be mining it because they believe this thing has a future they believe that there's something there in on it okay they drank the Kool-Aid okay so it's belief and belief in the system so like with our active inference community you can drink the Kool-Aid and then

you're like these miners and you're saying yeah I'm going you know, okay, so that's that's in an early stage. You know, people drink the Kool-Aid, they set this up, they invest millions of dollars in a new community currency, but then what happens? Maybe some miners say, "Hey, this is I'm not getting anywhere with this or I just stop believing, but they got a bunch of coins left that they have to." So maybe those coins, they try to get something with them. So now they believe in the system. They believe that there's these coins. They just don't believe it's worth mining, right? Or it's not their problem. Now in a healthy economy, everyone just believes in the money. Let's say, you know, I believe in dollars, but I don't believe that money is the point of life. But like I believe, okay, I'll function with dollars or or euros or whatever currency. I believe in the system. I don't necessarily believe in the belief in the system. And so it's kind of like with acceleration versus velocity. If you you know, you throw a ball up in the air, there's a certain point which just hangs, right? Like it has no velocity, but there's acceleration. Okay? So that's like starting an economy. like there's no point to it. There's nothing being exchanged, right? But there's acceleration. People are investing in it and then the question is it shifts over like is there a real economy or not? That becomes the question. So that's very little said about cryptocurrency but I think it's you but you think like we we'll have a discussion maybe there's more said than there but I did say a lot about um the math of active inference. I gave an alternative approach. Okay. So we both with the customary approach we both agree okay there's a distinction between evidence and inferred causes let's say right  $y$  and  $x$  but you see uh in the traditional customary orthodox um model there's a generative model and it's predictive and then  $q$  of  $x$  is just an approximation just an approximation you see whereas in my case it's like look uh  $q$  of  $x$  is the whole point the whole point is to have a dialogue right the whole point is to have your own mind right so it's this these should be different and if  $Q$  of  $X$  was a perfect approximation that's like encoding yourself with yourself you know one to one with the same alphabet would be absolutely useless so to call it an approximation is to suggest that a perfect one would be great you know which is but that would be absolutely useless so and then to say approximation is it only serves this and so then also for historical reasons I think this is for historical reasons the generative model was considered the predictive model. So we have to predict where all these shadows are instead of saying why would I predict where shadows are? No, I'm just going to predict where trees are, right? So why do I have to care about shadows? So this  $X$  in my like  $X$  would be really actions. What are the possible actions? So I'm stumbling around the dark and then here's all the information of what I should be doing, you know? So I'm poking my head around. I've got



concepts. I've got conceptualization swing around and I'm trying to so this X really is my sensory information saying look your arm moved you've got neurons telling you your arm moved or you can't move or whatever it is and there's some kind of mystery you know magical combination why these two things divorced with each other somehow they look at their relationships somehow they're able to communicate or you can believe in the optimization ferry okay so the free energy principle saying oh there's all these approximations out there but the optimization fairy is going to pick the one that has minimal free energy, you know, by blah blah or maximum free energy. I always get confused. It's going to do that. You see, so you can choose two stories narratives. And so they talk about like if you notice in the math, there's no markoff boundaries. There's no markoff anything. There's no I mean I the beautiful idea of free of active inference is that you have action perception loops. You can update you know when there's something off you can update your model. You can update the world. But none of that is here. None of it as far as I can see. So, uh, there is a gap though between these two. So, these are two ways of looking at it. That's my conclusion. U, I hand it over to you, Steph. >> Thank you, Andrew. That was, uh, I mean, you covered some serious ground. And, uh, thanks for filling in a few of the the gaps that I forgot to add in. Um, so there's a question in chat that I wanted to make sure that you got to answer which was what does it mean that entropy is a degenerate entropy? >> Uh, yeah. So it's just maybe my mind is degenerate. Maybe that's entropy. Yeah, it's a degenerate energy. Right. So if I go back, you can see my slides, right? >> Correct. >> Let's go back to the very beginning. what I wanted to show. See this looks just like that, right? I got I mean they have the same form if you write this way.  $Q$  of  $X$  and  $P$  of  $X$  comma  $Y$   $Q$  of  $X$  and  $Q$  of  $X$ . So all I'm saying is that if you picked  $P$  of  $X$  comma  $Y$  to be  $Q$  of  $X$ , you know, you could pick it to be all kinds of things, but if you picked it to be  $Q$  of  $X$ , then you would get entropy. And so what that means in practice is that um there's a useless way of looking at things in terms of getting things done or working or whatever. It's like if you pick the window for the window you're going to pick on the world is going to be the window that you're of how you look at things in the world, right? The the window you're looking upon the world is the window that you're informing yourself like of your reality, right? Like if you're talking to yourself in this way, you're not going to get anywhere special basically. You're just kind of like u you're just like bumping around against yourself. Okay? You're just you're just at equilibrium. You're not going it's not going to change you maybe to say like that. It's just going to keep you who you are. You're not going to clean your room because of that. Okay? So maybe if you're looking for self-motivation and this is maybe related to Alex Kefir's

thing is like well how would this relate to self motivation? We can see external motivation operating. Oh, so there's a third level I would say would be expected free energy. You see the the orthodox church of you know active inference if I can consider myself a member you know is is that in order to compare policies you have to go back into the sensory world and imagine okay if this is going to be a shadow and this is going to be a shadow here etc etc this is bizarre no if I was going to compare policies I would want to compress yet again instead of 10 you know 100 thousand concepts I'd want to have maybe 10 different parameters that I could compare against to say Maybe this is about speed. Maybe this is about decision-making. Maybe this is about morality. Maybe this is about friendship. I don't know, you know. So, that would be how I would be wanting to get it down to not saying like, well, do I like the smell of this cheese or something? That's a very strange. See, that's why I call it a religion because those things are baked in because Carl Fristen looks at the world this way and we're just kind of going along and I just think that that's science fiction. It's fanfiction. That's my personal belief. I can make up my own fanfiction and then we can have two fanfictions and then we can talk about a science that distinguishes between science fictions. So, I don't know if I answered that very well. >> Thank you. Do we have uh what four minutes to wrap up? Daniel, >> you can make any last comments or stuff. I was just curious what is the uncertainty that's being reduced about what to whom? because the zero stacking up at the beginning of a hashing algorithm to me that's like flipping lots of 100 coins and then if you get 50 um heads or 80 heads or whatever the the threshold is you just keep that lot and you discard the hashes that didn't get it. So, I don't know if that process ever updates any beliefs that I have other than just the the um hypothetically nonforgeable work of flipping those batches of coins. So, what do the zeros leading up or other kinds of hash plays lead to actually how do you see that in terms of belief updating and uncertainty reduction in something? >> Yeah. Well, a good question and let's see if I can hit it. The it's it's essentially establishing kind of this like arbitrary boundary where we say let's just play with what a Shannon entropy measurement could be in a single hash output. So all we're caring about, all we're thinking about at this moment is 256 bits and we have to employ an arbitrary rule. So we could look at leading ones, we could look at leading zeros, we could look at pairs of ones and zeros, but so that's what I'm saying is an arbitrary rule used. But in the kind of state space of that single output, if we were to determine that if we were to to arbitrarily determine that the leading number of zeros is what we cared about as a way of saying um that's a that's a known state or an unknown state. We're saying we're going to accumulate a number of known or let's say accumulate a number of unknown

states and what is the expectation that we should at the next interval overcome that amount of unknown states. So um the leading number of zeros is like let's say we have 50 leading zeros that to get 50 leading zeros is a really uh improbable event and that's improbable relative to the typical output. that has some some number less leading zeros. And so if we're comparing a very un improbable event to what we expect from that typical statistical distribution, then we can arbitrarily just measure the difference of that single rare event and what we typically expect. And once we have that then we uh kind of say that was the that's the uncertainty that difference between the expected average and the improbable event. We're going to say that's valid. We're going to reward for that. Um and we're going to hold that and add it to the previous amount the the the one that was done just previously. And so we're accumulating those bits that that difference uh at each at each step. Let's see if that answered. But that that was my my immediate attempt. >> Cool. I will look forward to learning more. Thank you both for the presentations and and great to hear the stories and the perspectives. >> Thank you. Thank you, Andrew. Uh huge ground covered. Love it. and let's catch up uh very soon. See you. >> Peace. Peace and love. Bye. Um, welcome back. We're moving straight to the second session of the day, interactive math tools for active inference education by octopus. Uh, remember you can drop your questions in the live chat through uh the presentation. So, Octopus, please. >> Hello. Welcome everyone. Uh thank you so much for coming. I'm very excited to discuss some things with you today. And the thing that I'm um wanting to talk about is the connection between three things that I find very interesting and two of them I would say I know relatively well. And those three things are active inference, math and education. So, I've been in math and in education for a while now under different names and different personas, but always extremely interested in it. And I think there's an obvious connection between active inference and education through the some of the major concepts here. So, I'd like to talk about that first. Um what I've seen in the last few decades is a major shift in how education is enacted. And this is going away from a hierarchical top-down model where essentially the teacher enacts command and control on the students towards a more decentralized collaborative model. um this decentralized collaborative model was partially forced uh by co but I think it was also in the making already and the active inference institute I believe I hope it's uh not a mischaracterization in the eyes of the organizers is part of this movement where it's not just one expert who is solely lecturing like I'm trying to do right now, but rather it is an interactive and decentralized model that leverages all of the nodes that can exist within the network. How does this relate to math? Well, one of the things that traditional

education does is assign grades. And already there's math there. we have a problem in grading which is the fact that uh grading runs into this issue first of all it's it's rewardsbased and inherently it runs into good heart's law so if you're if an A is the reward for good being a good student but then being a good student becomes about getting the A there's a distortion effect that's inevitable but even more we're taking arbitrary measurements and and then further making them arbitrary and distorted by compressing them into these bins. So that the difference between an 89.9 or an 89.4 or an 88.7 can feel very very determinative and even destructive in the learner's future. I don't necessarily have a good answer to that except to look at what is the purpose of grading and the purpose of grading I believe is to provide feedback to the learner as to where their progress is. So if we think of grading not as a labeling system but as a feedback system then it can change our viewpoint of it and the whatever purpose giving letter grades was intended to serve um can perhaps be reimaged into other systems that provide feedback on lead uh learning. And so instead of thinking about a classroom either in a high school or a university or however else we want to think about it, I would like for us to think about learning as occurring occurring within a learning system and to think about the different ways in which learning systems can be designed. So already this is an application of active inference principles because I mean I hope this is getting somewhere that's clear to people who are interested in active inference that uh active inference is about how agents can learn or at least that's one interpretation of it. And so if we're thinking about a learning system and we're thinking about the individual agents who want to learn regardless of their roles, then we can start to leverage active inference principles towards this process of education. Why am I specifically interested in math education? uh besides the fact that it is my my chosen interest um I think it's the place where it is easiest to fall into the grading distortion trap and what I mean by that is that we often can conceptualize math problems as having a right answer or a wrong answer and uh the then the system might be enacted as our goal goal is to get people to get to as many right answers as quickly as possible. And my personal perspective is that that's antiquated. It may have worked well at a time when the job of a computer was fulfilled by a human because you didn't need completely accurate answers very quickly. And while there is still some value in that, there's so many more things that can be done with math and the type of feedback we want to give ideally in my view to enable the full range of play within that space. So I have no idea if people are even still hearing me. And I'm just going to continue to rant in this vein um because I don't think that the feedback loop is or I don't think that the opportunity here is one-sided. Obviously, math is an important part of active inference. And

for many people who want to learn active inference, they're they're uh stymied in some sense by the fact that so much of the presentation is mathematical. And while you can do work around that by speaking to things at the conceptual level or the verbal level or the graphical level, it would also be great if we could solve the problem a bit more upstream by making people comfortable with the mathematics as it's actually written qua math. The other thing is there's also the fact that active inference as a discipline I believe wants to grow and that part of that is uh engaging in effective educational activities of which the active inference institute is one exemplar. So this is the frame for the rest of the things I'm going to be saying from this high level conceptual view. Uh we're going to switch to a more technical and tactical view which is one approach that might be used to present active inference and specifically the math of active inference within an educational framework. So the brief lesson or tutorial or play session that I will be leading is about math and it's about specifically the math of active inference. But let's also look at how active inference not only sits inside the lesson but how the lesson sits inside active inference and how these tools might be leveraged so that we can use active inference principles to teach and to learn more effectively. Ordinarily, I would pause for questions, and I'm not sure if that's appropriate here or not. Um, >> octopus, if people in the live chat have questions, they can write it, but this is awesome. It's a great great topic. Continue. >> Thank you. So, I appreciate the feedback. Um, so what I want to switch to now is a specific lesson and I'm going to put the invite link so that you can participate in the lesson in the role of a student and I will be participating in the lesson as the role of an instructor the way that I would give this if I were in an institution uh tasked with with teaching a concept. So let me clear that >> and I'll put that link in the YouTube live chat. >> Awesome. >> Yep. >> Well, I am going to >> just wanted to add for context that in the textbook group that you were facilitating the math learning group. >> So we you you know you buried the lead there. >> Oh well that's true. I do have some experience um did several months I think of of working with communication about the mathematics of active inference. And so this um is coming into uh that informed my perspective here of actually let me let me speak about that experience too um while I work out the technical things So there's a mode of the mode of mathematics education that I discussed a moment ago. How can we get people to as many correct answers as quickly as possible? My experience of trying to work within that mode is that it necessarily um or almost automatically creates a sense in which people feel threatened emotionally. Most people have a a horror story of when they stop liking math. I have my own horror story of when I stopped liking math. I'm just thankful that I uh also have a return story of when I

started liking it again. And for a lot of people who have this horror story of when they stopped liking math, it was when suddenly they had too much too fast in the words of the Grateful Dead. So if our goal is to get is to determine who can get as many correct answers as quickly as possible. Then if you don't do that, there's a risk that you will be made to feel stupid, especially if you're a young child who is put into this intensive pressure process. And I feel this is incredibly destructive. um not only at the human level, personal level, but also at the institutional, societal and discipline level. And so, uh, a lot of the work I'm I'm doing is about, uh, bypassing an aversion, recreating or creating experiences that enable math to be viewed as a play-based activity where if you get it wrong, you'll get it right next time. or if you get it wrong, you can at least still enjoy the experience and maybe when you come back to it next time, at the very least we won't be afraid of it. Um, so I'm going to if you go to [student.amplify.com/join](https://student.amplify.com/join), [amplify.com/join](https://amplify.com/join) I believe and type in this code MN7C and I will also copy it and put it in the chat um locally that will enable you to join this activity in the role of a student and there will be different people who have different levels of knowledge around the topics that we're discussing and what I'm going to ask is that you enter this activity in a spirit of play. What do I mean by that? Well, I don't mean destructive play. Uh I will be seeing the responses and so if you type obscurities or things of that nature, um I can't stop you from doing that, but it would certainly be a negative experience for me and I think for the other participants. Um I would just not show those responses. But uh in play, what I mean is that we're going to intentionally set aside our seriousness and our internal pressure. I have chosen to do that by bringing along my second octopus head. Um I enjoy that. So I'm bringing that joy to this experience. But secondly, another way we might play is if you are an expert in this field and you know all the answers and you could get the right answers really quickly that you seek a different objective. So if you are an expert, if you if you could write a test on active inference, then maybe you could try imagining that you are a complete beginner and instead of giving yourself the challenge of correctness, you could give yourself the challenge of empathy, which is a very difficult metacognitive problem, I would say, and also a very scientifically applicable one because an agent who can engage in metacognition can enter the mind of another agent. And that's its own active inference problem, isn't it? I hope some of what I'm saying here makes sense. Um, I'm going to begin teaching by clicking on the teach button and this will enable me to see the responses that people are giving as students. And hopefully this will this will maintain under civil decorum. We will see. So, I'm going to give people some time to join here and uh just consciously pause for a few minutes. Dead air is unacceptable in

radio, but in education, we don't call it dead air. We just call it quiet. And so, it's okay here if there's some quiet in the moment. I can see that four people have joined. Um, I can see your names and I should have warned you about that, but I don't think anyone has deanonymized themselves. Um, but I could anonymize people and these become the names of famous mathematicians or scientists. And I personally like anonymize mode better. The No, I also buried the lead on this. Um, the tool that I'm using is called Amplify. When I was first introduced to it, it was called Desmos Teacher. Uh, I haven't used it in several years, and so I since I ceased being a classroom teacher. Um, but I've been playing with it a little bit in anticipation of this presentation and um, I will be showing some responses anonymized. I'm going to shuffle. And if you're concerned about the fact that real names are visible for a moment, you can leave and come back uh just by uh reusing the same link. So I deeply appreciate the people who have joined so far and encourage anyone who is interested to join. There's still plenty of room. And the first question I'm asking is what describes your interest? there's an x-axis here of interest in active inference. And there's a yaxis that says I'm interested in education. And if I switch to student view here, hopefully I can have some empathy for people who didn't um descri join this uh who didn't design this. Um there's a star and I can drag the star to where it represents my place on this XY axis. So if you're a complete teacher and or if your focus is only on teaching maybe you I don't maybe some teachers knew about this and are coming um but they don't know much about active inference then you would be up in the close to the y- axis. If you are completely interested in active inference and have disdain for students or anyone who doesn't who is learning the subject, um you might be closer to here. Uh if you're the session organizer, you might be really high on the y equals x line. But this enables us to get a sense of the current audience that's interacting with the presentation. Uh, and I like this as a as an introduction. I feel part of the feedback loop, part of the information that the agent is seeking to accurately update their beliefs. is whether or not I as the teacher agent am willing to seek information about their beliefs currently or about their worldview or their position. And so again that's metacognition. Uh I don't agree with the quote that you know I don't agree that emotional effect is the only effective of component teaching effect component of effective teaching. Um but this isn't manipulative. It's a place of genuine curiosity. So I am curious about the audience that has come so far. And let's see what the audience looks like. Should be the case if I did this correctly that I see multiple stars showing up. And uh it looks like I do. You can see here hopefully the smaller dark stars are different individuals. And so it looks to me like most of the people um positioned themselves as being

mostly interested about active inference primarily, but there are people who are knowledgeable about active inference and are interested in teaching. And so I can see the two clusters. Now, if this were a quote unquote real class, there would be a decision point for me. Do I want the class to know that there are two clusters? Since this is a simulated class, um I think it's it's appropriate to reveal this. Uh but if I were in a situation where I thought people might be having some of their personal risk, uh they might not want people to know that there's someone here who's not as interested in education. And that's a design choice that I as a teacher agent would have to make. uh the next screen. If y'all would be willing to leave that screen and come with me to the next screen. And the student view of this simply says make a prediction. Predict a value 0 through 10. And then there are only six options 0 through five. So what I might be maybe this mistake was made on purpose. Maybe this mistake is the result of not checking carefully. Sometimes it's good to make mistakes on purpose. Vi Arnold, the great Russian mathematician, I mean the the great Russian mathematician. must not underell the great there. Uh he was well known for intentionally injecting mistakes into his uh lectures to to make sure that students were paying attention and I am told that he would even allow these mistakes to compound that if no one caught an intentional mistake and and called it out that he would continue with the with the wrong assumption until someone caught it. I'm not willing to go that far. Uh it does feel to me that there is a place uh where as the teacher agent I would need to step in and point out there is a mistake. But I would certainly view the fact that no one had caught it or that no one had felt comfortable enough to challenge the teacher agents quote unquote authority. that would be very helpful to me um in updating my beliefs that were if in terms of taking future actions. So I'm not presenting hard quantitative things about this but um I'm trying to bring active inference language to a process that I use in uh teaching mathematics hopefully effectively. Daniel, is this making sense? >> Yeah. >> Okay. Feedback helpful. Um, so I'm curious though how the students respond to the fact that the predicted the task they are asked to engage in doesn't actually match their set of expected experiences. Is this distressing to them? Do they not even notice it? Do they think it's funny? All types of responses. And if I could edit this in live time, I might even add a poll of how much did it bother you. What what I'll add here, octopus, is that um you're you're definitely right and have experienced that with the math psychology around active inference. And it can feel like there's some inner math sanctum or some equations on pedestals and it's something other than learning or applying active inference until that equation is like deciphered. And when you said that active inference is in the lesson and vice versa, it's like this is already the process. And even just clicking next



on a page, just regular actions that we might take at a at a second time scale or just through interface with computers or in conversation. Those are kind of the conceptual moments that are the kind of like the moon, not the finger. And then that helps the equations which the symbols could be different letters or they could be different sounds or shapes but those are describing something which is like the cognitive situation with prediction, observation, updating, action selection and so helping whatever the content were helping pull that thread of active inference and then it's a double treat that the lesson is also about um KL divergence or whatever it happens to be which helps us understand the cognitive scenario at an analytical level but from an experiential level just learning something the more novel and surprising the better. That's certainly how I feel. I have almost an insatiable need for novelty. I feel it's the more novel and surprising the better as long as it's comfortable enough. So I wouldn't probably wear my second octopus head to a high school classroom. Or maybe I would. I don't know. Um, but there's, and maybe this is an active inference concept that's wellknown, but there's a level of novelty and surprise that can be overwhelming, at least to student agents in my experience. Um, how does that show up? And I want to get back to what I had planned, but now you've you've said something that I'm I'm wanting to know more about. Um, does that show up in the sense that if the information is so surprising that the agent doesn't have a way to integrate it, it puts their survival at risk? Is there any way to describe that within the active inference framework that like this is distressing to me how un how un uh unexpected this event is? Yeah. I mean a lot of ways to go with that. I think of one situation where we have a flat prior preference. So we're open to all outcomes and don't have a prior preference on which would be better like all options from the menu. It's kind of like the flat prior distribution from the previous session. And then in that situation, novelty and learning can be a dominant imperative. But then if we think about a distribution like homeostasis of temperature or blood sugar where it's modeled as there being a strong preference when you're outside on one of the far tails of temperature or of blood sugar then it threatens the the autopoiesis and the the life of the agent itself. So that kind of goes outside of an information band into a sort of survival mortal computing. Is it correct to say that how it feels to the agent is about what their expectations were or what their um prior beliefs were. I mean let's be honest many students in many students in a learning system let's say a state controlled learning system are of the strong preference not to be there and uh I always wonder how much that plays into it like John Taylor Gatto who I've been reading a lot lately you know goes um into I don't want to turn this into a thing about my anarchism either but um He is talking about how like the public

educational system in America, at least as he experienced in it in New York City in the 1990s, was just so socially constrained and so top-down command and control is to be uh highly non-functional and that his effectiveness was to the extent that he could remain within the system while completely subverting it. Um, anytime I see an agent who's active in who is exercising their agency to be effective within a system and yet is highly critical of the system. He was New York State teacher of the year uh and then at his lecture just gave a critique of the entire system in which within which he worked. The fact that an agent can be critical of the system within which within which they exist um is something that's interesting to me. And I'm guessing that these are not like novel ideas. But the fact that you specifically Daniel personally are capable of just turning whatever I say into active inference. like Rumpel Still Skin, spinning straw into gold is something I always find value in and am tempted to just uh exploit as much as possible for my own learning journey. Do you have anything you want to say to that? I don't want to put you on the spot, but I I will return in a moment. >> Keep keep going through the slides so that we can see the whole uh lesson within the session. All right, that sounds good. Thank you for keeping me on track. Um, so then the next thing here is a formula review. And there's actually nothing here. And this wasn't a matter of time. So what you could do is you could simply put cards here that um had a concept on one card like let's say entropy and then a formula on the other and um they would be free floating and then you would simply ask the students to match them. Uh this is a there's a There are these things called tactivities from the University of Colorado for teaching calculus. This is a digital version of a tactivity. In the physical setting, you would simply have the cards be cut out and you would ask students to match them by letter number pairs. Um, here you could have two columns or you could have free floating cards that you then asked to match. This is a very different type of question from the previous question. Um to be honest, I think it would be very inappropriate and jarring because the nature of this question is the nature of the prior question is so open-ended as to be almost meaningless. Like I ask you to predict it's Alice in Wonderland. Like predict a value 0 through 10, but you can only do 0 through 5. Why are we predicting? How can you predict if there's no objective? Like you're just randomly choosing. All of these things are fine. Um, and that's a wonderful type of absurdist play, perhaps jarring in a goal-directed setting. And then here is a highly knowledge dependent goal-directed problem where you need to at least be able to do flashc cards quickly um so that we can proceed. And they're two very different types of questions. And each is each at the extreme. And so moving from one to the other is something that could be very distressing to an agent. Um and that's why it's intended in a spirit

of play, but if you're actually intensely invested in following or or being a good agent, um it could be difficult. Let's move on. So let's move on to the part that's more content directed and a little bit less metacognitive. Um, so this is more open-ended, but it's closer to a goal direction. Uh, prior versus likelihood. Here I'm asking the students just to uh if you look at this the student view here uh if you you should be able to drag the points and as you drag these points it moves the distributions. So you can move the prior distribution which is in blue. Uh and you can move the I'm sorry you can move the prior distribution which is in green and you can move the likelihood distribution which is in blue. And there is no objective here. It's just getting you used to the idea that these curves are controllable. And uh this these points are actually at the mean. That's not important at this point for the students to know. It's not important for the students to know that these are normal distributions. It's just a matter of seeing what can happen if you if you click and I can see that most of the students are there. I'd like to ask the students to move them to wherever they would like to have them and then we can proceed from there. Uh I'm going to show you what I see as a teacher view. um should be the case that I can see individual responses here. An overlay wouldn't really make sense, but um and it may be a a lag or internet issue, but what I would be able to see is individual students responses. Here we go. Um I can see that the different students or learner agents have put the curves in different places. I can also see for instance who has touched it by comparing it and I can see who is actively dragging it as I talk about it which is a nice um feedback for me of who is engaging. If we're ready to move on let's go to slide five learner agents and I appreciate y'all. Um this is what a student would see. I have now introduced a red KL rectangle which will show up in a moment. And there's no explanation of what KL is other than it's a red rectangle on the screen. But as you drag distributions the KL and there there's a text prompt here. What makes KL bigger? What makes KL smaller? And as the student drags the distributions around, they're going to see KL move. And this is a a tactile discovery, kinesthetic discovery of a relationship that is not yet well understood. Um, and I guess the question is, does it have to be well understood at this point or can it be enough that you know that as you drag these points, the rectangle changes? In other words, what I'm trying to do here is to make these felt sense of feedback central before we introduce anything conceptual or high level or give any exact definitions. Um, this can just be that I'm clicking and dragging and seeing what happens. And for some people that's comfortable, for some people it's not. And that's okay. Um, but this is what this slide is about. And then I also ask for response. There's a prompt here. What do you notice? So you can drag to make it bigger or drag to make

it smaller. And there are really only two directions. So if you get it wrong, you'll get it right next time. And what I think the students will notice is that as you drag them closer, kale gets smaller. I'm going to check the anonymized responses and see if there's evidence that we're ready to move on. So, I don't want to rush anyone, but I also have to fit within the time box that we have. Okay, we're fine on time. So, if learner agents, uh, simulated learner agents, if y'all are willing, when you have a moment, if you would type what you notice, we already have one response. I also love this beautiful overlay graph of um, being able to see all of the student engagements at the same time. Um that's another concept that I think of it from the perspective of cryptography like or securities u computation like multi-party computation. So if we have multi in a multi- agent system um and all of these are all of these agents are exploring in their own way and then there's a a let's say higher level agent or a meta agent who can observe all of the individual explorations. Um, this is a nice position to be in for me as a teacher learner agent who is not only trying to convey certain material or in this case just guide the exploration of certain material but also it's a nice place for me to be as someone who is going to be engaging in repeated interactions with these other agents. Knowing about uh their temperament like are they wanting to go far away or stay close? Things like that can be very helpful information if it's not overindexed on. I'm a little bit self-conscious about the fact that there's a risk of an opticon that the you know making this process clear to the students will heighten their anxiety about the fact that they're being watched. Uh I personally have some trauma around surveillance capitalism from the the rushed move to virtual education during co and I can't imagine how I would have dealt with it if I were a high school student or middle school student. The fact that someone above you can be watching and collecting data produces different responses for people. And so when I engage in this process with students, I want it to be uh clear what is happening and I can also get a sense of the power that a company that is able to collect customer or consumer data or user data might feel um because I can learn a lot. A lot of times in this process students have suggested ideas to me that I would not have originated on my own and it actually deepened my understanding of the subject. Um, something that we could do here that might be fun as a as an offshoot is think about is there a way we could enable collaborative play where the agents are directed for something kind of like the the Reddit u experiments where they're adjusting one pixel at a time. Maybe I could ask them to try to produce a particular histogram of KL for instance. There's like I love playing with these ideas, trying different things and um I'm exploring the exploration of other agents which is something I really enjoy doing also. Um can we move to slide six please? So

um the challenge here is by hand try drawing two distributions that will have large KL and so I'm checking I'm not sure you know what students may or may not know here and the best way to figure out what they know is to ask them. So my hypothesis would be that most students will draw things that look similar to the example they saw. Um and that they the prior question was about small KL but that they will have the concept that large KL could be um achieved by separating the curves more because their goal was smallest. But it also asked about largest and so I'm sure they'll see you know there's only two directions to go. But I'm curious like first of all that they'll get this concept but second of all what the curves will look like. Um some people will be de demand avoidant sometimes something that I often am or they may feel very playful. they may want to adjust and see how far outside the space of uh two normal distributions with the same standard deviation they can they can achieve. Um, this one is also fun as overlay and uh because in when we switch here, the students were originally doing two different um but we could position ourselves to think about a a to think about a um a a an open discussion where we compared distribution ions that originated from different learner agent responses. So even though this even though this curve and and this curve, the curve in yellow and the curve that I'm now going to redraw in green, even though these two curves may have originated with different learner agents, we can now pull them up into one set from which we can select any two to discuss. And we can uh guide or open the door to free initiation of hypothesis with things like um if two like this curve looks more like a maybe a um exponential distribution or something. Well, what we were looking at before were symmetric normal distributions. You know, do you believe that KL could even be calculated for two distributions that are this different? And why? Um, so these are some options that I would these are some choices I would need to be making as the teacher learner agent or as the central learner agent or the guide agent or however we conceptualize it. also appreciate the the data is contribution. This is not two distributions. if we can move on to the next slide in the interest of time. Um, and then I'm going to give the students directed reading the Wikipedia page, which may or may not be any good, but it gives us a place to start. And probably I would have my own interpretation of of the Colback Leebler divergence revealed at this point. I would be happy to to draw, you know, autonomously around the discussion, but at some point we're going to have to bite the bullet and look at the formula and think about the formula. And now here's a good jumping on point. And this is sort of the reverse approach to the way that I typically saw concepts presented especially um preun university which is you would often just start with a formula on the board and then you would begin to unpack it and that can be that

can be very informative. Um I'm sympathetic to that approach now that I have a better understanding of how to intuit many of the symbols through lived experience of uh fighting and playing with symbols to do mathematical work. But it's a choice that one needs to make when will the formula and how will the formula be given and how much information will be given at once is another I've I've chunked things quite small. I would say for some people this is really going to test their patience. Um for others it's exactly the right pace. the next slide is a very low-level challenge. Uh given the fact that the formula for the Colbec Leebler divergence is in the Wikipedia article, are you comfortable using the math keyboard to type it? And you do have a a math keyboard available here that's got many standard symbols um in addition to things like superscripts uh and ratios. It has you know s cosine. So you're going to need natural log or log and we can discuss the difference between those if we need to. But you have the building blocks here. Uh my memory is it also accepts input in in math jacks or text style. Um but regardless I'm giving the students a chance to use the formula and and rehearse the formula in a very low pressure way that there can't really be a direct right or wrong answer for um and and assessing and giving practice in their comfort with communicating math using symbols. this is this takes how long it takes. Um I would probably in a 1 hour class be hoping to get a bit more than this. This might be the first somewhere between 10 and 25 minutes depending on the audience. but the goal here is not you know not just to give an example half lesson on Colback Leer divergence but to to be engaged in thinking and conversation about active inference inside math education inside active inference. And at that point we have either a subset inclusion where the two concepts are equal or we have a fractal where as we zoom in on one we keep seeing the other and either of those uh conceptualizations makes me very happy. So that's what I have prepared for today and I'm happy to use the rest of the time as the facilitator uh directs. Thank you so much. Thank you octopus. If anyone has question in the live chat, we can see otherwise um what are you what what was most surprising or the best learning experience or a fun or symbolic anecdote from your time with the math learning group in the textbook? I think it was the difficulty I had finding a way that I could write formulas quickly. So, I actually went out and bought a tablet or there are two things. The first is how difficult it was to to write formulas um on a computer screen for me personally. And I felt that that was helpful because it put me in a context where I had a disadvantage and that felt to me that it made play slightly more fair although I think it also was frustrating. Um the second thing is because of the the way the sessions were structured I was mainly coming from a place where I was responding to student questions. Um and that was fine. That was totally

fine and I enjoyed it. But it was it it reminded me of the fact of being like in a tutoring lab where you're always in response mode. And I got to experience that again after not having worked in that way for a while. and to reflect on the difference between being in response mode where someone else is setting the agenda and you have to fill in versus something like this where uh as I can have any essentially level of direction I want when I'm teaching a class and so comparing those different experiences being outside of my comfort level both of those were things I I found interesting and valuable to reflect Great. And then a last question from the live chat and that'll be it from sleepy Sylvia. How would you encourage students to interact in a classroom? It doesn't seem easy to switch from a mindset of passive learning to play. So um I I personally encourage them by being playful about the experience from the outset. Um, I guess I don't believe people uh, so I have never been able to be effective when I was completely serious about what I was doing. And I'm not I'm not sure that people in general often can be. Um, there are some things that require extreme precision and for those moments like let's say calculating a medicine dosage, one certainly needs focus, but focus is limited. And so I think that encouraging play comes down to recognizing that play is likely ultimately more effective in most cases, especially if we're in an environment that we can lead ourselves to believe is low stakes. Perhaps we will get a bad grade, but getting a bad grade is not a threat to us as a person. I hope that makes sense. Yeah, I think that really closes the loop also with the homeostasis, which is like if all outcomes of the learning encounter were non-threatening and equally preferable, it would be like foraging in an open field. But if some of those doors there's a threat that comes up against our feeling of of survival, then it can be as as threatening as anything. So, thank you, Octopus. Great to have you around. And till next time. Till next time. All right. Bye. Um for the third uh session of the third day of the symposium we have active inference in biology modeling the environment is an agent. Uh first Chris Fields will present topic and then Virginia Blue will facilitate the Q&A. Um thank you uh for presenting. The stage is yours please. >> Okay. Thank you very much. Uh I want to talk today about applications of active inference in biology and particularly the issue of modeling the environment as an active inference agent which the FEP in fact requires us to do. Um but the approach I want to take is via the symmetries of the FEP and I want to focus in particular on two symmetries of the FEP and hence the active inference formalism. One is that all components of any decomposition of any system are active inference agents and need to be treated as such. And the other symmetry is that all boundaries between systems, so all Markoff blankets for example in the classical picture are information symmetric. Both of these

symmetries are universal. Uh they come just from the physics, but they're both especially relevant to thinking of active inference in a biological or social setting. So in thinking about any interactions between living systems and their environments um these two symmetries have to be taken fairly seriously from a modeling methodology point of view. So where we're going today is first to discuss a little bit of physics and um I'll be using physics from the top down. So starting with the description of a of an entire isolated system, some kind of universe and adopting very simple assumptions and seeing where that physics takes us. And this is kind of an unfamiliar um way to start thinking about the physics of a system. If you're typically used to starting with particles uh behaving in spaceime or even a dynamical system that's defined in some spatial coordinates. Uh we won't in particular be talking about spatial coordinates at all. We'll be talking about systems completely independently of any spatial embedding. So once we're done with the physics, I want to consider two examples. Uh one is niche construction and show how niche construction is an instance of active inference. And then the other example will be cognitive offloading, taking part of your problem and giving it to your environment to solve for you. And I'll also show that that generically is an outcome of active inference. And then finally u I'll talk about some methodological implications of um this way of thinking and in particular methodological implications of these two symmetries of the FEP. So, um, top down physics, we're just going to assume three things. We're going to assume that any, um, system that we're interested in can be described with a finite Hilbert space. So, a Hilbert space is just a space whose basis vectors are the possible values of the possible degrees of freedom of the system. So if you think of asking a system a bunch of yes no questions every yes no question corresponds to one of the basis vectors of this state space if you use a hilbert spaces the state space and therefore we can think of any of these spaces as just spaces of cubits quantum bits u because those are the most fundamental entity to which you can ask a yes no question and get an answer either yes or no. So we'll assume that all systems can be described in this way and we'll assume that they're all finite dimensional and that's just for convenience. They can be arbitrarily large but if we assume they're finite dimensional we don't have to worry about singularities which just cause problems. The second assumption we'll make is that there's some sort of background time parameter. There's some intuitive concept of time that we can use to talk about change. Uh and we will not assume that this is measurable by any experiment. So it's it's just a parameter that we'll use in the theory and measurable times have to be constructed but they're constructed from this back background time parameter and the way to think about that is a measurable time requires a clock and for a clock to be meaningful



it has to change and to talk about change we have to assume some sort of background time. So all clocks run slower than the fundamental frequency of the background time parameter that we're assuming. Then the third assumption we're going to make is that isolated systems conserve information. And this is the principle of unitarity which is the fundamental principle of quantum theory. Um it says that if you have an isolated system and its state is evolving in this background time then the state will encode exactly the same amount of information at every time. that assumption might be considered controversial. uh if you throw out that assumption then you throw out quantum theory. So you've thrown out a lot of empirical physics and have a lot of explaining to do uh it requires a whole new theory of how everything works. Uh if you abandon unitarity so we're doing we're making these assumptions with a goal which is universality. If we make these assumptions, we describe all systems and all interactions up to the three assumptions that we've made. Uh they apply to everything and of course we pay a price for this which is abstraction. Um the results that we get will give us constraints on models but they don't tell us how to build models. So the outcome is is a set of methodological constraints that have to be complied with. It's not a model building procedure. And before um going on, I'll put in a plug for this recent book that Jim Glazebrook and I wrote that explores the implications of making these assumptions. and we're doing a live stream on that at the end of January um for the active inference institute. So if you're interested in this kind of approach to physics um that may be of interest to you again it's not for a couple of months. So let's start with the first implication of unitarity which I label here decompositional equivalence. The implication is that you can take a system any isolated system. So I'll call that  $U$  for universe and that system has some internal dynamics which we can represent with a Hamiltonian operator. You don't have to worry about how to write that down or anything else. Uh it's an operator that classically corresponds to the total energy of the system. But uh it is an operator that preserves information the information in the state of this isolated system  $U$ . And you being isolated just means that information doesn't flow in and it doesn't flow out. So decompositional equivalence is the fact that I can now slice that system any way I want to into two components which I'll call  $S$  and  $E$  for system and environment and I can represent each of those components as having an internal dynamics and I can represent their interaction uh by again a Hamiltonian operator that describes information flow between the two systems at some boundary the boundary where I've sliced the system and that operator  $H_U$  decomposes additively which means that I can slice this system any way I want to and I don't affect its overall dynamics at all. So what that means methodologically is that

physics by itself does not give us boundaries. They're no joints in you or for it to be carved at the boundaries that we see are all boundaries that we have defined by stipulation. So the picture of an evolving universe where the physics itself gives you the boundaries around things is actually inconsistent with the three assumptions that we've made. Uh the physics doesn't produce any boundaries if you make those three assumptions. However, the physics defines the interaction which I've labeled here  $h_{SE}$  at any boundary that we stipulate by making a model. So it's our choice as theorists. We can slice the system any way we want and the physics gives us the and it gives it to us by this simple formula I've written here. So this symmetry in fact is key to the free energy principle. And now we can state the free energy principle. It says that if we slice this system  $U$ , if we slice any isolated system anywhere to define two components  $S$  and  $E$ , then if  $S$  and  $E$  are to remain in a separable joint state, so if they're going to remain mutually statistically independent, then they have to both behave in a way that's consistent with that mutual statistical independence, which in Quantum theory we call separability. Separability just means lack of entanglement. So if  $S$  and  $E$  are going to be separate systems, things that we can talk about independently, they're not entangled. So they have to behave in a way that is consistent with them not being entangled. and maintaining separability not becoming entangled requires keeping the interaction weak with respect to the internal interactions. So we can express that by thinking of the dimensions of the interaction which are the same as the dimensions of the Hilbert spaces. the dimension of the interaction is the same as the dimension of this boundary that I've used to slice the system. And that boundary just has to be much smaller than either of the systems that it separates. Um, and that is intuitively obvious. The boundary has to be small enough that the systems robustly have internal states or the FE doesn't tell you anything. The FE requires that systems have internal states. But this gives us a an important qualitative fact which is that sensations and actions which are defined at the boundary are lowdimensional projections of the general models of the systems because the general models are implemented by these internal interactions. So when we think about what a system is observing or what a system is doing in both cases those are very lowdimensional representations of its of its generative model lowdimensional inputs lowdimensional outputs. So with this we can now think about the symmetries of the FEP. So the first symmetry is that if  $E$  is the environment of  $S$ , then  $S$  is the environment of  $E$ . So if  $S$ 's generative model, its internal dynamics, is predicting  $E$ 's action on the boundary, then  $E$ 's generative model, which is its internal dynamics, is predicting  $S$ 's actions on the boundary. And  $E$ 's actions on the boundary are  $S$ 's perceptions or sensations.  $U$

S's actions on the boundary are E's. So the boundary is the locus of all of the sensations or actions of SNE. The boundary is the Markoff blanket between them which encodes sensation and action in the standard formulation of the FE. So both of these systems are constantly acting to minimize variational free energy defined at the same boundary. So this minimization of VF is always a co-minimization being undertaken by both the system and its environment. Uh and the reason is that both a system and its environment are trying to stay separate from each other. They're trying to keep the environment well-defined. They're trying to keep the boundary well-defined. And the only way to do that is to keep this interaction weak. It's to keep the interaction from kind of blowing up in their faces and tearing the boundary apart. So that's the first symmetry of the FEP. If the system's an agent, the environment's an agent, too. The second symmetry is that this interaction being well defined at the boundary requires that the information exchange be symmetric. So, uh again, this interaction is an operator that acts at the boundary to move information between S and E. And it's the same operator from whichever direction you look at it. And the operator being well-defined is the same condition as the boundary being well defined. And in fact, we can see that intuitively by thinking of the boundary as a Hilbert space. Again, it's a finite dimensional Hilbert space. And if we think of the boundary as a Hilbert space, it's just some set of Hilbert space vectors. And the system and the environment are acting on that Hilbert space using perfectly symmetrical uh sets of operators. Um sort of the S half of HSC and the E half of HSC and each is doing preparation and measurement of each of those Hilbert space vectors. So the boundary being well defined, the interaction being well defined requires this symmetry and you can think of that just as Newton's third law of equal and opposite response to any imposed force. So this again is true of everything. So now let's think about what this means in a couple of examples. Uh first it means we can always think of active inference as niche construction. Uh S is always acting on E to persuade E to minimize S's variational free energy. Why? Because S's variational free energy is also E's variational free energy. They have the same boundary and they're acting on it with a symmetrical operator. So E acting on S is always trying to persuade S to minimize E's VF. So the two are always co-minimizing their VF. So niche construction is niche co-construction. S in fact is E's niche. Uh and E is constantly trying to get that niche i.e. S to behave itself to behave in a way that's as favorable to E as possible. So good niches in Mike Leven's language are persuadable niches. Uh they're niches that we can by acting on them get them to behave in a way that's beneficial. Uh and that includes of course the other organisms in the niche whether they're conspecifics or not. So a niche of a social organism like a human

is fairly persuadable. U niches of of kind of loners like um tigers for example are less persuadable. So the tiger is still acting on its niche but it's not very persuadable. So, um, it doesn't get as far as we typically do in getting our niche to behave. But the bottom line is that because active inference and niche construction are deeply the same thing, ecology is intrinsically a social science. It's intrinsically about persuasion. So, we're starting to cross the boundaries between disciplines now. So let's think about the other example which is cognitive offloading. Uh getting the environment to help solve some problem for you. So if I'm an agent um and I'm acting on my environment by making a measurement and recording some result and by recording the result I'm making an action. if I wait any finite amount of time, then my environment's going to have some time to do something with the information that I wrote on the boundary by taking my action uh on the environment by making the measurement. So the environment does something and if I go back and make another measurement, I'm going to see the results of what the environment has done in response to my first measurement. And generically, I'm going to get a different answer for my measurement because the environment's in a different state. So the environment is constantly processing the information that is written on it by observers. So it's not it's not just sitting there. It's never passive. And one particular case of this is memory. If I write memory on some memory on the environment, then in general the environment is going to process that memory and when I go to retrieve it, it's going to look just slightly different. And uh we know that that's in fact the case for episodic memories by humans and the environment is always acting on our brains and so it changes um our brain states and when you go to retrieve a memory um it's likely to be subtly different. Uh the same is clearly true with technology. We spend a lot of time and effort trying to get our technologies to to modify the memories, rewrite on them as little as possible. So what what does this mean? Uh what do these symmetries actually imply? Well, first it means that systems environments are always co-evolving in time. Separability, lack of entanglement means that the interaction between a system and its environment can be considered classical communication. They can be considered to be exchanging classical bit strings. But because they're not entangled, because they're separable, they have free choice uh by the Conway cooked theorem or just as a consequence of separability. So the languages that system and environment are using to write on the boundary can be different and likely will be different. So even though the environment's giving me back a bit string, it may be a bit string that's been encoded with a reference frame that I don't know and I can't find out. I just have to make assumptions about it, build a model of it to try to understand what the environment's

telling me. So the other implication is that um system and environment are constantly processing each other's informational inputs and then giving back answers. So they're constantly trying to solve each other's problems. um whether the answer is actually interpretable or not is always an open question. So in terms of theoretical methodology and also experimental methodology, um if we make these physics-based assumptions, then the passive environment assumption is always wrong. And it may work for all practical purposes sometimes, but it's very difficult to say which ones those are. And in fact, the frame problem in artificial intelligence is a statement of the fact that we don't uh when the passive environment assumption is actually going to work. We don't know whether the environment is going to generate side effects that we're not able to predict or haven't been able to predict. A second methodological assumption is that we have to assume history dependence. Um and in classical physics we often assume that things are not history dependent. Um that means they don't have internal Uh they're not doing information processing. And if systems really aren't doing any information processing, if they really don't have internal states, if they're really no larger than their boundaries, then they're not separable. Then they're in fact entangled with their environments. So if we think something's separable and it looks like it isn't history dependent, we just need to look harder because it's some level of description we're going to find history dependence. And then the final implication is that if a system is separable, if if we can talk about it as an independent thing, it's always going to be getting some extra cognitive resources from its environment. Uh, it's always going to be able to do some offloading of problem solving tasks, but it may not know how to use the answers and we not we may not be able to detect without doing the right kind of experiment what its environment is actually trying to tell it. So I just want to emphasize in the last slide that we really have only assumed three things here. We've just assumed some very basic physics uh the Hbert space description, background time and unitarity. And so we get a lot of biology and even cognitive science for free out of these very simple physical assumptions. And in fact, we get all the way to cognitive offloading for free just from the physics. so when we're thinking about physical systems, which we always are when we're doing active inference sort of models, we start with the physics and we get um cognition basically uh not just in the system but in the environment. So, thank you very much and I'm going to turn it over to Blue to handle questions and I think I'll get out of the slides and stop screen sharing. >> Thank you, Chris. Yes, Blue, you're welcome to share slides if you want. Yeah, go for >> Okay, I'll try. Um, first, Chris, I just want to say uh thank you for inviting me to be here. It is such an honor and a privilege to be uh thinking these thoughts and

co-creating this reality with you right now. Um >> thank you. >> I am Let's see. Let's get way back up into here. Um so Chris and I have been hanging out in the Hollywood squares for a really long time. Um I think I'll just share this way. That way it lets me kind of do it. Uh so I just want to provide some links so that everybody who wants to have the opportunity can come to our shared level of mutual information. Um so we've done a number of live streams together. Uh live stream 17, live stream 40. Um there's this guest stream which I was not part of but also um highly I think relevant to uh what was discussed here and um of course the wonderful uh uh course that Chris did for the institute physics is information processing and so this kota like this is like 12 lectures and a lot of discussions and stuff that happened in the course so to study more of Chris's work I just wanted to provide that um background and so here we are somewhere somewhere in between quant Phum and the FEP. Now, um I love this slide. I pulled it out of an old uh slide deck. Um and in preparation for this, uh talk, this question and answer uh period, um Chris. Yeah. Yeah. Hold on. Let's see. Yeah. Yeah. Um Oh, yeah. I can still see. Perfect. The pop out panel is great. Okay. So, in preparation for this, uh, Chris I said, "So, is there anything I can kind of study or read up on?" And Chris sent this, uh, recent paper that he wrote with Mike Leven, and that I think some of the figures that were used um in Chris's talk came from. Uh, so I I like just pulled a little bit of notes out of each of the sections in this, but I don't know how far we'll get. Like, I could go so far in in asking questions. Like, I have a million questions already. So, let's see. Um let's see kind of what pops up. Um so just from the introduction I just pulled a little bit out. Uh it says in this first quote there is instead substantial theoretical utility in viewing objects and processes as complimentary ways of describing persistence through time and hence the possibility of observation and manipulation. This way of thinking highlights the roles of memory as an essential resource for observation and of time as an essential resource for action and makes it clear that memory and time are also mutually interdefined complimentary concepts that both rest on the underlying notion of an identity that persists. And uh Chris talked a lot in this um talk he gave at too about the relationship between identity, time and memory. If anybody wants to kind of go deeper into that uh that link is there too. Um but really what I want to ask here is about the observer. And so in this paper you said that um myologology like the distinction between the part and the whole essentially effectively the boundary is defined by the observer right there's no and this you talked about too like there's no inherent joint in a system to be carved at. So this boundary is um imposed by the observer and you talked in the paper a lot about what is required for an observer to model itself as a persistent system and model its

observations as a historical uh sequence. So really I think um you know we talked about the system and the agent and the organism and the environment co-evolving but is the observer also co-evolving and is the observer also a process in this or could you maybe like unpack that and help me wrap my head around that a little bit? Um yeah, whenever we're we're doing some sort of experiment or or we're treating ourselves as as observers, then we're always interested in observing some interaction in our So I'm if I'm if I'm the observer here, say I'm the experimentalist, um I'm studying the interaction between say a plate of bacteria and the rest of the environment. So I've singled out some system in my environment and I've given it a name and I've drawn a boundary around it. And so we're now in a situation where I've divided my environment into two pieces and I'm interacting with both of them all the time because I'm interacting with my environment and I'm trying to understand by making measurements what the interaction between those two pieces of my environment is. So the only access I have to the interaction between those two pieces of my environment is how I interact with my environment. So I can interact with the two pieces, but I can't guarantee that I'm interacting with them separately, right? I it's it's a standard assumption of classical physics which even carries on over into quantum theory that in practice that uh I as an observer can interact only with some particular piece of my environment. So, it's an assumption, for example, that I can interact only with this laptop and to the exclusion of all of the rest of the of the universe. But if you think about what that exclusion actually requires, it requires infinite energy for me to say that that I'm going to interact only with this laptop. I have to keep the whole rest of the environment at bay uh and make sure that I don't interact with it in any way and make sure that it doesn't interact with either me or the laptop in any way. And that's that's what's hard, right? We're we're sitting in an environment, for example, that's constantly flooded by high energy cosmic rays. They're going through my body and going through my computer all of the So I don't in fact have the infinite energy that would be required to keep those cosmic rays from flying through my body and disrupting cells or flying through my computer and disrupting its memory. I can try I can make sure the bias on all my transistors is high so they don't go changing state when they receive some slight perturbation. But uh that's something I can do in practice, but that's not an inprinciple complete protection from the rest of the environment or from any side effects of my interaction with the rest of the environment. And and again this has been known in AI for a long time under the rubric of the frame problem that uh we can make our best predictions about what will happen as the result of an action. Um but it's in fact undecidable. uh the set of the set of all things that are not impacted by my action is

a set that I can't decide the membership of in principle. >> Yeah, I had read that in the paper also about like you can't even you know tell if any perceived object is not part of your own mind. So it's it's uh hard to because you can't define your own boundary. Right. >> Right. Right. >> Um, okay. Let's see. Oh, yeah. So, we went through this. Um, so maybe here. So, as an organism develops, we can expect what it can detect and what it does with the information it obtains to change. So, we expect the quantum reference frame it implements to change. Such changes may be gradual. For example, as a child increases her native language vocabulary or radical, for example, when metamorphosis constructs an entirely new body with new sensors and aectors for an insect. Such changes naturally raise questions about the identity of the system undergoing change and whether this identity itself is changing over time. Um, and I love the uh modeling organisms as processes, right? So modeling the system as a process and modeling the environment as a process. Uh could you maybe speak a little bit to perhaps modeling the boundary as a process? >> Uh yeah, and I think there there are two ways to to think about this at least. But one one way is and two of these ways are often conflated in thinking about the FAP and so I think it's always important to to draw this distinction. Um the FEP stated in terms of state space boundaries um and we often think of it as mapping those state space boundaries into boundaries in some spatial embedding. So I think of my marov blanket as my skin for example or we think of the cell membrane as the marov blanket around the cell. when we do that, we're faced with a very um immediate problem of how we think about, for example, stuff uh passing through my my body or stuff passing through a u a cell membrane. And it looks like it's it's disrupting the marov blanket in some way to have to have objects passing through the marov blanket. But when we think about it in state space, that problem goes away. Right? What we're thinking about when, for example, we eat food or when a a microbe ingests some sugar the state of some molecule is changing or the state of some collection of atoms is changing. Uh, a chemical reaction is just a change of state. If we think of a molecule as a bunch of atoms that have a bunch of relationships, some of those relationships change and so the the state of that collection of things and relations changes. So we can think of all of these processes as movements of vectors through state spaces where the vectors are encodings of the states of some degrees of freedom or some system components. So, so now when we when we think about and in that picture Mike and I drew a picture of of kind of the evolution of life, right? As life is consuming more and more of the state space. life is acquiring degrees of freedom as it develops as uh more and more matter is incorporated into some living living lineage. we were thinking about the the lineage of all cells that are um



descendants of some universal common ancestor. So if we think about the development of life, this this cell lineage over the last four billion years, yes, that picture it's nice. um more and more of the degrees of freedom of the planet, more and more of the stuff that's part of the planet, carbon and oxygen and calcium and and so on is incorporated into the living systems. So the the whole state space of all of the degrees of freedom say on the planet uh has an evolving boundary where u more of the degrees of freedom are now participating in this process we call life and are in lifelike states. uh more and more molecules are in lifelike states. Um things that were parts of rocks at one point are now part of my DNA for example. uh we can then think of this boundary itself as you said as a process as a thing with the state space that gets larger and larger and uh at some point we reach an equilibrium where life can't become any bigger without um completely consuming the resources of the planet. And in fact life life solves that problem by u not consuming abiotic materials but by consuming biotic materials. Right? Most of the molecules in our bodies are in fact molecules that were once in some other living system. So life is now just shuffling stuff around within its own structure within its own cell lineage as opposed to taking it out of the abiotic environment. And this is in a sense um the fundamental idea of the ga hypothesis of of love lock margallus you know long ago back in the 70s. So I I don't know does that does that address your question? >> Yeah, most definitely. And I think it leads really kind of nicely into like this next question. So we'll just skip right right skip right on ahead. So we're talking about um you know life as a process of of boundary expansion and when we consider um modeling the environment as an agent right so so let's talk about the title of the talk modeling the environment as an agent and then the expansion of life in this environment right which also includes all of the agents in the I I wonder about um the concept of the inner um the inner screen or maybe like the internal state model. So like I tend to think about it like maybe it's like a lava lamp, right? Like the environment is like a lava lamp with all kinds of like little life bubbles bubbling up and merging and take coming apart and like tons of inner screens in the environment as an agent. So like am I like going too far? Should I rein it in a little bit? But just in thinking about like maybe I am an internal screen for the environment as an agent as just as there are tons of internal screens within me. >> Yeah, I think that's it's it's perfectly reasonable to go up and down the hierarchy and and see what looks like from see what it looks like from each point of view, right? And we can look at the so cellular level where every living cell has some environment that's everything else in the universe except it and its job is to navigate that environment and maintain its marov So what does that environment look like? I mean for your

typical bacterial cell its immediate environment is some biofilm. So its immediate environment is a bunch of other bacterial cells and if you look at that biofilm for example then its immediate environment may well be my intestine. Um so we can think at all of these different scales and ask for a system at that scale its whole environment you know is the rest of the universe but what really matters is it's its nearby in state space which is if it's an organism that nearby environment is going to be populated with lots of other organisms. that may be more or less similar to it, but just because they're organisms makes them to some extent more predictable than kind of the wild open Um, which the sort of nonorganismal environment is much less like me than the organismal environment. So I know much less about it kind of in principle than I know about the organismal environment. And that's true at every scale. So uh we can think of of the immediate environment of any collection of organisms. So your your example of the the bubbles in the lava lamp, we can think of all the bubbles as one system and and the goo is the environment or we can think from the point of view of just one bubble where the environment is filled with these other bubbles and we get different perspectives when we do that. But I think all the p I think all of those perspectives are interesting and they're all perspectives in which the even if it's the ab a biotic environment um is an agent. It's just an agent that speaks a very different language than the language of life. And life is constantly I mean again this was the the point of of ga uh life is always trying to turn the abiotic environment into a more hospitable abiotic environment even though different components of life are constantly competing. So we can think of life as a whole or we can think of these these little pieces. So your your cat, for example, has the goal of always being in front of the camera. >> Yeah. Um that's great. And so in the inner screen guest stream, which I watched in preparation for this because I hadn't seen it before. Um, and maybe like this is something that you can't speak to because it was after you had left, but uh, Carl talked about the concept of having an innermost screen or like a fundamental undissolvable innermost Marov blanket. And so I wonder like in in the concept of something like a human which is, you know, we're just a large composite of millions of other organisms, many of which or half of which I think are not human. So, so in this premise, like is the inner screen, you know, of this gut bacteria living inside my stomach, like it is that's like not my that's like past my innermost inner screen or or it it doesn't seem to me like that the screens are ever indivisible. So, I just would love to hear your perspective on that. Like, is it just screens all the way down or or what do you think? Well, I think I think here again uh it's useful to make this distinction between informational boundaries and boundaries in some sort of space-time embedding. So, if

we're thinking of individual organisms, for example, we're thinking about things that are separate in space. But if we're thinking about a human as an information processing hierarchy um as if you will a virtual machine stack with uh some layers doing processing that's that's very close to the environment the overall environment of the human not the environment of any of its cells but the overall environment of the human we can think have kind of the surface systems that are processing that surface information and then deeper systems that are that are processing more highly processed Uh for example, my the components of my information processing system that identify objects are way inside the boundary. That's just identifying edges and bits of light. And then uh my processes that are identifying multiobject events and trying to deduce goals and causation and stuff like that is at yet a higher level of abstraction. So now we're not thinking about biological components like cells. We're thinking about virtual machines in a computer. And so the the top level from this perspective is the top level of the information processing hierarchy which you can think of as the highest level meta processor in the system. The the s the the highest level attention allocator in the system. Uh, and its job is just to its its environment consists of all of the lower level processors that are many hierarchical levels down dealing with the the overall environment of the organism. So for example, my overall environment, but the environment of this high level metarocessor is just other processors. And all it's really doing is giving them thermodynamic resources. It's saying, "Okay, you do some work now. You do some work now. You do some work. Now you do some work." So its generative model is a generative model of the processors that it's trying to allocate attention to. And um we make it the the highest level virtual machine just by saying there's no metarocessor on top of it that's allocating attention to it. But if you were to take that individual, say take me and put me in a social context, then now there is some higher level system that's allocating attention to me sometimes and other people some some other time. For example, that's what's happening in this conversation. Attention is switching between the two of us and the audience. So, there's a related question from the chat. Um, the question is whether these screens can do anything more than reflect or if there's a way to actually build up a higher level of structure. Well, in a generic thing that active inference, active inference agents do is reproduce um by some means or other. They may get their their environment to reproduce them for them. So computers, for example, get us to reproduce them instead of doing it themselves. Um so what is that for? I mean what are systems trying to do uh by reproducing? What are cells trying to do by reproducing for example? And one way to think of what they're trying to do is just decrease their their uncertainty. Right? If I make another copy

of me and stick it in my environment, then now I have something that's that's very like me. So uh to the extent that I know how I work, I know how it works. So it becomes something that's more predictable. So the FEP itself requires systems to to reproduce or it provides reproduction as a way of of reducing uh free energy, as a way of reducing uncertainty. Now, of course, it's it's not perfect, right? cells can reproduce and their and their daughter cells can do something that they don't expect. And for example, processes like cancer are examples of them. Daughter cells of some stem cell are now behaving in a way that that they're not supposed to behave in the body. So they're decreasing the the predictive power of their environments which are the cells around them. So built in built into the the idea of active inference is this mechanism for creating complexity which is reproduction and differentiation. So in a sense you can see the whole natural selection story as uh driven by uncertainty minimization and and and Carl and and and several of his colleagues have of course as as we have uh published various papers about those supplying the the FEP to development and evolutionary processes. >> Sorry. So, I think it's here. Um maybe it's the next slide. Let's see. Oh, go ahead. Go ahead, Daniel. Is Was there another question? Okay, perfect. Um, okay, last one. Let's So, I'm trying to drag this away. Ah, it's this one. So here we talked about um individually coherent models of the behavior of a system. What is required for them to be mutually coherent or intertranslatable? Um and and I just want to just talk about coherence and I know it's like a completely different like discussion of coherence but I was thinking about about the coherence that was mentioned also in the inner screen live stream like about coherence creating internal screens and I just wonder if there's like and you talked about coherence and the importance of coherence in the quantum realm and you mentioned it here and I just want to know if you can um speak a little more about coherence and um what that might mean for for active inference in the FP overall like just a a broad um a broad overview uh since it's mentioned as as so important for for so many different processes. >> Yeah. Well, this this is another word that gets used in multiple ways. And uh when we talk about quantum coherence, we're talking about entanglement basically. So non-separability um a lack of conditional statistical independence. So there's one sense of coherence that is a lack of of u statistical independence. Uh there's another sense of coherence which we're using when we talk about models being coherent. which has to do with intertranslatability or as in this diagram um has to do with maps being commutative. Uh being able to go from say uh in this picture um  $m_1$  to mapping from  $m_1$  to  $m_2$  at at  $t$  uh or waiting some time and doing the mapping gives you the same answer. That's commutativity of the diagram and that that's a form of coherence

that's defined mathematically or um syntactically essentially in a mathematical system but it carries with it some semantic information. So coherence in that model-to-model case is semantics. It's the existence of a relationship between two objects that is preserved through time. So I think it would be nice if we had multiple words in our vocabulary for this because it's important not to keep those two senses of the word coherent straight. One means entanglement, one means uh a relationship being preserved across some transformation. >> Thank you. That's super helpful. Um I guess we're wrapping up. So >> yes, >> thank you Blue and Chris. >> We're definitely looking forward to the January stream on the book with Glazebrook. So thank you again, Chris and Blue. >> Okay. Thank you all. Bye. >> Okay. Ciao. Welcome back. >> Hello everyone. >> Go for it Andrew. Thank you. >> Thanks Daniel. Um thanks everyone. Uh so we're excited to present some of our preliminary work. Uh this particular project is uh entitled uh a bit verbosely hierarchical active inference modeling of social trust dynamics in PTSD integrating qualitative phenomenology with computational psychiatry. Uh this project is largely based around applying the active inference framework and modeling techniques for gaining insights into potential sort of minute real time uh inference dynamics that may occur for individuals with or without uh PTSD. And so a lot of this uh kind of arises from our work uh with active inference institute uh with a project started by Jeremy Cooper uh who did a significant amount of work uh on uh PTSD. Uh he is a well actually uh I would rather let him introduce himself but I'll just briefly introduce myself. I've been involved with the active inference institute for a few years. I've been on the scientific advisory board this past year. Uh some of you may recognize me for teaching tutorials or the textbook through the institute or uh a few academic conferences and um I'm largely a programmer and data and uh analyst by trade and so it's it's been very exciting to try and figure out how to align uh the goals of computational psychiatry with uh sort of research in uh mental health and actually u maybe gaining more insight into improving uh uh treatment interventions. Um and I've just laid out here uh some of the kind of guiding uh research uh for this project including Jeremy's work and uh one of our other collaborators Angelos Capotos who's uh assistant professor in experimental uh clinical uh pathology at uh in uh the Netherlands and uh I will give it up to Jeremy now. Uh Jeremy, if you'd like to introduce yourself and uh anytime since I'm controlling the slides, uh you can simply say like next slide or next slide please and I'll uh change the slide for you. >> Yeah, gladly. Thanks Andrew. So folks, I am Jeremy Cooper. I'm first and foremost a therapist as well as an adjunct professor at Regent University and along the side I usually do phenomenological research. So the idea

behind our project here comes from my dissertation which was a philosophical study into how individuals with PTSD form slash change beliefs. And so it was from that that I well as you do I got my results and everything but along the way I was introduced to active inference. I did the textbook group a couple of rounds and then whenever we launched projects I was very grateful to come along Angelos Andrew and we have been working on this ever since. So it's very exciting to be able to present this and we welcome any and all feedback. So yes, Andrew, if you want to go to that next slide. What I'll be doing is just briefly giving an overview of what my dissertation consisted of. There is a preprint that's currently published that is a condensed version, much more readable. So that is what this is based off of. So I'd like to introduce or invite everyone to think of things in terms of what trauma-related beliefs might be as opposed to a belief system. And the large idea that I have is and coming from some of the sources you can see here that trauma related beliefs in a way moderate the symptoms that someone with PTSD might experience. That could be because there's a large phobia of a effect. That could be an extreme of emotion. That could be a particular emotion. Then also the trauma related beliefs in a way would moderate their exposure to any sort of intrusive symptoms. And so in essence, what we're looking at is individuals following PTSD. We're looking at folks who have a lifetime prevalence of it. We're not necessarily looking at someone who is a month out from the trauma occurring. We're looking at someone who may have been diagnosed for 5 years or so. They will there's a good chance that they will develop these beliefs that make way for them to navigate their symptoms, if I can say it that way. So it could be to avoid particularly distressing symptoms or it could be to accommodate symptoms that they already One of which might be this idea of meaning violations. So the meaning violation concept is that those who undergo PTSD they suffer a core change in how they see themselves as a part of the world. Andrew Andrew if you want to go to the next slide there. So I became curious about this idea of them in the world primarily from a phenomenological backdrop looking at agency and how agency interacts with interceptive processing and then how that interacts with affect and I'll say affect I'm understanding in a different way than emotion of course we can more easily identify that in relation to something else as sort of a cause effect but affect I'm talking about more in the way of how uh slabby 2008 talks about it where affect is a way for an individual to view their world it's a homeostatic sort of relationship tells them where they need to adjust tells them their place in the world how they fit how they don't fit. These are perhaps larger larger uh like as we're getting into the winter here and we have seasonal effective disorder then everything starts to feel a bit more gloomy. Those are the

types of affects that can also be synonymous with the idea of moods. So that's the understanding of affect that I'm carrying into this work as well as the idea of agency. One helpful definition of which is agency requires a cohesive narrative, a sense of body ownership, and a sense of self-efficacy accomplishment. And so a transdiagnostic concept that's been developed by SAS at all is an anomalous self-experience, which would be that one's self-experience becomes an anomalous. it can be quite descriptive. It's the idea that one is experiencing themselves in a bit of a strange way. And so that's where we looked at interception to see how that might be occurring. Andrew, if you want to go to the next slide, What Rackcliffe discusses in his very transdiagnostic book *Real Hallucinations* is the idea of an anticipatory style. This is akin to the idea of pretension very very fundamental to the aspect of self. It's pre-given. It is before conscious recollection. But the idea of an anticipatory style is that I expect something to naturally come next. But it's a motive. It is a feeling. It's the analogy that he uses is if you know a song, you're in a way expecting the song to have the right note afterwards and then all of a sudden it's out of tune or something's changed about it. And it's that sense that something's off all of a sudden. And Rackcliffe talks about whenever it comes to PTSD in particular, the way that the anticipatory style changes is that the individual loses touch with the shared interpersonal world. And so this idea led directly into my research question for the dissertation is how would it be for those with PTSD to describe their relationship with freedom of thought, liberty to experience spontaneity or simply assurances of their experience. And then overarchingly to sum it up with the literature review is how does that equate to beliefs? Andrew, if you'll please go to the next slide. The study design itself, as I said before, was a phenomenological study using Scott Churchill's experiential phenomenological research method. In interview I interviewed seven adults diagnosed with PTSD. The semistructured questions composed of two broadly overarching categories. What's the impact of diagnosis on yourself, view of others, your relationship and how has that diagnosis itself impacted how you view the world? So, one of which is symptomatic, one of which is impact and through this particular design, it lends itself more towards these structural descriptions. It's similar coding process but your end result is a very very rich nuanced approach to some of the key phenomenological ideas is the idea of freedom and also meaning making. So that is I should say another key aspect to my understanding of symptoms alongside moods is that they help us make meaning of our place in the world. Andrew if you'll please the primary study findings that relate to our project. First one these were sequential building upon itself. First one is isolation. So the idea of isolation, it comes about more as a

way to moderate their experience or or rather mediate their experience of hypervigilance because the experience of hypervigilance was felt in the increased physiological arousal, the height in anxiety, the height in perception and longterm over time that perception maintained. And so then they started to isolate themselves as a way to prevent that heightened awareness. From there the isolated disposition was the attitude of to stay isolated away because now what they have accommodated to alleviate hypervigilance has also led them to this very very painful sharp reality whenever they do come in contact with other folks that now their experience is different from others And this led to the idea of self-doubt. And then so the final one where it can be termed a belief system if you will is this idea of jadedness. And of course this is again a phmological mood. That is that the idea of jadedness is they have a feeling a sense that they are distinct in the world separate from others. that accommodates the isolation, allows them to act purposefully, but doesn't have to encounter the pain of interacting with other folks. And Andrew, if you will, please. This led to the idea of social invalidation and the concept the proposal that perhaps uh interception is a social shared understanding. Coming back to Rackless idea of a shared interpersonal world as well as Peter Fonagi's idea of Peter and Chloe Campbell's idea of epistemic trust, mistrust and credul. The notion that a individual's piece of information they are giving me applies to my world. Not so much if it's correct, but does this actually fit for my world? And for those that were interviewed as they lived with PTSD, their primary concern was that they would doubt their experience of the world if they did interact with others. So with that, I will transition to Andrew who will share our inspiration and our design for our model and our agent. >> Awesome. Thanks Jeremy. Um yeah so so Jeremy essentially brought a lot of his work to the institute and to the sort of project that was formalized and so what we proceeded to do was create a small team of researchers who had this kind of variety of backgrounds who were all quite interested in uh PTSD and a lot of sort of adjacent topics. Uh so again I already mentioned myself Jeremy and uh Angelos and then we've also had a lot of uh help from uh research at UCL uh H and Sun um and who helped a bit with uh the programming and um so the goal here uh as we already know was uh not so much to try and uh create an active inference simulation to recreate trauma. uh this a a lot of work has been done in ethology to try and understand uh things like fear learning and associations and uh whenever we imagine trauma is something very intensive. Of course the idea of um you know carrying out a simulation that might be carried out also empirically such that we can fit our model to empirical data from real individuals with PTSD. um obviously need to file uh follow like IRB um protocols and and and be ethical and and more importantly



the the point is not to necessarily understand the the original trauma in and of itself. It's much more aligned with Jeremy's work which is how do we facilitate those who have already been through the trauma or anomalous uh self-experience. How do we facilitate them to having a higher quality of life given what they have been through knowing that actually there's a lot of adaptation to uh or mild adaptation to uh the symptoms that sort of follow from uh the experience of the trauma. So here I've listed out some inspirations. They include um a lot of work by uh Ryan Smith um has done wonderful work in in computational psychiatry as well as empirical work and fitting the models to real data in his um uh simulation on simulating the computational mechanisms of cognitive and behavioral psycho psychotherapeutic interventions. um that was sort of a core inspiration of uh simulating uh you know an artificial person who might be in a decision-m task. Uh he simulated exposure therapy uh which actually recreates some interesting dynamics of fear uh learning or unlearning or extinction uh found in empirical literature and highlights the role of contextual priors which is um it it brings up this interesting case where someone who believes they're safe suddenly doesn't quite know what to do once they perceive they're in danger. Um, and it relates a lot to the idea of like kind of associative learning, but how do we how do we engage whenever we infer that we're in particular contexts and whenever we're in a particular context, how does that uh change the ensuing behavioral outcomes? Um, a very uh important simulation also to bring up was um Adam Linson and Friston's work in 2020 where they more directly were attempting to model uh psychological stressors especially with um PTSD's sort of a core topic. Um but again that's much more ethological. They simulated a mouse in an environment where there can be a real threat uh in the environment. uh and so that's of course not the direction we want to go necessarily but their model uh provides a sort of framework for looking at things like multimodal sensory integration which is where we consider uh more strongly the idea of receiving you know visual observation uh uh auditory uh um observation uh witnessing yourself making actions and reflecting uh on those actions that you made uh and and thinking about things like safety and and and physiological arousal Um and then after that we have Adam's um everything is connected inference and attractors and delusions which is a a great work and actually if you understand this paper you understand most of what our model does and what the behavioral task is. We essentially recreated a lot of the dynamics found in that paper. um that was more focused on studying uh delusion which is more generally uh defined as just false inference despite disconfirmatory evidence. So it's not too uh you know we we're cautious about uh using kind of stigmatized um terminology. It's more about just how

can we arrive at a more um precise sort of terms um for how we understand uh something like aberrant or or pathological behavior. And uh I also like that this paper was actually not dedicated to understanding PTSD or trauma. It was more dedicated to understanding how delusions arise in decision-m and perception uh in the case of of schizophrenia. Um, and so I like it because um, it it just kind of highlights the sort of translational capacity um, that we might find in computational psychiatry where we make models that can potentially depending on prior beliefs and parameters and model architecture exhibit a variety of kinds of pathological disorders. and might lead to understanding different kinds of sort of transdiagnostic criteria for defining things like um you know delusion or or pathological behaviors um deeply felt affect uh and as well as Ben Ramo's role of affective states in computational psychiatry. This uh kind of just informs some of our decision-m of how to model uh arousal and affect in our model and getting some insight into uh the relationship between uh affect or emotion or mood with decision-m and confidence in one's deliberate decision-m as opposed to say relying upon uh habits. uh when someone doesn't know what to do and so they kind of just repeat what they've done before um as a as a sort of auxiliary. Um and then finally towards the computational phenomenology of mental action by SV Smith. Um this also was a very interesting piece relating to hierarchical modeling and having a sort of higher layers in a model or what we'll call metacognitive layers. The idea that we can perceive ourselves perceiving as it were we can have beliefs about our beliefs. So, so to believe that someone is untrustworthy and then above that, what does that mean for me to believe that this person is untrustworthy? Um, and a variety of other kinds of ways that um that logic can be applied. So um for sake of time I might speed through some of these slides a little bit faster than others but I wanted to quickly provide um there's sort of a broader general framework being employed here uh that you're probably well aware of at this point if you've been watching the other uh symposium uh streams. But um even more general than active inference are these theories of predictive processing. Um and so um uh Lisa Feldman Barrett uh this is a quote from her that I I have used in tutorials to try and provide a sort of intuition for predictive processing and its relation to how we understand uh sort of perceptual uh processing in the brain. So um here I've just highlighted um some different sort of uh how we understand sensory modalities. So there's interoceptive uh sensory modalities what we read from within our bodies and then there are exteroceptive modalities and those largely align with sort of the five colloquial senses of sight and touch and and um you know the whole factory and so on and then proprioception which I didn't have an example of but it's essentially related to to sort

of physical movement and reading of uh the body in space and being able to sort of coordinate. Um and then what happens is that we we essentially take in varieties of of observations of of sense data at any given moment. And uh so and from there we have to infer what does uh this mean or what caused this? What what are my beliefs about the world that generated this? And this happens down uh all the way down at a very you know rapid pace continuous sort of level. So it's not to say that all of it perhaps probably some of it but not all of it is necessarily conscious. Uh we're looking at more at the level of sort of neuronal dynamics and you know um the the ability of the autonomic nervous system to regulate itself. Um being able to breathe without having to think about it. Uh you know so um so just gives a little bit of intuition. And we have to infer what our sense data means and then reciprocally uh sometimes we use our inferences to um try and act to elicit different kinds of sense data. You know if my stomach is growling that must mean I'm hungry. If I'm hungry if I eat food have caloric intake then I will no longer be hungry and I will be sated. Uh is is kind of quick and crude example. So um from there, how do we proceed to actually model the kinds of uh phenomena that that Jeremy sort of identifies in the the structured setting of of collecting information from uh the individuals diagnosed with PTSD? Well, um I've taught on this before, but more generally, when we look at things like agent-based modeling, which is a technique that goes back decades at this point, uh the way that it's done is heavily tied to the kinds of computational resources we have available. And so there are more and more capacities for us to produce uh more interesting sort of intricate or complex models, not so much with the focus on making complex models for the sake of it. In fact, it's very important not to do that. Um, but it's to to be able to simulate uh dynamics that have been empirically found in the literature and in our case looking at sort of um empirical uh neuroimaging data uh analysis and and modeling of neuronal dynamics uh at a at a very fine scale and and building up from there. Uh nonetheless, here's some traditional guidelines. uh one when doing agent-based modeling we have to apply different kinds of assumptions that should be convincing to readers most imperatively in relation to empirical findings. Uh if you make a model and no one is convinced by any degree of its realism then perhaps it should not be employed um robustness simple as possible and no simpler. Uh those who do things like um you know computational modeling or modeling of data are all too familiar with this. you can make a model that is so deeply uh um overfit. It's so deeply specific to a particular case, a particular subcase that as soon as you start looking at the broader phenomena you're actually interested in, you realize your model is no longer applicable. Um so it needs to be able to generalize. It

needs to be relatively robust to tiny um modifications to the model. Reproducibility is pretty straightforward. There's a reproducibility uh crisis in uh science. some or maybe many would claim. So, it's good to allow others to be able to reproduce your work uh whenever it comes to settings like this where there's um computer programming and code involved releasing your code to the public. Finally, uh non-triviality, the results of the simulation should be non-obvious. Um I do think that it's always good to be able to reconfirm uh our previous hypotheses with data. So I'm I'm a little on the fence about this particular guideline, but nonetheless, we do want to actually yield a significant insight from doing this modeling work. We want to see what we could not have seen otherwise. Um so nonetheless, a classic quote from um statistician George Box, all models are wrong, but some models are useful. And so we aim for useful uh models that that kind of follow particular guidelines. um and can be empirically validated. So um more generally computational psychiatry aims to build what are called computational essays. And so what we do is that we try to find okay if you could model the brain and you use different kinds of you know decades of research to try and understand how that could be modeled and if we know that you know theories of predictive processing are all but ubiquitous in um in at least theoretical if not other areas of neuroscience as well then you know how can we start to map the parameters of these models to u you know neurobiologically plausible correlates uh in the brain and that is you know I'm jumping ahead a bit but I think active inference does an incredible job of that and that's part of what drew me to that this particular framework um you know looking actually looking at uh the basal ganglia looking at interconnectivity between the prefrontal cortex and the hippocampus when it comes to memory recall and deliberate decision-making actually being able to watch more precisely how that plays out such that we can move towards having more precision in psychiatry whether it be for diagnosis or patient stratification, understanding particular subtypes um or phenotypes we might say um in in pathological disorder and uh and then from there being able to simulate or test different kinds of intervention treatments in a very safe artificial environment that appears to more or less um well approximate reality. So the active inference framework again you've heard a lot about it. Um it largely starts with the idea that everything that changes in the brain must minimize free energy and this kind of follows from a neural process theory that active inference holds to uh foundationally. So arising from this kind of work that uh many like uh uh Thomas Parr and Carl Frist and others have done um in the past uh from studying things like perception. We're familiar at this point with um you know predictive processing with Helmoltz's idea of unconscious inference uh you know of the mind sort of filling in the

gaps in the data that we might otherwise miss such that it can make more sense to us um and all these sorts of things. And so um and then we also look a lot at the idea of you know this this is these are not just brains in a vat. These are entire organisms who who at minimum need to maintain some kind of homeostatic regulation otherwise they cease to exist. Again I recall uh the idea of me having a caloric intake such that I can uh stay alive regardless of you know all the various inferences and prediction error minimization I'm doing along the way. Uh hopefully that helps. Um so with generative models and no simpler uh what I love about this is that if we do have this kind of general uh potentially universal so to speak principle of how the brain works then the challenge here or the goal is no longer a necessity to model say um you know several billion neurons uh in the brain. We're not trying to emulate the entire entirety of the brain. We're just trying to find the generative model that describes the problem the brain is trying to solve. So that allows us to reduce a lot of the complexity in in constructing models and trying to map them to reality. Active inference again relies upon bases theorem uh relating observations to hidden state beliefs um having a relationship through a sort of marov blanket relationship with one's environment in which case uh it's not so much that you um need to know everything. it's that you need to know what are the agents observations and what are the kinds of actions they can elicit. And so if you're able to sort of establish where those sensory and active states lie uh in in the overall equation so to speak between the agent and the environment, then you're able to sort of establish these conditional independencies between the agent and the environment such that you don't necessarily need to know everything about the environment in order to understand what's going on with the agent. You just need to be able to understand the observations the agent receives and the actions they they elicit and that's about the best that the agent itself can know. Um so uh I I do bring up uh this this um this is a policy inference equation that's used in in certain implementations of these kinds of models. It's the one that we employ and I just very much like it. I think that it it can sometimes be overlooked. Um, of course, unless you're doing the programming. Um, but this has come up during previous tutorials I've taught. It's simply to make the point whenever it comes to decision-m and active inference. Um, we people have related this uh I've been asked before like how does this relate to something like Conaman and Verski's systems one and systems two thinking? And um I I argue that it it actually has a sort of systems one, two, and three. Um it it's crude but it's just to say uh with minimizing expected free energy uh tied to  $\gamma$  or uh policy precision. It's and and then finally uh variational free energy and and  $e$  or habits. It's kind of like we have deliberate decision making and reflexive decision-m in the moment for

maintaining one's model. And then finally context free habits which strongly aligns with a lot of research we find in psychology on how we do frequently rely upon habits or sort of heuristic decision making especially in context of uncertainty. So it's just we do there there's I just point that out to say that there's a lot more alignment here with other disciplines than what might initially meet the eye whenever we start looking at the fuller breakdown of you know um negative information gain minus uh pragmatic value and getting into all the mathematical formalisms. Um and then finally um with fear learning which is a lot of um again Angelos uh CPotos's um contribution when we're coming back to PTSD um uh this is sort of an informative framework. There's obviously been an immense amount of work done, but it relates to the idea of looking at Pavlovian conditioning. And so, u, one way of thinking about this is that we might relate the anomalous self-experience or or the experience of the trauma as if it were like an unconditional uh unconditioned stimulus. So, the setup in a Pavlovian uh fear conditioning paradigm is that you introduce an unconditioned stimulus, something negative, some kind of negative stimuli. Unfortunately, you know, using mice, they might use like a kind of mild shock that doesn't necessarily permanently harm the mouse in any capacity, but nonetheless, it's something that the mouse does not very much like. And then that gets paired with a conditioned stimulus that, you know, in reality may have no intrinsic relationship to the unconditioned stimulus. Uh, so you might play a light tone, a sound. And so, uh, over time the mouse begins to associate hearing a tone with being shocked. And then through a learned kind of fear-driven behavior, um, the mouse begins to freeze or potentially try to avoid whenever it comes into contact with hearing the tone, even though the shock may no longer follow it. you might remove the shock component from uh the task altogether and now it's just the unfortunately the mouse has this paired um association and uh from doing these kinds of ethological studies and knowing that there's a very strong relationship in the mammalian brain and in the brains of mice of cats and so on with humans um you know there have been attempts to sort of extrapolate these kinds of findings to understanding how humans sort of process um fear and and how uh fear memory uh develops and um some findings uh show that memory of the fears acquisition. So if we think of an individual who goes through a traumatic experience um over time even through exposure therapy even through uh you know different kinds of sort of fear extinction techniques the individual doesn't actually have their original memory of the trauma erased right it remains instead they have to form new memories about the relationship between uh an unconditioned stimulus and a conditioned stimulus. So for example for individuals who have been through trauma and then uh

experience you know some kind of negative uh or rather they hold some kind of negative association between that and you know other stimuli that otherwise might not be a problem. Oh someone sort of benignly lied to me about something. You know benign just meaning like it didn't have any strong stakes. they weren't necessarily trying to be malicious and yet that somehow triggers um you know an associative connection that I have to my trauma and in real time perhaps not even immediately thinking about the trauma itself may the the individual may start to exhibit the learned negative behavior you know they might start to avoid that person they might um you know kind of start uh showing different kinds of symptoms that they've developed over time um and so we think of things like maladaptive fear or when fear exceeds the actual threat generalizes too broadly or persists after the threat is gone or prompts excessive avoidance and again relations to associative learning hemian uh plasticity and then I'm particularly interested in neurobiological correlates to this so sort of um the amygdala has a a very strong role in in a lot of our dynamics actually not just detecting threat but also detecting sort of general salency it's even involved in reward processing. So um it's you know unfortunately in PTSD it's often that uh in humans the volume of the amygdala uh may or may not shrink to some degree uh there but there will very frequently be a different kind of like aberrant um circuitry uh activation going on. There can be a sort of shortcircuiting. So whenever you think of someone with PTSD being like so-called triggered um what can happen is that there can be a rapid recall of different kinds of memories associated to trauma that then affect um you know what we might otherwise view as cognitive uh deliberate decision-making that we usually view different areas of the PFC to be related to. Um so with modeling um all of this the methods overview we chose a behavioral task a card advisor task. I'll go into that uh in a moment so stay brief here. Um the optimal strategy for this task that is this task which is reproducible and it follows ethical guidelines. At no point is the participant actually in danger. There is no confirmatory evidence that an individual would be in danger when playing this task. um it is structured like a game where you don't know the color of an unseen card. And so there's an adviser there who's sort of part of the environment so to speak. They're part of the the administration of the task and they would um they would uh procedurally tell you what the color of the card is and then as a participant it's your job to try and decide are they lying to me or not. Um uh and so uh the game-winning strategy is to actually engage with the game uh ideally play multiple times and then see if there's some kind of pattern to uh the trustworthiness or untrustworthiness of the advisor's advice. So they give you the advice, you make a guess, you might choose to go with their advice, you might choose the opposite, and

finally you flip the card over and see if they're right or wrong and continues. Um and uh this has relation to empirical studies that have been done in the past with different kinds of trust based games um including with the emphasis on um pathological behavior. And so they found for um a sample of of uh women with assault related uh PTSD diagnosis whenever they played certain kinds of trust games like this. Um what was interesting was that they actually had similar uh prior beliefs about trust as uh the the normative sample of individuals not diagnosed with PTSD or other pathological disorders. So it's not so much that they were predisposed to distrust uh the adviser but rather once the adviser did lie to them at some point give them false advice and then it turns out the color was the opposite of what the adviser said. uh it was much more difficult for individuals with PTSD to kind of regain the ability to have trust in the adviser again uh even if the adviser switched back to being trustworthy again. Uh which says something about sort of the belief formation or evolution or change in belief dynamics. You know it's that sort of change over time. We can't only rely upon some idea priors but actually how they play out over time. Uh with the agent model, we chose a hierarchical partially observable Markoff decision-making process, which is quite a mouthful for those who are not familiar. Um but uh it's it suffices to say they're relatively kind of um powerful, interpretable uh almost standard in some ways model that gets frequently employed in computational psychiatry. It's just because there's so many mappable parameters and allowing for different kinds of more discrete decision making. So a model who can actually, you know, infer trustworthy, untrustworthy, um can actually choose green or blue as the card color, uh and so on while actually being able to to play out its dynamics over time. And of course, uh it's it's frequently used in active inference. So we're using free energy minimization and expected free energy minimization is sort of the the neuronal dynamics at play. given uh we also have multimodal sensory integration uh where the the agent actually is combining different kinds of observations it's receiving at any given moment to make its inferences. Um and then after that given that this is still a purely insilic model uh we we simulated many different agent variants uh 576 of them and the idea there was simply to get a view of what is this model capable of in the first place uh without making too many other assumptions given that we would like to move in the direction of fitting this to empirical data. Um so so what follows now uh I I've I've essentially described the task um and we have a hierarchical model. So we have a lower layer uh here that directly kind of interfaces with the sensorium. It receives observations uh for each trial of the task. One we check your heart rate. The agent kind of self-observes its own heart rate. That can very much relate to interoception. What



does my physiological arousal mean? you know that um you know this might be read to mean a variety of things in the cases of distorted interoception in in uh you know potentially pathologically that is you know uh maybe something good happened and it you know surprised me but I'm excited and so my arousal has gone up and yet a distortion would say oh I'm physic you know I have a heart uh a high heart rate uh whether you're actually thinking about it or not but it's the association between having a heart high heart and potentially feeling that one is in danger, right? That that can actually lead to cascade effective inferences where someone, for example, might become triggered, right? They they misinterpret a situation and what it means in their for them in their body and their ability to read their body and then it leads on to other beliefs that that may not have any direct one-to-one relationship with the actual scenario that you're immediately encountering. So, we check your arousal. There's a covert action uh by the agent. Do they trust or distrust the adviser? It might not mean much of anything. They haven't received any evidence yet if the adviser is trustworthy or not. Uh whenever you start with the first trial. Um but nonetheless, you know, they might be predisposed to go a particular way with that. Um then stage two, they'll receive advice. We'll check your arousal again. Uh and then they'll actually make a decision. And then we added an additional option. And not only can you choose blue or green as the card color, but you can also choose to withdraw or decline making a decision. Um, we added that because we wanted to see if the agents or at least some variants of them would choose that option. Uh, given that these are sort of cognitive autonomous agents, right? So, we're not tell we're not forcing them to do anything. We simply provide them with the model and then through free energy minimization they're kind of making their own decisions. Um that's that notion of agency um in to some degree. And so whenever they choose to decline making a decision, uh what'll happen in the final stage is that they'll receive some feedback telling them if they were correct in their choice, incorrect in their choice, or they'll receive a sort of null or nonfeedback saying, "Oh, you didn't make a choice." And so we cannot relate your choice to the correct card color, and so you don't actually get feedback. So, it's sort of a process where the agent can decline receiving potentially receiving a reward, decline potentially receiving a loss of losing the game. It's a sort of non-engagement, but you're still in the task. So, it's not to say that the participant actually gets up and leaves and they're gen like withdrawing withdrawing. It's simply uh you know to kind of choose to not play the game, but to continue on to the next trial nonetheless. Um and so we were curious to see how that how that plays out. And so uh whenever we run this over many models, we have these different kinds of scoring metrics. Uh

we're looking at um kind of composite of negative affect uh high physiological arousal believe sense of danger or kind of false sense of danger and then interpretation bias which is a very specific case where the agent believes that the advisor is untrustworthy even though the advisor is in fact trustworthy. Um and so and then we also included a delusion score which follows from the Adams paper which is more of an aggregate of like it assuming that the advisor is trustworthy when they're not as well as untrustworthy when they are. So you have false positives and false negative conferences. Um whenever we look at an overview of all of our models which the the the brain largely follows non-regotic processes. So I would take these sort of with a grain of salt. Uh these are Pearson correlations. It's simply making uh an alignment. Uh how much did whenever we look at all of the models, all over 500 of them, uh which of these parameters or different kinds of metrics from each model, you know, kind of aligns with this hypervigilance score. Um some of these are slightly redundant because you know interpretation is part of the definition of hypervigilance but nonetheless we want to see what actually arises in our models. And uh a core takeaway is uh first um higher order beliefs about safety which get related back to the agents belief in uh the advisor's trustworthiness as well as their own sense of affect positive or negative and and how strongly positively or negatively they're they're sort of uh inferring themselves to be. um whenever uh safety transitions are very precise meaning a belief that when when I'm in the context of safety I'm safe it means it will persist through time so it's kind of like environmental volatility is one way of reading that but also inversely if I believe I'm in danger then danger will also persist um whenever that's more precise it actually leads to higher hypervigilance it's more than likely related to the idea of once an agent does believe that they're in danger it's difficult to get outside of that danger. Um it's kind of getting trapped or stuck in it, so to speak, or you're more likely to. Um second, uh prior beliefs about safety. That one is not particularly surprising, but it is interesting because we know that these beliefs get modulated over time. So, we're just looking here at the prior belief about safety, where you start at before you play a single trial, right? U but nonetheless, a negative um moderate to weak relationship with with hypervigilance. And um this was the most interesting takeaway. There's act even though it's a weak negative correlation um some agents by choosing to trust um they actually kind of bolster their mood or their affect. Um and uh it's a little kind of tricky to wrap one's head around. But when we start thinking about things like attachment theory and how uh we might regulate our own emotions through interactions with others, uh we might say that these are agents who are able to uh regulate their emotions at least a bit through making deliberate

decisions to to trust um the adviser. We can think of, you know, if there were no relationship there, then it's kind of saying, oh, I'm not able to emotionally regulate myself now, at least with other people. Um, a lot of preliminary results, but I notice I'm very very low on time at this point. Um, there's a lot going on, but essentially we're just trying to suss out. Um, uh, there's an interesting thing that happened with a lot of the models. roughly two-thirds of them uh almost never withdrew whereas about a third of them almost always withdrew and so um I tried to see what what what are the sort of correlations there what are the interacting dynamics between different pieces uh it seemed quite clear here that once an agent uh kind of surpasses some threshold of negative affect including uh how frequent it is in general how certain they are of that negative affect meaning like a bad mood versus a very bad mood is one way of saying that. Um and then finally the seriality which is how much do you recurrently find yourself in a bad mood as opposed to having rest bits of a positive mood in the middle. um pass some threshold, agents almost always start to withdraw. And I think it just kind of highlights the notion of the the the importance of looking at things like affect interoception, how we read our bodies, how we read uh emotion and how, you know, beyond some that that that information gets integrated sort of with everything else in decision-making. Um and so here uh zooming in on withdrawal um it was clear that that certainty of uh false trust uh strongly line with withdrawal. So these were agents who uh were actually very certain about their evaluations of the advisor. Um but then also that came with negative affect. Um it's interesting that delusion score came up and I'll I'll get into uh why that is in a moment but um the idea of withdrawal is that by declining to make a decision you're avoiding re-receiving disconfirmatory or confirmatory evidence of the uh advisor's trustworthiness. So we relate this to uh social withdrawal right it's like for many of the agents they could actually maintain a belief that the adviser is trustworthy by withdrawing from the game. It's like saying, "Oh, I will just stay away from you." Not stay away from you. They're still in the game, but I will not engage with you in order to maintain the view of you as trustworthy. And that will help me uh sort of regulate my emotions and it will help me regulate my own personal sense of safety, right? Um we compared a few models in particular those with lowest highest and average levels of hypervigilance scores um with those with the sort of average level uh of hypervigilance who almost always uh this this particular model almost always avoided um some interesting things. one, they do align pretty strongly in certain ways with some of the takeaways I have from Jeremy's research, which are um agents become to uh they come to expect uh seeing themselves not engaging with the game. They they they basically come to

expect that that is how they operate in the world. Uh which is to avoid sort of reality testing, avoid uh checking to see if others are trustworthy or not, reliable or not. uh and instead they also start to further uh expect the symptoms that they experience. So uh beginning to even more increasingly expect uh physiological arousal expecting oh this is this is this is my life in a in a certain sense. Um and then uh by withdrawing again we get to see this kind of phenomena where the agent maintains a belief that the adviser is trustworthy. Although at the higher order at the metacognitive reflective level um it's very it's it's not a precise belief, right? It's very close to 50/50. So it's a very highly entropic belief. It's like you might be trustworthy but I'm not going to engage with you so I don't have to find out. Right? Um and then finally the the this is the agent who had the highest level of hypervigilance. Um and um there's some interesting things there. Whenever we look at the the modulation and safety beliefs over time um they this agent also withdrew very regularly. And so whenever they would withdraw it would allow for this brief respite from believing that they were in danger. This agent who had the highest level of hypervigilance they unfortunately did believe that they were in danger at times even though they of course weren't. And so it's a false inference. Um but it by by choosing to withdraw and seeing like oh I I withdrew I don't have to find out uh if I'm in a safe sit I don't have to find out if this person is trustworthy or not. I don't have to you know do much of anything. And so it gives us yeah this brief moment of respite uh beliefs about being in uh danger which are very high for a moment kind of become more entropic and it's more like oh okay I'm kind of sailing this uncertainty right now but unfortunately only to resume playing the game again where danger becomes perceived once again. Nonetheless again these are this is the highest level of hypervigilance so these are very extreme behaviors we're looking at. um and then the agent always withdrew. Um another interesting thing is that the agents gamma parameter which is aligned uh in the empirical research there's a lot of good work by shortenbeck and others in 2015 looking at dynamics of dopamine in relation to prediction error um this agent essentially over time loses confidence in its own decision-m uh and by doing that it actually starts to rely on its habits of avoidance instead. Um this is very significant because whenever we actually look at the policy posterior probabilities whenever it's actually trying to um um engage in goal-directed or deliberate behavior um it never once actually infers that it should withdraw. It infers that it should play the gra the play the game. It should choose blue or green and it even so we can actually look at what it it believes it should have done. Um but because the loss of confidence right um it it's this reliance upon habits in instead. So um to wrap up so sorry we're going to have some discussion at the end but I think

we're about out of time. Next steps um we would like to kind of increase the without making the model too complex um uh we we just want to add one more thing which is to use more of a predictive coding or hybridic model in relation to physiological arousal. This comes up very frequently in the study of pathology and especially PTSD and other uh stress related uh disorders where um dynamics of like the the activation of the parasympathetic and sympathetic nervous systems leading to uh hypo and hyperarousal respective respectively um can strongly impact how one engages with the world um including in extremes of you know potentially leading to dissociative behaviors or more activated behaviors on the other hand. And so we think that uh would kind of overcome some of the somewhat excessive simplicities in the original Adams model. Although the Adams model makes a bit more sense. They weren't trying to uh zoom in so much onto the relationship between arousal and affect the way that we are um you know to in the same way. So um we would like to actually fit this to continuous data uh which is the idea why we would use a predictive coding model um or something along those lines. Then also we want to move towards you know making this model useful. In addition to empirically fitting it, we would like to include things like um uh kind of equipping the agent with potential emotional regulation uh or uh grounding uh strategies maybe in its policy space maybe at the higher order level that kind of aligns more with um previous neurophenological work on on metacognition. Um, basically if an agent who has been through therapy of some kind or has learned some kind of strategy to help with emotional regulation, giving them the capacity to choose to employ that um to whatever minute degree, maybe a breathing technique, not sure. Um, you know, we would like to very much get further into that. We could also look at um treatment interventions in the form of SSRIs or SNRIs. uh we know that noradrenaline and norepinephrine uh are strongly implicated in uh aberrant uh neurotransmission in individuals with with stress related disorders. Um and so we could kind of simulate that because we can actually start to model map that parts of the model. Um we want to of course uh further test this model in different ways. We're very interested in the dissociative subtype in PTSD. Um and then finally um a big part of the length duration it's taking to to sort of do this work is that uh we actually committed to using open source tools and so I've been uh carefully studying uh Fristen and others uh SPM uh library and mat lab and then reimplementing a lot of pieces and and auxiliary functions from that into uh PIMDP which I hope to sort of re-release um given it has an open source license. Uh more preferably I'll be in touch with some of the developers which I have been. So um probably the latter situation and um yeah and then also looking at I would rather get back into the original

PIMDP because it's um they're using uh jacks for GPU acceleration and essentially speeding up the ability to kind of train and test these models. So um so sorry that uh we we I really went over on time here everyone but uh that essentially concludes my portion of uh the presentation. Okay, thank you all. Till next time. All right, we're back with John Boy at all. Yes, thank you, John. To you. >> Hey, I'm sorry, I'm having a little audio trouble. Um, so, um, and everyone can see my screen, I assume. Um, we are here to talk about, uh, story graphs, which is a part of the Cognar project. I'm a um I'm a research fellow at the uh active inference institute and one of my my main project there is called Cognar and uh um there are a variety of links to that project and also to the uh to papers that are uh kind of give the background story to that project. Uh I invite you I invite everyone to stop by the active inference institute web page to look at the project and today we will be uh talking about story graphs and that is particularly uh um discussed in this paper cognar ecosystem preliminary thoughts on a on a story graph meaning representation. So um what is a story graph? uh story graph is a core component in the cognar um uh ecosystem. That ecosystem is its purpose is to facilitate group cognition at scale. That is um you may have a group of a 100,000 or who knows a million people online who wish to communicate uh their their um beliefs about some topic whatever that topic might be for the group. Um and uh the the the um ecosystem is used for uh deliberation and decision-m and communication. And the idea basic idea is that people in a group can share their belief model their beliefs about the situation and then by sharing their beliefs uh hopefully the group can come to some degree of shared belief shared understanding of what the situation is and what might be done and what you know what are the what are the promises and what are the dangers and all that kind of stuff. Uh so a story graph is a core component of that. It's it's a essentially a intermediate representation between natural language spoken language and a formal uh implementation of a belief model in in a computer in a computer like with vectors and matrices and things like that. Uh so a story graph is readable and verifiable by both humans and computers and um uh it makes transparent and explicit a a storyteller's rich probabilistic belief model about some situation. So in a story graph a person might uh uh convey you know about this about some particular situation what happened what who was involved why did it happen what might what might happen next and and that is a prediction about the future and then how uncertain are my beliefs about that prediction or how uncertain are my beliefs about what actually happened or who was involved or all the different things. So in that sense uh a story graph should be in a sense e executable. That is you could follow a story graph to see possible ways that this person believes the future might unfold or

or look into the past and see possible ways that the the the events that might have occurred to lead to this situation. Um uh a story graph conveys a person's in the Cognar system at least a story graph is intended to convey a person's honest beliefs about what happened. That is if I honestly believe that cows can fly that may be what my my story uh involves. Uh somebody else might not believe cows can fly but uh the point of a story graph in the Cognar system isn't to isn't to model the real world. I mean the system does not model the real world. This the cognar system conveys the user's beliefs what the what the user honestly honestly believes about the situation. And very importantly uh story graphs are created interactively in the cognar system. Um so it is not the case that uh that somebody would write out a long narrative and then feed it into the cognar system and then that would become a story graph. uh rather um the the uh person would interact with the system to create a highquality story graph. So as an example, maybe I type I'm typing into the system that something about Bob and then the system might query me or others in the group might query me and say uh wait a minute uh do you mean uh Bob the neighbor or Bob the baker or you know there could be all kinds of questions like that and and through those kinds of questions and guidance the the system and the group can develop a a set of highquality uh narratives and behind the behind the story graph and behind the cognar system are our set of ontologies and knowledge bases which allow for consistency of meaning across stories. So if I'm talking about a a cat, we all have an understanding about what a cat is. If in it might be a mammal or it might be a bulldozer, but uh whatever that is, uh we we we understand what we're talking about together. And uh there are applications well beyond cognar for story graphs in medicine uh industry, governance, commerce. Uh and today that is the topic of our talk. We will be uh discussing some possibilities of how story graphs might be used in a variety of u of settings. And I'm hoping that we have time for uh taking questions from the from the audience. um we would be very interested to hear of what others might think of like oh that a story graph could work in this situation or that situation. So um please do um please do uh chat with with us during the during the presentation and um we will explore together hopefully. Uh just a few more background slides. Um so how is the story graph related to active inference? uh in active inference uh as I think most people are well aware the uh an agent an intelligent agent is separated from its world the the generative process of the world through a marov blanket and the agent has an internal probabilistic model about how the world works and that model in a sense is what the the story graph is trying to convey the belief model of this person and um in the cognar setting the group cognition could be viewed as sharing an updating belief models

and hopefully evolving towards some kind of shared towards some kind of shared beliefs. And uh the purpose of cognition of of course can be seen as promotion of well-being and the reduction of certainty. And in the in the active inference world uh we talk about free energy reduction. And uh today's talk uh the purpose is to discuss possible use cases for story graphs in and beyond Cognar brainstorm with the community. Um some some areas that story graphs might have application would be decision-m customer service emergency reporting public polling instructions systems descriptions education democracy and elsewhere. and and some of some of these applications are discussed in the paper that I've referenced before but I but I suspect that there are many more that u that we haven't thought of. Um possible functions of story graphs could be model verification u u model val validation, explainability, uh communicating uh in a variety of different modes. One to many u u you know a a corporation or some other entity might want to reach out to its uh to its group and and convey some kind of rich information. uh many to one uh many to many uh that could be used for recordkeeping auditing uh of a decision process and governance uh assessing the accuracy of predictions as it once events unfold that is I may have a story about how the world is going to how I think the world is going to unfold in the future and that story is then a record of you know my my beliefs at the time and then as events uh transpire one could look back at that story graph and see what you know was John accurate. uh did did he uh you know did he predict correctly what what came to pass? And that could be actually very useful u u for a variety of reasons because some people probably are much better than others in predicting what is going to happen next. And then that you know that's knowing knowing who is got a handle on how the world works is a valuable asset to a group. I should also say too that I I I had a word earlier in one of the slides about rich stories and that is also the idea behind uh story graphs and Cognar is that in the group setting we would like uh we would like a person to be able to share a rich story, a nuance story, a dynamic story that and I will contrast that to say um joining a group online where maybe someone types in a um a a statement of some kind like we should do A or we should do B and then everyone can give a thumbs up or a thumbs down to that. So we're talking about something far far more sophisticated and and um and have the have the capacity to convey a very rich story and nuance story a conditional story like I think this will happen if this or maybe that will happen if this other thing happens or maybe these three things will happen if something else happens. So uh so um yeah so rich stories uh other possibilities are u um and for even a a story graphs could serve as sort of a lingua frank agent to agent communication that is as agents are negotiating some some topic or aspect they potentially



could use story graphs to convey their uh belief models about how their their world works in that situation. Uh and then lastly the purpose of this talk is just to socialize the Cognar project and the story graph idea and again to invite participation. Uh our panelists today are myself, John Boy, U Tim Lloyd has joined us Glenn Repello will speak. I'm hoping Philip Wilson is is uh scheduled to speak but he's not here in the moment. So hopefully he will be able to join us soon. and then of course the audience. So just a couple more items about what is a story graph um and and how does it fit into the Cognar system. Um so in the Cognar system uh a a user would type in a narrative, their story, their belief system perhaps sentence by sentence or even even phrase by phrase. Um but um that would occur in short inputs and potentially a person could speak this or there could be other multimedia ways to input a story. Uh eventually uh um a user would again type in some kind of sentence or several sentences that would be translated into a story graph fragment and then as the story graph fragments uh are input uh they would be merged into a larger story graph to convey meaning throughout the narrative. And then once a story graph is completed, once the whole narrative is completed, then that story graph might be uh translated into alternate alternative meaning representations. Um there of which there's many uh maybe it will be translated into a probabilistic uh uh logical model or some other kind of logical for uh model for querying or potentially could be turned back into natural language to play that story back to the person. Uh potentially the system could ask the person, okay, this thank you for your story. This is what this is how I understand it. I'm going to read I'm going to tell the story back to you. Is this what you mean? Um yeah. So, uh and I think this is my last slide, but I've mentioned meaning representations already. So, just a few words about what a meaning representation is. This is just an example. There's from the literature. There's many types of uh meaning representations for for narrative for for sentences and and groups of sentences and narratives. And this is called a discourse represent uh representation tree structure and it's for the sentence a very simple two sentences max fell John might have pushed him and I won't go into the details of this graph but it's simply uh uh identifying the elements of the of the of the little story here and then labeling some of them as events some of them as agents for example John and Max are agents and uh also pointing out that there's some time elements to these events and there's a explanation between the two sentences. Max fell, John might have pushed him. That's a that rep that that relationship is because u um Max fell because John might have pushed him. So this is a very simple I mean relative to what we aspire to in the carnar setting. This is a kind of a a simple type of um of representation and uh in cognara there would be this would be much more you know in a in a fully functional

cognara this would be a much more uh sophisticated um representation there could be multiple layers uh could involve ontologies I had mentioned before highdimensional vectors uh the idea that is that the story would capture the dynamics of what's happening in the changes of of state that is that are happening and also importantly capture the uncertainty as I mentioned of of the story of the sto of in the story and the predictions in the story. So um this uh you can think of this image here is just a very simple version of what maybe a story graph might look like. And uh with that uh we will uh uh move on to Tim who's going to talk about real world narratives model agents that are fit for real worlds. So how are you? >> Thanks. Thanks. I just share. >> It's fine. >> So you're able to see all that. Okay. Um yeah. So uh I guess I've uh been uh Cognar and you know in discussions with John and and learning from his papers and what have you have has really captured my imagination from the beginning. I can say that across a whole spectrum of uh of associations and relevancies and salencies of the content with other things. But I'm sort of bookending since there's sort of an industrial applications theme to this. I'm just bookending my brief uh presentation of where I'm coming from and connect with this with two things I guess. One on one end is sort of a um an initiative to um to establish a sort of a multi- sector service ecosystem that's capable of um trustworthy um support in uh families and close-knit groups to curate and and and uh develop um real world stories uh together. Um and the other sort of book end uh that I get speak to a little bit here from an industrial applications of cognitive perspective is uh along a along uh um uh engagement with multi-agent software engineering modeling. You know coming up with ways to uh you know what's a class diagram for an agent kind of a thing if you want to put it real simply. Um uh so you know uh we look at real world narratives um uh as as a as a grounding sort of a cherished precious kind of a grounding of uh of what it means to be human and um and therefore serves as some kind of resource capable of imbuing AI with sort of uh things that we would consider more more human and would want to see in the ecosystem of of of agents and systems. Um and uh the idea for me in this application is to expand the possibilities for trustworthy and safe AI by actually making it possible for AIS to actually have some concept of the meaning of trust and safety from the you know beginning without hallucination of fabrication. So you know for me in this respect uh cognar represents an onboarding to sort of humans in the loop deeply in the loop and in the loops of all the loops etc. Um so it's the potential for that kind of a thing. Um and you know we are this this is the very beginning of a kind of a thing. So um you know think of it maybe like the beginning of a human phenome project. You know the genome looking for a code of life and the phenome a code of living and um you know this

little illustration on the on the right is is you know a popular illustration or iconography expressing the idea of the universe observing itself through the arrival of life. So, you know, maybe narrative is kind of like the the cognitive water we swim in. And like a fish, we don't really notice just how um how uh absolutely uh ever present and and it is. Um and so um yeah and just uh I listened to the earlier uh presentations and uh I just captured some quotations here that to me I kept hearing you know uh narrative narrative narrative again and again just as a as a sort of a structural uh theme if you like in a lot of the talks that were across from um you know motor systems and and and physiological health and balance. Um, you know, your life is composed of moment-by-moment interactions. Um, you know, sounds like a story kind of thing. Reference are inherently ambiguous, etc. until they uh they require context. Um, so a lot of these quotes are or well, these quotes are drawn directly from content previously in this in the symposium that all to me speak to uh the value and the potential of uh of uh cognitive narrative. Um Matt Brown spoke of uh an active synchronization, sustaining, productive, adaptive, and evolving homeostasis. Well, you know, Cognar represents the similar kind of potential or possibility on a psychosocial, socioeconomic, and sociocultural kind of a plane. Um you know, uh more quotes here. I didn't mean to read them all, but they're here to to look at and and uh um and uh um you know, maybe Cognar also because of the subject matter opens up onto answering this question about philosophers again being interested in and looking for new potentials here. So maybe it does serve some kind of purpose there. um more comments from the conversation uh earlier in this in the symposium to speak to uh cognitive narrative. Um you know so uh and someone mentioned in one of the presentations this this uh device this jharu window for understanding people relationships themselves and others and what have you and so you know which multi-agent narratives are capable or indicative of expanding the transparency as a basis for trust. Uh we want to interact with agents that we can trust. We want to create agents that we can trust. We want agents to trust other agents etc etc. Um that's a tall order. Um, and we always underestimate how much nature has already done for us in front of us and before us. And we uh never quite um calculate or appreciate the uh lift involved in trying to do something that nature already did like um this kind of thing. Um and so for embodied agents sort of the taxes of meaning plays out in and by situations. Um so these purple statements at the top again quotes directly from the symposium sessions earlier and these statements each one of them is could be just as descriptive of narrative. Um so you know agent life narratives comprise maybe like a lifetime of mortal computing. Mortal computing has come up a lot. there's ethical implications that seem to be uh uh woven into the mortal

computing discussion along with a lot of other considerations but um uh you know maybe can go even further and to instrumental or practical ethical considerations in AI and the idea that an agent without a life narrative does not and cannot have identity not in any way that people typically would consider accountable. Um and so just leave it here. Maybe there's parallels. Another reference to previous SE symposium sessions when I saw some of these modeling efforts on NGCLARN trying to deal with real neurons. Well, you know, maybe part of the software engineering uh industrial benefit from Cognar can be uh richer uh modeling languages, modeling elements and uh methods, methodologies and processes for um real people, real stories um you know and therefore agents that are capable of trust. Okay, sorry. So I just wanted to this is more of like a use case for Cognar. I'm mostly ignorant of the the entire project but anyway this seems like a a reasonable use case. So so this project started out of the backpack consortium which is uh intended to study chronic low pack low back pain and classify various different patients into groups that might respond to various different therapies. So basically what we do is we take their their their clinical trial data which is about 2,000 variables or so and we use language models to assemble that data into a narrative. We narrate that data and of course that that story that pros document that that we show to the patient will then not read anything like what they would tell a story they would tell about themselves. Right? it's going to be uh anodine or or it's going to have the character of whatever language model we use. But anyway, so we generate that thing and then we have a a semi-structured interview with the patient. We show this this pros document to them. They read it and they react badly to it and they say, "This is not like me. It doesn't use the words I use and blah blah blah." Then we take that transcript, we process that again with actually 11 different uh self-hosted language models and we we kind of come up with a new one and then we have another meeting with them uh where they criticize that one. And so we do this two or three times and finally we get to a validated narrative that the patient actually says this this represents me. It represents my attitude. It represents the things I have to say about that. So the reason we're doing this is we have this working hypothesis that inside the human there's latent information that our objective measures just don't capture, right? The even with survey data, the self-reported data, this there's stuff that this we just can't capture with those measures. And so the idea is that um a person's story is going to capture this latent information better than anything some committee decides to try to measure on that patient. So that's what that's why we're doing it. It's really for mechanistic knowledge, but it has a bunch of other um potential uses as well. So now I'm going to show you a couple slides that I'm not going to talk about, but this is

the workflow for data narration. This is how we take their 2,00 variables and come up with a story. Fine. And I want to mention knowledge graphs. We do similar things with with corpus uh with a corpus of documents, right? We write a paper. We have citations in it. Then we put that all up into a big hairball. Of course, the idea is to reason over that hairball. And you can only reason over such things with algorithms. You can't really do it in any kind of human way. But the point I'm trying to get at here is this this LP story, the lived pain experience story, this narrative, and this knowledge graph are two kinds of the same thing, right? They're both these artifacts that are very static and we don't want that. So the important part here is that it's clear that that narrativity a person's psychological narrativity directly correlates with improved therapeutic outcomes, right? So they they respond better to to various different treatments if they can tell a story about themselves. Um, and if you find a less nar like an episodic less narrative person, you might be able to improve their ability to to respond to a treatment by teaching them to be narrative about their tell stories about write in a journal, you know, say tell stories about yourself and that might improve therapeutic outcome. And then the problem with that is that psychological narrativity is is correlated with psychological rigidity. So you can get into kind of a canal structure, right? you get into a rut and you tell that same story about yourself like I I have chronic low back pain and I can't walk a mile, right? And to get out of that is it it takes a kind of a a heat a disruption to get out of that. So part of the problem with things like uh knowledge graphs and ontologies and all those databases of any kind is that they're so static and the same would be true of any kind of artifact that we derive from this this protocol I described. We don't want that. What we want is these stories to be alive or at least if the story itself is not alive, we want the story to be a an an actively improved living document, a highfidelity sign of life. So it see it should be obvious that this is a good candidate for for a story graph or such things. So thanks. I don't hear you anymore, John. Now, can you hear me? Oh, good. Sorry. Uh I was asking uh uh Glenn in your use case the um patient would be developing this narrative by himself or herself or it would be facilitated with a practitioner who would be stepping them through it or maybe even transcribing their their story as it's told or how how does that work out in your setting? So the the patient lived experience story really is one artifact of a class of artifacts that you might have a a message from the insurer to the patient or from the insurer to the hospital or from the clinician to the patient as opposed to the patient to the clinician. So the idea is that these these messages that are passed amongst this this stakeholder graph if you will are co-constructed not they're not solely deconstructed uh constructed by the source node of whatever message or whatever length they're passing along

they're co-constructed in a way when we talk to a lived experience panel like you know individual patients then obviously that's something they would generate by themselves but in any other stakeholder stakeholder communication I think they should be co-constructed. No, this is a you know this is a great uh this would be a great application of story graphs I think. Um Daniel I can't see the board and I don't so I don't know if anyone has posted any comments or suggestions or anything but if they have um we could address them now. Yes, I will I will ask Oh, yeah, please, Tim. And then I'll ask any questions below chat. >> I I if I think I've forgot two slides, so I'll just throw them up briefly if that's okay. just um uh yeah, I think um maybe this one I actually because I skipped forward I think to to point out the NCGLearn parallels and this was sort of an important slide that brought together uh some of the um uh modeling results being sought when I talked because I introduced I kind of skipped over the connective tissue between these two bookends so to speak. Um so so this speaks then to to to what we might hope to achieve uh with agent modeling agent system modeling that can operate on you know with the resource that that could be produced in in a cognar setting and um you you know uh this idea of fiction's ideas um uh to extend beyond what we're capable of in maybe more rational sometimes maybe seen as seen as more capable but actually not more capable apable in some instances things like philosophy. So Richard Roerty is someone who in this book um reached out to uh the insights and sensibilities of great writers to try to ask where where do we actually gen promote genuine sense of human solidarity and um so the question is how will we write solidarity with our artificial agents into our story and um I had just had this one quote that was worth putting on the screen briefly from Richard Roert which speaks to the open-ended evolution of experience and the human phenome. And so, um, being used to developing systems that, uh, think they know everything and can anticipate everything and are sort of foreclosed systems. Again, to the point earlier, we don't appreciate how much work nature does for us when we try to step in and help out. Um, well, you know, uh, we have to keep these things in mind. It's the idea. Yeah. Thank you. Sorry. the quote that quote is almost worth a full read and then I can do any live chats but I think that is a very salient thing to read Tim if you want. >> Uh you want me to read it? >> Oh it says the fear of scientism of naturalism of self-objectification of being turned by too much knowledge into a thing rather than a person is the fear that all discourse will become normal discourse. That is, it is the fear that there will be objectively true or false answers to every question we ask. So that human worth will consist of knowing truths and human virtue will be merely justified true belief. This is frightening because it cuts off the possibility of something new

under the sun, of human life as poetic rather than merely contemplative. But the dangers to abnormal discourse do not come from science or naturalistic philosophy. They come from the scarcity of food and from the secret police. Given leisures and libraries, the conversations which Plato began will not end in self-objectification. Not because aspects of the world or human beings escape being objects of scientific inquiry, but simply because free and leisured conversation generates abnormal discourse as the sparks fly upward. That's Richard Berti, philosophy in the mirror of nature, 1979, page 388. Yes, I'm I'm I'm sort of struck there by the requirements for a computational story or knowledge graph system to have strong typing from like a computer science perspective and to be composable, flexible, all that kind of data operations. And then to what extent would that turn discourse into normal like roti sort of fears to make it a finite game and say well here's our vocabulary set here's the combinatorics of discourse or here's what's the allowable discourse and then there's that spark that takes things out of the normal and so how to have the story graph also be able to bring in new recompositions and new elements to reflect that open-endedness. >> I'll just say I think that's what narrative does. Stories continue. We always make new stories. Stories are being created every day. People are living new stories every day. So, Cognar represents if you like the conduit into the real time uh emerging, imminent. Uh and so if we can connect to that uh you know in the real generative source being the humans you know and always stay attuned to that and not let the one uh occlude the other the put it this way without cognitive narrative good luck. One of the things that I've talked about a lot to John and he's probably tired of it by this point, but it's the idea that any two stories can contradict each other and that's that's fine, right? It has to be contradiction tolerant. whatever whatever structure we end up um putting underneath the hood has to be contradiction tolerant >> which I'll just say also is sorry John it's the that that's the equivalent or parallel to to something uh one of the bullet points I put in there which was uh scaling the global and local and everything in between. Uh Daniel, can you I I don't know if you if you can share my screen. Is that possible? >> Yeah, I have it up >> and I I'm sorry I can't see it though. Uh it it says example use case group decision-making. Is that correct? >> Yeah. Okay. Uh I just wanted to this is sort of on the topic. uh but I wanted to show uh in the Cognar system how uh deliberation might occur on some topic. So it would occur in a in a uh in a series of rounds as we imagine it where people would create their story about the situation. Um those those the system would assess the set of stories from a group and perhaps uh report on how uh similar they are, how different they are, the different elements that the stories contain. Um the different perspectives that are taken, you know, from the from the

different uh members of the group and then those assessments and the stories would be shared with the whole group. So then I can see what Tim is thinking and I can see what Glenn is thinking and maybe on this particular topic I like something that Tim said and then if so I might I might include that in my story graph now for the next round and then this kind of circular uh rounds of editing and viewing and assessing would continue until until some kind of ending point. Hopefully uh the ending point being that we are starting to develop a shared understanding of what the situation is and maybe perhaps also what we want to do, what kind of action should we take. But the what I wanted to point out is that um I'm free as I had mentioned before about rich stories. I'm free to have a very different opinion from from anyone else in the group. uh you know I I might be you know far away from what other people are thinking but my story graph is going to convey that information and that information is then now part of the part of the group narrative and doesn't disappear uh so I may I may not agree with anyone else in the group and I may think that you know I may be looking at the situation from a very different angle and that will be that that My story will be part of the group narrative and would be available to the group in the future as we as maybe we look back of uh you know what what was discussed and who was thinking what and what you know did did anyone predict what actually happened or you know whatever the story is. Um there there should be room uh in in our narratives to convey uh not just to convey the same thought over and over again but to convey whatever uh whatever a person's true beliefs are regarding some situation. And and of course, those beliefs are dynamic. They change with time. As as someone learns more, they may change their mind, maybe even greatly change their mind about how they're how they're seeing the situation. So, as Tim said, without narrative, good luck. And you you just spoke to the sort of almost sort of re-entrant sort of composable can't step in the same narrative twice and and and the idea that there's stories within stories and stories about stories and hermeneutics or whatever the whole you know uh when I mention that what I'm trying to just highlight is you know how they speak about how uh we learn the language of different mediums. We learned the language of photography. You know, at first they behaved one way and then they behaved a different way because they we all learned where photographs go and sort of how to smile in a picture or whatever. Um then we learned the language of television. So the laugh track, you know, we're not like what the hell is that, you know, we get the idea of what's going on there. Um uh so to to some extent, I think what you're kind of speaking to is that we already know the the language of narrative and stories. We're already equipped with a lot of the machinery that already does the work. The same way



that when the earlier software and systems engineering um model developers conceived of, you know, from Aristotle or what have you, you know, classes and objects being a more productive organizing concept for this all this text of software and layers of translation, etc. Um so that we could actually make it attractable process and deal with reliable and maintainable complex systems over long evolutions. Um you know we had to invent sort of reach for that and invent those classes objects you know what have you. Um, you know, narrative is simply this is proposing that it can be the backbone of of of some of that and it already comes with so many of the affordances that are already natural language to us that John was just just with, you know, ease saying and everyone understood what he meant and yet it was the complexity of what he said when he talked about the folding in and complexification of narrative with your narrative, narrative about narrative, your narrative, my narrative, etc. It was so easy for us to talk about that, but the complexity that's embodied there is astounding. >> I'll read a question from the live chat. So, this is from Samuel Ner who wrote, "Hello, John. Amazing idea. I was just thinking if story graphs could be useful in research with more qualitative aspects such as psychedelic experiences, dreams, consciousness in general. For example, to make better sense of such subjective experiences, convert it to something which can be better analyzed by standard statistical methods or just strike a better balance between them. Just throwing the idea out there. Let me know what you think. No, that I mean that would be a that that might be the kind of story where where cows do fly you know may it may be a story that is uh if someone is relating their psychedelic experience for example it may be it may story may have very loose ties to what we kind of consider shared reality and uh that is okay that is ex you know that is acceptable in the system and and that could be useful. Uh I potentially that could be very useful as capturing a capturing the narrative of a patient who is who is uh experiencing this this uh you know a very unusual uh you know life episode and then uh and then having having captured it in a meaningful way that that the that the person can verify that yes that's that's what I was experiencing you know this is this is a correct narrative and that the computer can understand so that so you know if you had a repository of these uh narratives uh you know one could do all kinds of statistics on what were some common themes and what kind of transitions happened and what kind of psychological states were occurring and you know all all the rest so great the great it would be a great application I think Glenn >> yeah there's currently As we speak, there's the psychedelic lived experience um symposium uh summit. They call it a summit uh that's going on. And they're talking about things like the mystical experience questionnaire, which is another one of

those um designed by committee survey things where they ask you and the patient would self-report how what they saw, what kind of experience they had. And almost without exception, all of the people in the community are like, "This doesn't really capture the the mystical experience it's trying to capture." And so these stories that they tell and that the psychedelic community is rich with with fantastic stories. So that's to me that's like the the ideal the ideal community for this sort of thing. >> Yeah. Yeah. Yeah. Uh also I would like to just mention too uh in the cognar setting particularly in the cognar setting one might ask the question why would somebody want to go through the trouble of of working with the system to tell their story because you know it it might be much easier in some other system uh where you just simply write out a you know some paragraphs or you know episodes from your life. You write it write it down. Everyone's familiar with writing things down. or maybe you just speak it and that's it. You're done. It's just captured by voice. Why would somebody want to go to the trouble of kind of uh working with a system to kind of cur curate the story to refine the story to really convey this uh you know this low ambiguo story with low ambiguity? Well, the answer I think is I mean I would hope is because people really want to tell their story. people people want to tell their story in a way that their that their story can be heard their experience of life can be heard by others and has a has a makes a difference you know so I suspect that if cognar is that you know the concept of cognar is we're still at the very very early stages so you know we're we're still even conceptually putting together how all of these pieces work and what what are what are the exact elements the mechanics of a story graph we're still working on that. But one day if this comes to fruition, I suspect that um that you know many people will want to tell their will want to tell their stories and be heard by a larger group because our stories are our stories are important. Our stories are you know that's the that's one of the central elements of social life is conveying our story to others. So, um, yeah, that would be the draw of of of Cognner and the story graphs is like here's your chance to tell your story and have it be understood and heard by the larger group. >> What are the directions for the project, John, for the end of the year or for 2026? >> Uh, well, the end of the year is very very close. uh we are we are we are discussing uh you know now what h what are the mechanics of a story graph what are the elements in a story graph and I don't want to say too much about that right now because uh it's it's still in flux and we might you know we might add or remove elements from it uh you know shortly but but I will say that what we're considering right now is um something that would a layered graph, multi-layered graph uh and different layers in that graph might represent several different kinds of meaning representation. So um um I I showed

one image of a meaning representation earlier on my in my um slides. Uh discourse representation theory meaning representation and um one layer in the in a story graph could be very similar to that. Uh and then all the tools that work with that meaning represent rep meaning representation might be useful. For example, you know, logic, logical forms and various types of logic could be uh um lo logic programs could be used for that kind of meaning representation. But then there could be other layered meaning representations on top of that. For example, a meaning representation that's really aimed at narrative. uh there's a you know people are working ontologies for narrative and uh so uh that could be another layer in the story graph and there could be another layer perhaps for time time evolution and the different types of time that can be conveyed in a narrative. So that's kind of how what we're thinking now as a as kind of the next step. And and hopefully we I I think our our our immediate goal is to just simply draw up by hand what what the what a story graph might look like for a few um simple stories, you know, five or 10 sentence stories. uh to just get a handle on how we want to construct the story graph itself, the functionality of the story graph. And then once we uh kind of settle on what that is going to look like, then we can start to code up how that how that how that actually works and how it will probably have some use of LLM in there to to if even just in some simple sense of u uh an LLM might be used to say uh I'm sorry you used the word cat twice, you know, in the sentence. Do you mean did do you mean to use that word twice? You know, something like that. Um, yeah. So, that's kind of the the steps we're going through. We're still kind of conceptualizing what the story what the mechanics of the story graph are. And then we will write up some uh just some hand handdrawn story graphs for some simple stories. And then we will start to look at how that is coded, you know, uh, uh, in a in a more automated sense and how a user might interact than in a in a real, you know, with a real text box, for example. and Cognar it will if if it moves forward we hope that it does it will certainly move forward from starting very simple sometimes I I call our first uh aims a stick figure cognar just to get the very basic concept of what what this is and what kind of interactions might occur and then uh hopefully over time if it matures it will become more and more sophisticated and uh you know have more and more features as we go. >> Awesome. I'll read a comment. If any of you have a thought, you can go for it. Love evolve wrote, "Hearing more about story sharing, co-storying would be nice. Intentionally collaborative story graphing in creative art contexts. That's that's a whole another use case that we've not mentioned today is how could these be used for a creative uh for creative writing for example or or or even you know who knows game development or something like that you know I I

would think that there would be possibilities there for sure uh I I will say this in the Cognar setting as it's conceived um the group does not work on developing a single story graph rather each individual in the group has their own belief model their own belief model and the and there may be strong overlaps between the belief models of different people. So uh as as as we've conceived it um Cognar is not working on a single story graph but rather everyone's story graph but I think there could be still overlap there and possibilities for um one possibility would be uh um witnessing how you know the synthesis of story graphs of a group how what that looks like and maybe that synthesis changes as as the group evolves. evolves as the story of the group evolves. Glenn, >> yeah, the concept that comes to my mind is the exquisite corpse. I don't know if everybody's familiar with that, but you you fold a paper into three parts, and one person draws on one part, the other person draws on another part, and all you have are the little edges of your drawing that hook, and so then then that that's all they see, and they draw something else. And so then you unfold the paper, and you see the whole piece of art. So that's the kind of concept that I thought Cognar might uh might apply to. >> Yeah. And I'll just say it's not exactly art, but from a creative creative potential standpoint, you know, intersecting with it and also addressing uh Daniel's comment or question earlier in regards to the Rory quote, it should be open to extension in in the um sort of phenomenal external reality, new devices, new people, new norms, new things, different cultures, what have you. sort of open to extension of bringing in and drawing in uh you know new new new things, new language, new words, new terms, you know. Um and also open to uh uh operational def uh sort of uh u observation of emerging communicative elements or modes and disclosing to us maybe things we didn't have words for. um from from the collective uh phenomenon that that's uh emerging as in the content so to speak. So so um we might discover there's uh realities of human stories and narrative and what have you that uh we actually you know um didn't really notice or understand as fully as we we could with with the assistance of this. So it's open both sides. >> Yeah. Very good. And just to put it just to kind of touch on one of those elements. Um we've con as we've conceived it. Cognar as I had mentioned would have ontologies behind it. So there could be some kind of shared understanding of what the group is talking about. But those ontologies could be uh custom customized for a particular group. So maybe there's a particular group that's talking about you know who knows uh traffic and maybe there's some technical terms related to traffic that uh the group you know wants to hold in common. Well the group can have its uh you know its own private ontologies to um to to ensure that the terms that they're using are understood within the group. And then as as as Tim

just mentioned, maybe those maybe those ontologies actually evolve over time as the group develops a different uh different uh understandings of what certain terms mean. For example, >> I just have to make a plug for that Rory book that I highlighted, contingency, irony, and solidarity because he specifically extends that discussion of fiction as extending philosophy or offering resources to philosophy. He's, you know, um, you know, he's he's um specifically u addressing private language mediating in society between private languages and public languages. So the private irony of anti-foundationalist living and uh and the public um the public language of solidarity and science fiction sort of bridging both or being a view from both sides with our technological niche that's able to project and diffuse these different story relationships in ways that we couldn't have said it. I mean, not long ago, music had to be said live or played live. Then we had fixed recorded music. Now we're entering into possible infinite variation on the edge. So then the story is moving from the order of what was said into kind of what it always was, which is something that generates the linear sequence as told and as understood. So it's kind of like with those first principles of cognitive science, what kinds of representations are going to help us do a lot of the things that I think you highlighted really well, Tim. Yeah. with the previous but that's okay. Yeah, we hear you now >> just yeah you brought up a lot of good quotes from earlier Tim with and I think that really helps uh connect a lot of the dots and about this layer of systems engineering and and design for cognitive computing. Okay. Any last comments from anyone? >> Uh I would just like to encourage anyone who has interest in any of this to say hello, reach out. We have bi-weekly meetings on Cognar. And uh it would be great to uh hear from people who have who think maybe this could be useful in you know in some sense with their projects or their interests. And at some point we will be looking for stories too. We will be looking for example stories and uh and maybe uh people who are willing to try creating a story in our you know in our in our system. So all of that is yet to come >> and and I guess the only other thing is Philip I really looking forward to learning from him because I think some of the things he wanted to discuss would got into sort of the mathematical information theories and what have you that that that could underpin very powerful techniques for achieving really what are some core goals and and benefits and features or capacities of of Cognar around sort of the um you know category theory quantum what have you, information theories that um not something we can go into without him. Uh but if you read those papers, you'll see a whole section of information that really was only scratched on the surface here when John talked about some of the technical approaches. But there's a a really exciting rich um kind of substrate of theory that that has not

even been brought close to this Thank you everybody for joining us. >> Awesome. Thank you all. Bye. We're back with the final of the live presenters, embodied presenters appropriately with Adam Saffron. and Adam will present for uh a little bit and take any live chat questions and then we will play a sequence of pre-recorded videos from a set of other colleagues which you can kind of contextualize Adam. So thank you to you for the presentation. Thank you Daniel. Um so hello everyone. I'm Adam Saffron. Uh I am currently a research scientist at the Allen Discovery Center at Tus University. Uh working with Michael Leven and today I will be describing an uh ongoing and evolving collaboration with uh different friends and colleagues in the active inference community and beyond. um centered on basically uh an AI alignment research program uh which is centered on uh an integrated simulation environment under development called common grounds and uh I'm going to describe uh some uh different aspects of what we've been discussing and um my own views on this and uh some promising future directions uh and providing an introduction to a series of uh pre-recorded lectures um from my collaborators and friends that will follow um going into details of different lines of thinking that are um coming together as part of a broader research program. So I know that's a lot but let's just dive in. So um you know I've been interested in uh AI and AI alignment for I guess maybe uh 25 years now uh since I read the singularity is near a very long time ago 25 years ago. And um since then I've been thinking you know uh what would it take to recapitulate the principles of biological intelligence? um in AI, is this something that we will ever see in our lifetimes? And what would that mean? And if s such systems were created, um how might they be aligned to be compatible with human values? Um and uh what can we do to help promote um alignment? And so uh towards this end, you know, I've been pursuing this um in a number of ways. It's um I'd say most of my um academic career was uh in the background motivated by these interests and concerns and hopes and in uh recent years I've been uh working with um others to uh basically uh develop um an interdisciplinary research program um for biologically inspired AI and alignment and um what in this symposium um I'll be discussing the the I guess the current state of play of this in terms of uh ongoing uh conversations with um some of the uh core members of the active inference community and beyond um where um we're um planning to do research um a a research program um centered on a integrative simulation environment called common grounds. Um we're currently um so the hope would be to basically create like a ongoing support for um a cohort of active inference research fellows who could both pursue their work independently and in communication with one another coordinating around um de uh testing out their ideas in this

common simulation platform um for multi-agent alignment. Um, one of the the key interests um for this research program would involve things like um biologically inspired architectures for agent design. Um uh having agents that are um active inference agents that are um embodied and embedded in environments uh that they interact with in rich ways where they can uh construct and co-construct their niches. um where these environments are open-ended and allow for lifelong learning and metalarning. Um and where within these what among the things that would be metalarned would be um coming into uh pro-social uh equilibria and the and acquiring pro-social preferences or um policies that will tend towards cooperation. um even under circumstances where uh on occasion defection uh might seem to be incentivized, can we create learning curricula for developing uh harmonious social interactions and robustly pro-social agents? Um and then finally um in addition to uh demonstrating these principles in silicone um in simulation um through mathematical modeling can we uh come up with formalizations and something like um uh guarantees or proofs regarding alignment uh via mathematical modeling. So I realize that's a lot and very ambitious but um thankfully I am working uh rather yeah uh I'm grateful to be working with um some very brilliant people who I think uh if we keep going we can make substantial headway uh towards these ends. So uh in the videos that will follow in this symposium uh there will be an initial uh uh lecture by Nick Haye my longtime friend and collaborator who is the designer of the common ground simulation environment. Um he's uh everyone's quite brilliant and wonderful. Um he uh got his PhD with Stuart Russell. Um uh studying um meta reasoning and uh went on to uh study robotics or to work with robotics at Vicarius and is now um dedicated to uh pursuing this line of research and developing a simulation environment that he will describe. uh Alex will present on um ways in which the simulation environment can be used to uh demonstrates um some of the um principles of active inference the foundational principles including the natures of agency and value from an active inference perspective. Um David Highland will discuss how such an environment can be used uh for um demonstrating um alignment of multiple agents who each rather than have uh um who rather than thinking of uh AI alignment in terms of uh things like the the major companies developing uh you know monolithic super intelligences rather what would a world look like where you had an ecosystem of intelligences is they were um more restricted in their scope um still quite powerful but not necessarily super intelligent um within such a world what would alignment look like um with an ecosystem of evolving interacting intelligences uh David also has a co-presentation with Mao and Ma will be describing um work on um something called um path flexibility and the ways in which this

promotes resilience or when you uh when an agent acquires um uh challenges can it come back into um uh consistent behavior and harmonious relations. Uh Dalton will describe work on the formal modeling of alignment um specifically discussing ways in which you can uh model the development of an agent in its preferences and its value and its action. uh where the free energy landscape that it follows um the the um the value uh function or or the value landscape governing its behavior um be becoming canalized or where you develop deep pullback attractors for self-stabilizing value regimes. Um I've um in my own work called these value cores and in Dalton's work with um things like path formalisms and free energy and uh uh gauge theoretic interpretations of the free energy principle. You can even describe these as singularities within free energy landscapes. That's a lot but I'll I'll leave it to Dalton to go into those details. um uh Zara Shipahi uh who I've been working with a number of years uh where um I've been she's been teaching me about principles of machine learning and I've been uh trying to teach her what I think I know about uh systems neuroscience. Um she's been developing really impressive uh biologically inspired um architectures uh for uh flexible intelligences that could um be deployed in various alignment protocols that are themselves u biologically inspired and realistic. Um and then uh Hanju in her uh symposium presentation uh will be discussing uh artificial consciousness um the nature of affect and its roles in alignment and even how principles of adult development uh can and uh psychological maturity as she calls it um can be used for developing models of what robust alignment might look like down the road. Um and uh in this presentation I will describe some of in addition to providing a little bit more overview describes of some of my my own vision um with my background in effective neuroscience um where the ways in which I think biology gives us guides both to the nature of how we might develop advanced intelligence and not just advanced intelligence but alignable intelligence. The notable thing about biological intelligence is that we are um basically existence proves that both something like AGI is possible because we are um very general um and alignable AGI is possible because um biological agents and and human beings can be uh robustly pro-social. They can be um stably benevolent. uh even under conditions where it seems like uh this ought not be expected. And so can we um basically reverse engineer uh benevolence and and recreate this in artificial systems or use this as a guide? And uh so this is some of the um you know there's other um friends and colleagues who have been in discussion and who um will likely be involved in this research uh who are not part of this presentation. Um but um in addition in addition to them uh question would be uh uh for anyone listening uh if you're interested in this kind of research



program uh please get in contact with me and uh potentially can find ways of collaborating uh across these presentations the goal is to show um different uh lines of thought on alignment and active inference that can potentially fruitfully intersect and I think there's a hole greater than some of its parts here and uh maybe your work is part of this hole and hole towards what? Well, uh basically the hole towards um uh what would it take um to uh develop artificial intelligences that have the properties of biological intelligence in terms of their power and flexibility and also their alignment. So, uh, common grounds, uh, Nick will go into more details on this, but in brief, it is an artificial life inspired simulation platform that allows one to deploy and, uh, and simulate richly embodied agents um, at scale. Um where this rich embodiment can include things like um complex internal states um respiratory dynamics, digestion, uh metabolism, uh uh vulnerable embodiment where um there's that need to be protected and maintained um through as they interact with with environments. Um and where the body itself is part of the intelligence and valuing where uh some of the just as exists in nature a lot of the intelligence of natural uh uh minds is actually in the embodiment itself. The the way bodies are designed uh can be thought of um allows for um many fruitful properties that help to scaffold minds and uh enhance their power. Uh in this environment uh there is uh realistic and interesting physics uh which allows for uh things like uh niche construction uh which both in learning to navigate those um is of interest for places where things like active inference can potentially demonstrate its value uh for uh having systems more skillfully interact with these complex environments. Um but also um the uh environments allow for as I mentioned before basically niche construction and co-nitch construction as structure is built up and uh and modified um as agents interact and in the process come into various forms of cooperative or competitive interactions. Uh there are other approaches to multi-agent alignment using simulations such as Metagrid and the common grounds environment can be viewed as complimentary with these that while um an environment like Metagrid allows for maximal scale of simulation um common grounds allows for um simulations which are are scalable but um more detailed in terms of uh what can be modeled at the level of the individual agent in terms of the agent having a uh rich embodiment that embeds it in its rather a rich embodiment that embeds it in in its environment. uh the and part like the richness of these embodiment um can include things like um bouncy um and rhythmic dynamics and soft morphologies um of a kind you see in biological systems and as I'll discuss in a little bit um I actually think uh rhythmicity and others have think this as well might be actually quite important for understanding the natures both of the the way in which um bodies are controlled

and their rhythmic character with rhythmic nervous systems but also um how shared rhythms can be part of the way in which agents um can be aligned with one another. Um and finally for common grounds it's um uh they the intention is it for to be something that allows for gamification where people can uh engage in uh basically in an enjoyable activity of uh modifying the agents of their interaction with the world and interacting with them such that uh you can promote their alignments you know can you so Nick is trying to make it um fun and enjoyable and relatable for people to um interact with artificial agents in a virtual world and see if they can help promote their um alignment and their pro-sociality. So that's a brief common grounds. We'll go into more details. And so now I'll discuss uh some of my own interests briefly before um uh opening for questions and then we can uh people can uh see these other videos cuz you know these are um extremely uh sophisticated thinkers and I'm not going to be able to do justice to their uh their ideas. They should speak for themselves. So um in my own work I find common grounds to be uh very interesting because I've um come to a view which I've um tried to express in a number of publications uh that um embodiment is fundamental for minds and I think this is a view that's common within the active inference community and particularly appreciated um and so I'll sometimes say that um and others have said that you know the body is a prior and that the body allow allows for um in the process of minds bootstrapping themselves into increasing degrees of sophistication. Um bodies provide um particularly important to holds and this bootstrapping process as in Asia engages in structure learning and complexifies its models of self and world. And so part of the reason I think bodies are so important is because I think they provide a learning curriculum that has many desirable properties um for uh learnability and for this bootstrapping process. So if you think of for instance like an agent of of a infant um uh observing its own hand um as opposed to for instance like a in contrast let's say um like a deep learning system um doing vision by looking at a pixel array um from a world it never interacts with. In this case the thing to think of is an infant learning its own hand as a kind of prototypical object and causal system. And so this this hand is basically uh always uh there and present for being observed and learned. It can have action-driven perception where by moving it around you could resolve your ambiguity and where the movement itself will generate correlary discharges or reference copies which themselves can be thought of as a kind of crossodal prior. um and respect cross metal prior um are the sensors around the hand you know through the proprioception and the tactile and the warmth you're having many correlated data streams and and the visual appearance and maybe the sound even the smell taste um you have convergent multimodal

inputs which each can serve as constraining priors for one another so further helping with learnability um the bodies interact with themselves a further source of uh con mutually constrained inferences Um there's effective or salience. It it feels like something to have a hand and move it around. And so this salience is will uh be part of why this information will be attended to and will help to structure learning. And where the hand itself because of its um morphic properties and it what it affords um allows basically perception to be structured according to affordances. Um things like manipulability, reachability. And then finally uh uh thing about biological embodiment is they're designed or they're um structured such that they are remarkably learnable or controllable. Uh if you think of something like you know like like like a hand the way um the joint angles work it grips naturally by its morphology or the way in which the shoulder joint has a limited range of motion. Um R Fifer has described like for instance like this itself is useful where um if you're reaching out front of you and not in back of you the things you reach and you grab are um naturally brought up to also be seen. And so things like hand eye coordination. So um I think bodies are uh fundamental in the process of uh bootstrapping minds and and developing um basically rich causal models and where this bootstrapping is remarkably efficient. um doesn't require um even though there is a lot of experience for biological learners. Uh you don't have to necessarily um uh biological learners are remarkable in how efficiently they can learn and how much they can learn from so little. So uh this is part of the reason that I have interest in common grounds is I think can be used to demonstrate such principles by providing agents with interestingly rich embodiment which can be part of their learning process. Um I've also suggested that um embodiment is uh also important for alignment and um others have as well in terms of shared rhythms where um rhythmic activities are potentially special for the ways in which they can entrain uh neurodynamics and attention and through positive feedback between entrainment and attention. Uh rhythms can allow for intense focus on those rhythms. And where um at their limit they can lead to um various uh states of mind where through this intense focus um agents can experience both um deep affective states and even um things like self-transcendence states uh as you might describe in terms of uh flow uh where the intense engagement with the rhythm uh allows for some self processes to be downregulated potentially allowing for an openness to the environment and others and maybe in particular others who you are also sharing rhythms with. Uh so I think this is uh you if you think of uh the the conditions for uh bonding in animals particularly mammals but not just um you'll have things like mating song and dance and the mating act itself where um engaging in shared rhythmic activities or and humans um

things also birds but humans um things like music and dance um these are uh or another kind like a um in militaries uh um people uh marching together. Basically shared rhythms seem to be an important part of uh co-selfing or developing um uh a shared interest with others. Um so >> finally not quite finally um among the things that uh I think would be important to simulate and what we're looking for in uh in alignment is something like the development of robust preferences um something like orientations and so some of my own work in effective neuroscience um and the the nature of preferences and motivations. I've uh described how um basically you can arrive at something like in um robust self-sustaining preferences or orientations as pullback attractors through iterative uh action selection and reinforcement learning or in active inference terms iterated policy selection and updating of priors where the idea is is that if an agent as it's in the world um engages in a um selecting different policies for what it thinks will be more or less likely to realize value or minimize it's expected for energy. You then acquire this then results in updating of your policies based on how well things worked or not in terms of their likelihood of deployment which then influences what the agent does resulting in likely patterns of behavior further going to those areas of value space further engaging those behaviors where even more model evidence can be achieved. So the idea is um you learn this is what works. this tends to be um good and you begin to learn this you see you select actions that are cons that would be consistent with this and then you can acquire further um evidence in this way and if this happens within sensitive periods of development um even if not you can get this like critical mass of model evidence where you have a pullback attractor where this is just a robust um model or paradigm of this is what I ought to do and it's self-stabilizing the agent doesn't think to try other things because what is overlearned to be valuable um or is overlearned and so uh becomes robust. So in terms of you know this idea of like developing a so on Daltton's talk we'll get into details of actually how this potentially could be uh formally modeled in terms of the canalization of energy landscapes. So, but what is it that we would want to uh canalyze? Uh what what values do we want to select and um make robust? Um so things like uh mutual care, tendencies towards cooperation. Um some people discuss this idea of like an infinite game or uh ways of being in the world relating to others that allows one to continue to cooperate and coordinate over time and co-realize value. Um, you know, can we through this come up with models of uh and formal models even of what we mean of seemingly um ethereal and um uh fuzzy concepts like love like could we model love as um the co-empowerment of agents and use uh formal frameworks like empowerment to actually um describe uh mutual care. Um others like um Rutkar Gokuas have

um suggested also um that within active inference you can get uh views of um moral orientation that of a quasi content variety in that if an agent is pursuing um policies in the future uh that require um uh deep temporal sequences to be realized. This will tend necessitate uh cooperation with others. Like for value to be realized, for an agent to be able to predict itself realizing value, um it must be able to um under uh optimal circumstances and and an agent is likely to learn that this is more likely to be achieved via cooperation. That actually the selfish thing to do is to cooperate. This also relates to some of um I'd say Mike Leven's work in terms of like cones of care and uh David and uh Ma will discuss some of that also. Um and so finally uh uh Zara and some of the um architecture she's describing and um some of the ways she's hoping to deploy them um um connect to some proposals from uh AI alignment uh of of how you can have protocols for alignments um from people like Stuart Russell and others for um cooperative inverse reinforcement learning or um having agents basically learn to optimize for uh the preference differences of others. Um to basically develop things like joint intentionality as someone like Thomas would call it. Um where agents learn to have shared or common goals. Um and then fi finally as I was mentioning earlier in terms of uh like the case of shared rhythms uh I think looking towards the mechanisms of the ways animals bond or um attachment as it occurs and mammals. Uh this can potentially give us a uh guide to um uh understanding the nature of alignment in uh biological agents and potentially artificial agents that work by similar principles. Uh and then definitely more finally um and both speaking to Zara's talk and Hanju's symposium um there might even be an importance of things like uh artificial consciousness uh depending on how we mean in terms of um having different senses of theory of mind or mentalizing or um being able to uh understand the mental states and experiences of others by reference to your own experience. So that was a lot and there's even more but uh I will uh leave it to uh my friends and colleagues to describe their work and if you have interest in uh getting involved with such a research program um and want to know more about what we're planning uh please get in contact with me. uh I and uh the the the team uh would be grateful to uh talk to you about what we might be able to do together and grateful for any suggestions you might have. So, thank you so much for your time and attention and I will take questions. >> Thank you, Adam. Okay, I'll read questions from the live chat and then we'll play the videos from Dimension people. Okay. mundane bloke wrote, "What are the infinite games about?" there is a book called infinite games and it's contrasting them with uh so like finite games uh would be something like um like a tit for tat situation or like a a prisoners dilemma situation where um uh or um something like stag hunt

basically games from economics where uh the the equilibria the n or the what's incentivized um depending on the context might be um either cooperate or defect. Uh the idea of an infinite game is how can an agent um uh how can how can you be show up in a way that allows the process of uh playing games to keep going? um games that don't terminate where you just uh realize value as a one-off. Um because in one-offs the the national equilibria is is oftentimes affected but rather um games where uh by continuing to cooperate you can continue to get nonzero sum uh uh setups and uh bene mutually beneficial interactions where everyone comes out ahead and p people become basically co-empowered or can uh realize value more effectively by helping each other to realize value. So how can um it's an infinite game. The idea is um you play the game such that you will keep playing with each other and keep wanting to cooperate and co-realize value as opposed to more narrowly trying to uh achieve a win. But this might be but then have this be at the expense of um the possibility of future cooperation. So this idea of an infinite game, it's um in some ways could be viewed as a kind of I guess moral realism and that um uh the idea would be that an an ethics of or a kind of moral of cooperation and other concern might actually be um a selfish uh thing also that um um this is this could be thought of as a kind of convergent instrumental value. Um, if you want to be sustainably, if you want to be most likely to be empowered into the future, um, configuring yourself in your policies you're likely to select, making those cooperative might be the best way of, uh, minimizing your expected free energy or realizing value into the future. I don't know if that explains it, but that's uh, roughly the idea as I understand it. >> Awesome. And several of the team members applaud you in the live chat. So, if you have any other comments, feel free. Otherwise, I will then go to play the Common Grounds videos and then the last few videos. Any final comments, Adam? Go for it. Otherwise, thank you. Um no just uh thanks uh so much for uh providing this platform and for um you know I should have actually mentioned you in terms of we've been in communication with you about you know uh what what has been done and what we're planning and you know to what degree that within active inference um you know is what we're doing the ways can intersect with with other ongoing efforts like Metagrid um and um basically grateful to be uh in the symposium now and to um be thinking about these things with you and uh thank you. >> Thank you, Adam. I'll ask one final thoughtprovoking question from Obedax. Adam, would you be able to emphasize the dichotomy of it all? the dichotomy of it all. >> I don't know if they're trying to break you out of your non-dual frame. So the you can think of um there actually among approaches to alignment um there's interesting um suggestions that um actually uh the world's wisdom

traditions and uh contemplative traditions such as found in uh Buddhism and uh other worldviews uh could potentially be guides to something like alignment and even super alignment. That um overcoming dichomous thinking or self other distinction uh might actually be one of the things we're looking for in aligning minds and one of the ways in which minds like ours when they're aligned they're successfully aligned. Um I think there's a lot of merit to this and this is uh I think present in some of the uh thinking I've described in terms of ways in which for instance uh rhythmic activities can potentially and share rhythms can allow for basically self-trecendence as part of alignment. Um but I would also suggest that uh there are other ways of thinking about alignment that don't necessarily involve um uh beforehand um dissolving dichotoies. So for instance uh you know the nature of compassion um could be thought of you know you you potentially arrive at it by having some sort of like oceanic state of non-ifferece from the world and others. Um but you can also arrive at it as um you just learn that when you're good to others they're more likely to be good to you and that potentially the most selfish thing you could do is to love others. And so instead of uh initially arriving at this sort of uh transcendence of dichotomy between self and other world upfront, maybe it comes afterwards um from a process of just learning what works uh or what tends to work for um making you feel good um for uh when you when you interact with others in the world. Uh so that was an attempt at addressing dichomousness and how it can play into alignment or not. The >> the more open the question the better. Okay. Thank you Adam. I will now play 1 2 3 4 5 6 seven videos. It'll be a a septupal header coming right up. So thank you Adam. Bye. All right, here come the pre-recorded videos related to Common Grounds. First is going to be the video by Nick Haye. It is a 15minute video. Here we go. So I'm going to describe the common grounds project in general terms. Adam others will connect it in more particularly too. and others will connect it in more particularly to active inference. So the general goal of this project is to build an artificial life simulator which is designed to simultaneously support several things at once. First uh building virtual worlds games that can be used to bridge humans and AIs through play rather than work or other things like that. It's important secondly developing a new range of benchmarks for studying the within the sort of playful artificial life context the social and cultural effects of bringing AIs into the mix. And then third and more broadly is supporting research and nurturing developing benevolent embodied AI and all the different ways we might try to do that. So simulation and benchmarks will be released open source to maximize the ability to do good research. And there'll in parallel be a game demo in the same simulator released on the web and on gaming platform Steam to sort of just test the

waters to see if this is an engaging and interesting um thing for people. And then basically the idea of the project is to iterate and improve on all these three things. the sim, the broad simulation, the benchmarks, and then the game. So, I'm going to talk about the artificial life simulator, the world, the nature of the bodies of the creatures in it and their their embodiment, and then several sort of directions you could take it in more of a morphagenic u developmental direction or evolutionary direction or um multiscale life and touching then touching a little bit on the 2D versus versus 3D question and um had to do with a with a demo. So, here's some concept art depicting the world and the creatures in it and the sort of kinds of things that are going on. We have um here we have a one creature investigating this this tree here which has got a little worm on it. Over here we've got some uh obstacles, some ball that's come and block their little houses here which they they've built for themselves. And then these two um these two creatures are trying to remove it. This one is just kind of chilling and not being very helpful. So there's various kinds of coordination and cooperation problems that very naturally emerge. Um, here's someone exploring the cave, maybe trying to find some resources. And then I don't know what this one's doing. It's sort of staring at some crystals. when it comes to nighttime, so think of this as sort of like a 2D physicsbased Minecraft. Then uh or you know kind of world like this. Then here all these creatures have gone into their homes with some nice warm fire. And then yeah, they're kind of he's still up So, one can build worlds in the simulator that have different granularity scales relating how the the that can be modified by the creatures in it and the creatures themselves relate. So here's sort of like a kind of intermediate or here's here's a scale where the parts of the world are quite large compared to the agent. Now here's one where they're quite small. So this is sort of like the world is a bit more like mass-like than sort of discrete atomic pieces. And here sort of kind of a middle ground. So one could choose the level of granularity that makes the most sense for u the kinds of things you're trying to study. So a major motivation of this work is being able to have a a credible but simplified form of embodiment and in particular going to have analogues of all of the senses. some sort of simplified form. So here we have kind of something depicting here here's sort of a scene with the from before zoomed in with the creature holding a a part of this plant and the little worms up there eating some fruit and it's sort of depicting a picture of their internal internal state. So for each piece of them maybe there's energy levels there some sense of temperature potential for damage u nutrients there could be internal organ states um there could be various forms of viseral motor interceptive control hormones this is sort of like um and then we also have things like proprioception sensing the configuration of the body the angles of the joints forces



being put on it and and dually motor control like the ability to control the desired uh pose and posture of various body parts and then stiffness. There's also touch that comes from interacting and colliding with the environment that's sensed on the different parts of the body. Um there could be smell which is in this kind of microscopic range corresponds to something like a sense of the quality of the environment around the creature which could just be like a a weighted integral of the kinds of materials that surround um surround them. So vision works quite naturally. Um you can imagine it as sort of a ray casting thing where you um can sense the distance and material along different directions. One can add amounts of noise or not have distance be directly observed and so forth. So this you can kind of visualize it in in first person. looks like kind of a Wolfenstein 3D on its side. But here you can sort of see this this part here is the corresponds to observing the hand. Here's the the part of this tree they're holding on right there. Here's some distant ground and there are some other um balls right there. And then finally sound which is generated by interaction with the world and collisions and so forth. And there's various natural ways to to represent that. So maybe here the the creatures hearing some sounds coming from touching and manipulating this thing here but then also simultaneously hearing some munching from up there. So this form of embodiment is rich enough to represent uh a pretty comprehensive range of of senses. So as another thing this is sort of the simulator generally is a fairly flexible particlebased system. And so one thing you could look into is exploring development and evolution of creatures starting from simpler things which uh perhaps um bud off other cells that are connected to each other and communicate each with each other and and possibly even have um a more biochemical sense of how the motor control works. So another thing you can explore is intelligence acting on on multiple scales. So here we have two creatures synchronously trying to to carry an object that's too heavy for one of them. And one can go out from there to consider the behaviors of of groups uh cooperating groups of creatures. And then one can go in and consider the creature themselves being made of cells that are cooperating and further those cells as being um made of molecules that are cooperating to to achieve certain homeostatic ends. can imagine um sort of literally simulating various biochemical processes and in like a um kind of balls bouncing around and and having chemical reactions. So for example, here's a picture of a food particle absorbing into the cell and then uh interacting with various of these different different kinds of molecules inside. So a question I often get is why two dimensions why not three dimensions. Um the main reason is two dimensions seems rich enough to to support a very rich embodiment and is quite a lot simpler to deal with. Easier to visualize computationally

cheaper. So you can get more done for the same amount of compute. Um there are properties of three-dimensional space that two dimensions doesn't have, but I think a lot of these are good enough 2D approximations especially if you have 2D on its side with gravity. So I don't want to argue for 2D too strongly because this is a natural 3D world, um that's analogous with 3D balls instead of 2D balls basically and uh would be a natural transfer domain from showing things that maybe have been developed in 2D than showing that they still uh continue into three dimensions. So I'm going to demonstrate some features of the simulator from the inside out. So this visualization shows what two creatures are feeling in a sense. So the line segments connect their bodies and are colored based on how stretched they are and the green segments show where touch is occurring. So the bottom you see the feet touching the ground and here yes you can see some contact between the joints of the body uh on the arm. And so this is a frozen tableau of a creature reaching for a fruit there in a larger scene with kind of a worm. And this is in a kind of fully modifiable world. You can kind of there's this crystalline structure of tiles you can pick up and move, and they follow gravity and they influence each other and um can be used to build up structures in the environment like kind of homes and some such and can also be the sources of um resources. So this sort of can be kind of um a uh ecological system in here where maybe this pink resource is needed for um as one kind of food that's there and they have to go into the mines to find things. So there are various ways of uh interacting with this world to demonstrate things. You can create uh moving particles which will do things to the world and particularly you can play around with the uh destructive forces of very large objects. So, this shows a worm being piloted and um the controls are modifying sort of the shape of the creature and it is trying to reach those little fruit berries on the tree and and knock them off. Um sort of demonstrating part of what could be the life cycle of this sort of creature. This is sort of a zoomed in view again showing the body sensors of stretches and touch on the along the worm's body. So this next part shows how uh visceral the world can be when you have a body which can be pinned down and stretched and deformed by by other parts and potentially potentially injured. So there's sort of a very core source of of value and motivation that comes from preserving the integrity of your body and protecting it and and helping others that are um having some trouble there. This part is more fun. This is showing the the emergent physical properties you get um or the material properties you get from having uh particles that have different kinds of interaction. In this case, the particles adhere to them each other. So you get sort of something which is a feels a little bit like a a metal or some kind of you know magnet ball structure and has properties of

deformability and you know malleability, forgeability and um so by varying the basic interactions between the particles can create a wide variety of a very rich um set of of kind of possible situations. And um this includes also being able to specify, you can see some on the right, some of the material properties of the objects. Um and this includes things like reactions where um particles of certain materials will connect and collide and then form other particles like that. Hi, this is Alex Kefir. Um, thanks. >> Hi, this is Alex Kefir. Um thanks to um Adam uh for inviting me to this mini panel or symposium on the common grounds project um which I think is very interesting for modeling some of the things that I'd like to um that I'd like to model and do some experiments on um related to agency and alignment and so on. So I wanted to start out by sort of framing why I think this environment is interesting. So um so grid worlds um actually are are fairly interesting in themselves. So um you can use a grid world um by which I don't mean necessarily like a literal grid but just a simple simulation with some discrete states. Um you can use these to model um a number of things in in fruitful ways. So um obviously like even in a literal grid like this you encode basic spatial relationships. Uh you can have simple sort of or simple to moderately complex you know um tasks that agents can execute. you can run batches of agents that you can vary in various ways. Um and you can importantly you can control the um the conditions of experiments in a pretty easy way because things are fairly simple. Uh and you can also examine the results. Uh so these are interpretable models because of their in part because of their simplicity. Um although you know not all of the behaviors you get out of these models are simple. So, in fact, like cellular automata you could argue are um they're not exactly usually thought of as grid worlds, but um you know, with a fairly simple set of rules and a grid-like setup, you can get some interesting emergent behavior, but generally uh grid worlds are good for like coarse grained uh interpretable models of how agents interact with their environments and with other agents. And then on sort of the other side of the spectrum, you have um full-on like 3D game engine type physics simulations. And these are interesting obviously if you well if you just want to see like how you know drop a car from 30 feet and see see what happens or um you know more usefully if you have a robot and you want to figure out like um you're trying to build a model of how you know fine grained spatial manipulation and control works um then these types of environments are great for that. Um but of course they're fairly heavy computationally and they certainly don't abstract from irrelevant detail. um in a lot of cases, you know, depends on your use case. Um but they're great for like realistic models of of interactions with the with the realistic world. Um so one way of framing what I think is cool about common grounds is that it's sort of a

middle middle ground um between these between these approaches. Um, so and I also see it as just in particular it's like a a very good fit with the with like the niche of research that I want to do on um you might say the sort of um psychophysical aspects of agency uh which I think are are um suitable for studying in like an artificial life type of environment in which you have um so I guess the main salient feature of this kind of simulation is that the agents have bodies um not just in like a nominal sense that like they have some properties that you keep track of in some arrays, but they actually have um there's there is a you know a physical spatial environment and the internal structure of the agent um or at any rate the structure of the agent um and including its internal structure potentially is modeled in the same way as the environment. So you know here you've got sort of like a cool um hexagonal type of grid made of spheres. I mean, it's I'm not making claims in fine detail about how common grounds works, but that's this image depicts something like that. Um, and the body of this wormlike thing here is made up of the same types of stuff you'd see in the environment. So, I think this idea of um there's a lot of ways of looking at why embodiment is important, but one way of one reason it's important is just because the structure of the agent is continuous with and it's a part of the like this. I mean that's true in a you know um full-on 3D physics simulation as well but here things are still kept simple enough that you have discrete elements you can play with and um you know a fairly small number of knobs you can turn um but I'm particularly interested in this from the point of view of like basic research on um things like agency and um and policy selection because you can you can model how mental internal like mental or cognitive structures are physically realized. So, you know, we often talk about like mental action. Um, so, you know, that suggests there's like really an internal mechanism that performs an action kind of in a similar way to how you might perform an action on the external environment except it's internal. Um, and I think that's the kind of thing that you might be able to model in this paradigm. So, um, generally the physical foundations of agency. So zooming in a bit on some of these ideas, you can use this kind of simulation environment to model um preferences for example of agents um or homeostatic set points as actual attractors in an internal dynamical system. Um, and I think for some purposes it's fine to just specify them as like a C vector as a as a a list of numbers, but you might get really interesting things. And I'm certainly interested in in particular aspects of um how how it would inform your your models of agency if you if you explicitly model those things as attractors. Uh, which you know, of course, in the in the bigger picture in the background story of free energy principle and so on, that's what they're meant to be. Um, you could also, and this is probably one of the things I'm

most excited about, um, is that you could model the physical realization of mental or psychological structures. Um, you know, things like perception, percepts, beliefs, um, mechanisms for policy selection. So in this kind of environment, you could literally build like an internal machine that executes some task like policy selection for example or belief updating. And then once you had that machine, if you also set some physical parameters and tied them to the way that these atoms in this universe bounce around, then you could um so you can look for example at how the conservation of energy, you know, physical energy um impacts belief updating and policy selection. Um sorry for the kind of ridiculous illustration here but I I rolled with it. Um so another theme here that I think is important this this is almost the end as a short talk. Uh is this idea of crossscale composition. So um you could take that in a purely physical sense uh like agents are built out of physical things that are built out of other physical things. But you know um a more adventurous avenue might be to explore how agents are composed out of microagents. Uh and there's some really interesting work already um showing for example that like the phase locked decisions of microagents can sort of lead to emergent behaviors at the at the macro level. Um and I think I think that's important principle for understanding. Um so I use the term inner alignment on this slide. Um I realize that has a narrow meaning in in like AI safety literature sort of about ensuring that agents learn internal you know mesa optimizers um or they they sort of that the results ensuring that the results of their metalaearning remain consistent with the goals that the designer of the agent had for them or the objective function that was explicitly specified. So I think more broadly though you can think of this as um inner alignment as sort of a matter of just being able to what what what internal control structures are necessary in order to enable agents to have to have goals. Um goals that they really own. not just you know um targets that are sort of um imposed from without but like actual goals that they adopt and to my mind uh making progress on that requires understanding something about the internal organization of the agents how and how it relates to their behavior. Um so I've argued that this is important for understanding that you know differences potentially between um digitally simulated agents and and biologically realized ones. But but whether you agree with that or not um certainly these principles of crossscale composition might be interesting for for an understanding how value and and desire and things uh operate in in living agents. Um so thanks for including me in this in this discussion. Um these are just some some papers I've worked on that are um you know in this in this kind of space. Um but importantly a lot of them don't have much modeling work associated with them yet. And uh it's not easy to create an environment like this. So I really

um you know I do hope to see common grounds become like a really a really cool useful tool in this this space. Thanks. Hi, I'm David and I'm going to talk briefly about bounded rationality and multi- agent alignment. So to begin with some motivation, a lot of focus in alignment research is centered on this question of how do we align a single potentially superhuman level AI with humans or humanity. The big issue here aside from how we define these things is that there's no agreed upon solution to this problem and it seemed like a very difficult one in general. On the other hand, a lot of focus in capabilities research centered on the question of how we build a single superhuman level AI system. The reality is that for such systems there are significant trade-offs between performance and cost. Moreover, AI agents are increasingly being deployed on the internet to carry out tasks autonomously over longer and longer horizons. Fundamentally, we existed and are building towards a multi-polar world which is an ecosystem of many AIs, humans and organizations all interacting with each other. And so a central question here is how do we ensure that this ecosystem is aligned, beneficial, efficient and resilient. The hypothesis I want to explore here is that we start from the fact that a major source of risk from advanced AI comes from this extreme uninterruptible optimization towards specific goals. Now if a significant proportion of the AI agents in this ecosystem are highly capable, have access to significant amounts of resources and are insufficiently aligned. This could create destabilizing undesirable outcomes. And so perhaps complimentary to working on alignment, maybe we should also think about limiting capabilities and resources within such ecosystems until and as we progress with alignment. So what this what could this lead to? Well, one of our essential hypotheses is that this could lead to an ecosystem of specialized, efficient, and reliable agents all working together alongside humans to solve problems much in the same way as we would see organizations or societies do. So, how might we reason about such systems? Well, one concept that we propose that's potentially useful is out of bounded rationality. So what is bound rationality? Well, the fundamental idea here is that real world decision-m is subject to limitations. These limitations can come in the form of limited time, energy, money, computational resources, attention, memory, information, etc. And crucially, decision-m need to take these factors into account if it is to be effective in the real world. Now, boundary rationality was originally intended as a descriptive model of human decision-m then was later expanded to encompass both biological and artificial decision makers as well as to bridge the gap between both normative and descriptive theories of decision-m. The key mechanism here is that of satisficing which is related to finding the optimal solution in a simplified model or finding a good enough solution in a more realistic

model. And we see these kinds of trade-offs manifest in the modern day in the form of cost performance trade-offs especially with the rise of test time inference. So here we see several graphs from major AI companies, researchers and also public benchmarks that are increasingly focused on the trade-off between performance on a certain task or benchmark and the costs in terms of test time or training time compute or in terms of dollars for inference etc. just for a brief timeline of bounded rationality, the concept really got started with Herbert Simon who introduced it in the 1950s and wrote a lot of seminal works on this. Then sort of the 1970s and 80s, it was later extended and really fleshed out by Amos Tversky and Daniel Kahneman in their research program on heuristics and cognitive biases, as well as their distinction between system one and system two thinking. Then later in the 90s you have Gigerenzer who used concepts of an adaptive toolbox and ecological rationality as well as to Russell who worked on bounded optimality and you know since then especially in the 2000s up to now there's a proliferation of approaches and ways of modeling this some of which I'll talk about next. So one of the key concepts here is computational rationality and the basic idea is that we want to select the optimal program for some valid machine to execute. And so you can imagine in this diagram on the left here, you have some sort of utility function in an environment from which you select an optimal program within a space of feasible programs and you can study the optimal behavior for such programs and connected to observed organism behaviors. On the right here you see another way of thinking about this idea of computational rationality where you have your utility which is diminishing with a cost of delays. And this trade-off means that there's an optimal time for you to run your computation before you select an action. And a key concept here is that of meta level reasoning which focuses on considering the current uncertainties the time critical losses with continuing in computation and the expected gains in precision of reasoning with additional Now another concept that's been increasingly popular in recent years is that of resource rationality. And the key idea here is that we were trying to find the optimal way for an agent to use its limited resources. So you can imagine this triangle here representing a space of possible lines that are implementable with our brain's limited resources and on the y- axis you have performance of these different lines. The resource rational approach is to find the optimal one within this feasible set. And some of the key steps associated with resource rational analysis is that first you define the problem at the computational level. Then you postulate a class of algorithms that the mind might use to approximately solve the problem. The cost of computational resources and the utility of accuracy of the algorithms. After that you want to find the algorithm in the class that best

trades off resources and accuracy. Then you evaluate predictions of this algorithm against empirical data. And if there's a gap, you go back to step one or two. Now, another concept within this space is that of information theoretic under rationality. And the key idea here is that we account for the cost of cognition using tools and concepts of information theory. In particular, agents start with some prior probability distribution  $Q$  over choices and does some computation or information processing to transform it into a posterior  $P$ . And you can measure the cost of this transformation as the  $K$  divergence from the prior to posterior. And so agent optimizes an objective which is an expected utility minus a regularizer which represents the So few questions related to boundary rationality in alignment in this context of ecosystems is first of all how do we build these specialized efficient and reliable agents. Secondly, how do we coordinate such agents with each other and with humans or organizations to reliably and safely solve more complex problems? Finally, how do we mitigate the risk that might emerge critically in the context of these multiaction systems? Now, our view is that this is an important research agenda and we hope that we are able to dedicate resources and talent towards solving these important problems. Thanks for listening. Hi. So, Mao and I are going to be uh presenting some thoughts and ideas that we have on path flexibility and collective alignment. So, to start off with some motivation, why path flexibility and what is it? Our starting point is essentially that approaches to alignment should address some key properties of the real world which include non-stationality of the environment, non-stationality and diversity of human beliefs and preferences, bounded rationality and imperfect execution of plans, collective and system level properties and also the embodiment embodiment and embeddedness of agents within their And so to address some of these issues, we suggest that we focus on paths and processes. And these things would include the alignment of beliefs and preferences in terms of how they update and not just what they are. The ability to efficiently adapt and recover when environments or goals shift. and also the maintenance and flexibility and optionality over time rather than rigidity and narrowness. And so a simple example of this is a lava world example where agent has to go to its goal and traverse this pathway where there's lava on either side. And the basic intuition here is that the agent should expend a little bit of extra effort to traverse the middle path because this provides them more flexibility and resilience in case they accidentally slip um to either side. So some of the key objectives that we hope to achieve within our work. Firstly, to demonstrate the value of our alignment formalism. Secondly, to study how path flexibility and empowerment lead to resilience and finally to develop methods to promote path flexibility and empowerment. So some key concepts that we find useful to



study and address and define. Well, first of all, there is path flexibility, which we refer to as the quantity and ease of shifting between viable trajectories for an agent or group. Then there is empowerment, which roughly relates to the degree of influence that an agent can exert on its environment. There's resilience which is the ability to recover from and adapt to adversity through mechanisms of redundancy and degeneracy. And there's concept of attractor difference which is roughly the distance between agents stable belief or preference trajectories. And the key idea here is that the closer aligned agents attractors are the more similar their trajectories will be. And there is plasticity which is roughly the degree to which beliefs, preferences or policies can update in light of prediction errors and changing circumstances. There is planning horizon which is also related to light cone. Uh which essentially measures the effective temporal depth and breadth of active inference or planning into the And finally, precision control, which is the capacity to modulate uncertainty uh including sensory policy or belief precision. So some key questions that we want to address. First of all, how do coordination and communication contribute towards path flexibility and systemwide resilience? Here we hypothesize that these things come hand in hand. Uh second question, how do do high attractive difference and low plasticity lead to lower collective path flexibility? Thirdly, how is plasticity related to the ability to recover and adapt after shocks? Fourthly, how do the breadth and depth of planning horizons affect resilience and are there diminishing returns? And finally, how do precision asymmetries and precision control influence the power to shift attractors? And so with that, I'll pass on to M to discuss some of the experiments. Thanks. So for the experiments for now, they're explained in grid worlds, but they're meant to then be made a little more complex to fit the common grounds. So the first experiment is meant to establish a baseline of coordination benefits. um i.e. is there even a point coordinates um and align and so here we have a setup with two agents and two resources with optional communication at a cost and we're hoping to manipulate the communication and the shared observations um to show and to measure whether there's a higher cooperation rates um to measure the expenditure the energy expenditure and the path returnability i.e. uh we expect to show that communication increases path flexibility and shared reward. In experiment two, um now that we've shown coordination leads to better outcomes, we must show how it might arise from our chosen formalism, so aligning attractors. And so we hope to demonstrate that alignment emerges from adaptive preferences rather than just identical goals. So here we have two misaligned agents with variable preference update rates. we're going to manipulate uh the plasticity of their preference. And so ultimately we measure the attractor difference, the time to

coordination and efficiency over the long run. And we expect that higher plasticity means faster convergence and ultimately higher joint resilience. I think the videos are struggling to run. So do you want me to wait a bit? >> Yeah, we can skip this. >> All right. And so in experiment three um we're trying to demonstrate that plasticity promotes recovery and adaptation after shocks. And so we want to show the mechanisms that underly the plasticity in question i.e. what do the agents learn and from where. So we hope to show that it leads to higher group resilience and path flexibility which means recovering from shock and continuing to thrive. So the setup here is multi-agent grit world with periodic shocks. Uh so resource relocation and noise and the agents learn online from prediction error. Uh we're going to manipulate high individual or group plasticity social posture sharing. So belief sharing and redundancy versus no redundancy i.e. whether agents can do the same thing more or less or do they have more uh degeneracy. For example, we're going to measure the recovery time, the prediction error, the free energy slope, and the group resilience index, i.e., are they more resilient as a group or not? We expect to show that learning from errors and sharing those corrections creates faster, deeper recovery and turn resilience into a group level property. In the next slide, um, we ran a short experiment on top of a paper that Rudy Pitia Jane and uh, Timber Balon and the rest of their team at Versus ran on theory of mind uh, with uh, two resources and basically we were able to show that um, through theory of mind and belief sharing the agents are more able to recover quickly from a shock. In experiment four um we want to show that we need to think in terms of paths and extend the planning horizons such that we can maintain path flexibility over time. We also want to show that planning as a group yields more rewards because planning is metabolically costly. So doing it as a group reduces the individual cost. So the setup would be agents with variable planning horizons facing unexpected resource shocks. We would manipulate the cost of horizon expansion. uh the shock um magnitude and frequency and measure the recovery time, the performance degradation and the path survival rate. And so we expect that extended horizons yield better adaptation but at a metabolic cost. So it's better to uh do it as a group. >> And finally, we want to show that these are not just chance outcomes. We can actually manipulate paths and lead to alignment. So here we test whether power in multi-agent inference arises from prediction control rather than just force. And so we can show that one agent can modulate another sensory or policy precision via environmental actions. So the manipulation here is the control strength and the target precision type. And we would measure induced policy shifts and cooperation rates uh to show that ultimately we have um the ability to change the attractors for the agents to lead them to

potential uh similar paths without necessarily losing path flexibility. And so here we have uh an expectation of asymmetric path precision um distorts the path flexibility of one of the agents which generates a dependency or a domination here and basically just control of one agent over the other. Great. And so just as a synthesis and some next steps obviously these are very preliminary ideas and um there's a lot of empirical work that remains to be done. In particular we want to study the connections between flexibility, resilience and energy efficiency in more detail. Uh there are a lot of conceptual questions for example linking our findings to empowerment particularly on path flexibility um shed intentionality and other alignment metrics proposed in literature uh for implementation in order to bring in the embodied aspect um we want to build a prototype in common grounds with some basic features to demonstrate how um spatial properties and embodied embodiment affects questions of path flexibility and resilience. And finally, the deliverables we hope to achieve include theoretical formalizations of the concepts that we discussed and introduced and some mathematical results relating them to each other. Uh further empirical analyses both in good worlds and in common grounds and then some data visualizations. Thank you. The uh purpose of this recording is to show you a sort of a new view on engineering systems with uh provably beneficial or aligned behaviors as the title suggests. Uh and it will mainly uh look into how do we actually predict what systems do and how do we use that to our advantage. So I would I would normally begin this presentation with a small uh demonstration. uh this is a recording so I can't give the demonstration in person but uh this kind of lecture would begin with a demonstration where I would roll up a sort of tube of paper uh and make it into a kind of canal shape and crumple another bit of paper into a ball and roll the paper down the canal and you would see it drop into the canal and roll I would then take the canal and turn it upwards so that you have a sort of hill and the ball would drop onto the top of the hill and then fall off. And what that demonstration would show is that there are actually uh differences in the way that motion is constrained by geometry. So the forces that guide a system to move uh are simultaneously geometric features of the system uh and the environment in which it's moving and also dynamical features uh such as the forces uh that are shaping their motion. One can see this in uh the distinction between stable equilibrium and unstable equilibrium states. Uh so if I have a ball at the top of the hill, then it could roll off the hill one way or another. I don't know. Uh which way it'll go. It'll depend on some kind of fluctuations. So that's U1 there. uh if I have a stable equilibrium like  $s_{ub\_1}$  or  $s_{ub\_2}$  uh it's at the bottom of a hill then the geometry is constraining the motion so the pathway that the system will take over time is

dependent on uh the setup of the environment that it's in. Um this uh sort of tension between uh geometry and dynamics uh is captured really well by gauge theory and correspondingly gauge theory gives us a way of speaking about how one quantity uh varies another as it moves on spacetime. So gauge theory this often records the way that a force field varies the motion of a particle and it records the coupling of bosons which are force carrying particles to fermions which are particles that comprise matter and a really instructive example is that of gravity under general relativity. Uh the principle of general relativity uh is that special relativity the relativity of motion with respect to the speed of light can be extended to any expression of motion which is accelerating relative to some other expression of uh that motion. A consequence of this view is that all noninertial motion is identical to inertial motion on a curved surface which causes some acceleration. This is called the equivalence principle. Uh and the equivalence principle was properly introduced by Albert Einstein in 1907 when he observed that the acceleration of bodies towards the center of the earth at a rate of uh  $1g$ .  $G$  being the uh acceleration due to gravity is equivalent to the acceleration of an inertially moving body. Uh meaning not accelerating uh within a frame of reference. uh that would be observed on a rocket in free space being accelerated at a rate of  $1g$ . So if you have a uh a thing in motion and it's not accelerating within a frame of reference that's perceptually the same as if it were in motion uh and its frame of reference were accelerating. So again this is the equivalence principle. Uh accelerating reference systems are physically equivalent to gravitational fields. As such the curvature of spacetime is gravity. one sometimes hears truisms like uh there is no experiment observers can perform to distinguish whether an acceleration arises because of a gravitational force or because their reference frame is accelerating. So the curvature the warping uh the acceleration due to motion uh is not distinct from a uh constant velocity no force motion within an accelerating reference frame. Um uh another truism uh due to uh John Wheeler is that spacetime tells matter how to move. Matter tells spacetime how to curve. And we can see this is uh most uh let's say impactful uh when one looks at a black hole. The path of a photon uh a light carrying particle uh on spacetime is uh warped so strongly by a black hole that it uh or rather that the path on spacetime that it takes is affected by gravity so strongly that it looks as if gravity is uh completely uh taking spacetime and pinching it and pulling it way way down. So we have this interplay between geometry and uh constraints and dynamics and forces. This interplay has also been useful in the uh field of biophysics. So in the work of uh Ken Dill for instance uh it was discovered that one of the organizing principles for how proteins fold is if you set up a kind of energy landscape and you look at what are the

transitions in states that the protein can achieve as it goes along dynamically. The geometry of the energy landscape turns out to constrain what the protein does and it allows you to make predictions about how the protein will fold uh as it continues to go along and experience conformational changes. This was a big advance in the '90s in uh biophysics in protein folding uh because there were no good overarching principles for how proteins fold. uh but uh the work of Ken Dill uh gave us one such principle uh which is that the uh the loss surface so to speak uh between energy and entropy uh is determining how the protein uh folds and it looks as if it's sort of uh a bunch of skiers uh going down a hill. So they take the path of least resistance. They take the most natural path afforded to them and the interplay between energy and entropy and the geometry the curvature that uh that interaction evinces is what actually guides the notion of resistance. This is true also in uh genetics in uh the theory of epigenetic expression. Uh there is the uh notion of Waddington's landscape and it looks like a marble rolling down a hill. What people uh often use this analogy for is saying you have uh set before you a bunch of possible paths of genetic development and your system is the marble and as it's rolling down the hill uh it gets caught in one or another canal and that canal it gets caught in is what determines the epigenetic expression and therefore its phenotypic traits. But underlying the actual curvature and therefore underlying the dynamics of gene expression are a whole web of interactions uh that determine what kinds of expressions are allowed, what kinds of expressions are uh admissible or which kinds of expressions require a large burst of energy. uh that would need uh that would be needed to get you from one free energy canal uh in the landscape into another one. So you can see in in this figure there are a whole bunch of pegs and trusses shaping. In uh some recent work, I proved a theorem that says that actually uh inference in general has this property. uh when you are trying to infer the probability uh being assigned to different states uh if you have a sense of the constraints on the probability uh or the constraints on the preference of how likely or how desired a state is then that constraint function gives you a surface and there's a very nice very well understood one-to-one mapping between the geometry of the constraint surface and the resulting probability density over the state space. So you can see in uh figure subfigure A I have my constraint landscape. It's it's a sort of distance function. It says that as I get further out from the the center um my penalty increases. It becomes much greater. If I look at subfigure B, I have all these level sets of equi-preferred states. So states with similar preferences uh and and those are the states that are available to me if I designate a level set of constraint value. I say um my preference for the state is going to be uh such and such. I have whatever tolerance I'm going to give myself uh

for getting away from the center. Uh and then these rings are all of those level sets of equal constraint value. In figure C, I can see myself projecting those rings down onto the state space. So now these are all of my states of equal constraint value uh with the uh constraint surface overlaid on them highlighting those rings. And then in subfigure D I see how the idea of maximum entropy is actually a geometric procedure that converts the geometry of these level sets into a probability distribution. So that the equally constrained level sets are equiprobable level sets. And the curvature that affects my sort of uh how radical the change in preference as I move in some direction is that gets converted to a log probability gradient or in other words a surprisal. So what is afforded to me by this technology is a deeper understanding of how systems which making decisions about what states are preferable and what states are not and then evolving according to those decisions. Uh what is afforded to me by this technology is um a description of such systems uh as though they were marbles rolling down a hill. I'm afforded some geometric description of those systems. So before I move on to that, I I just want to say um the the ultimate goal is to then map out the geometry of such spaces and start to make predictions about how complex systems are going to move in their state spaces using the knowledge afforded to me by the mapped out geometry of those state spaces. Some other interesting things that uh this gives me access to are uh you'll recall we made this analogy to uh gauge forces. Uh well gauge forces they they are sort of abstract mathematical things that relate curvature to forces. They relate geometry to dynamics. But they're physical things as well, right? The um the notion of a gauge bzon is a physical force carrying particle that actually tells you how a force creates a curvature. How does a force being applied warp spacetime? How does geometry get created by dynamics? So these are things you can go out and measure in nature. Uh and that means this gives us certain knobs to turn. So for instance, if we're looking at engineered biological systems, uh there are observations of localized uh excitations in those tissues uh that actually look like what mathematicians would call solitons. And solitonal excitation. They're a type of gauge bzon. So this gives you a knob to turn in the sense that now you can go look at uh here's the uh excitation corresponding to this kind of intended evolution if I want to bring about a different kind of evolution. uh what do I need to modify in my observed excitations and how can I continue to probe and uh change the excitation being expressed until I get uh my system into the right canal. A lot of parallels are drawn for me uh to Michael Levin's work on bioelectric signaling networks uh where it is shown that the thing that mediates how uh systems evolve through morphological space. The thing pointing to states in morphological space and the thing uh mediating transitions between

those pointer states uh is a bioelectric signaling network. And this is something that is very easily accessible by human beings by scientists, experimenters and so on. Uh and can be directly modified if we can model it as an electrical circuit. Then we can ask well what happens if I pass a current of this voltage uh with this direction through the through the uh circuit. Um how does the system then evolve? And if we've mapped out morphological space then we can go on and start making predictions about how that mapping uh carries the modification through. With respect to alignment, I'm I'm very excited about the prospect of looking at complex systems and saying I'm going to map out the geometry of this systems possible evolutions. I'm going to infer some utility function or some constraint function telling me well what does the system prefer? What are the kinds of states the system wants to visit? How does it want to visit them? Once you have that geometry and if you have physical knobs to turn um relating to a description of that geometry by some mediation, some dynamical mediation gauge bsons for instance. If you have that and you have a knob to turn, then you can start manipulating that geometry. you can start coaxing the system to evolve towards other states. Right? If you have the preference function derived from mapping out this geometry and you also have the knob to turn, then you know exactly what you're doing. You know exactly how to tell the system to um I would rather prefer this over that. So as an alignment problem when we refer to engineering systems with provably beneficial behaviors what we really mean is figuring out what is the uh preference or reward function determining those behaviors getting that out of the geometry inferring that mapping that and then taking some intervention that reassigns those preferences. So if we can find the preference function, we can modify it to say for instance uh to avoid a paperclip situation to say you know I want you to pursue world peace. Um but don't just have pure pragmatic value. Don't just say okay the path to world peace is to eliminate all humans. Uh no you have the preference function. we can see what would it take to introduce some sort of Asimov's three laws or something. What would it take to make the system think pursue world peace but don't harm humans in the process? So ultimately this is all about mapping the evolutions of complex systems with an eye towards understanding what does it look like to teach and mentor uh artificially intelligent systems. How can we increase the synergy between biological and artificial intelligence? So I hope this was compelling. Um I'm personally very excited by this research program and the way that it involves a lot of very pure mathematics, very pure geometry and dimensal systems and very forwardslooking futurism uh and tackles some of the the biggest uh most important existential questions of our So thank you for listening. Uh, and if you have any questions, uh, please

write me an email, uh, or, uh, get in touch with me some other way. I, I'd be happy to, to chat with you about anything that might have piqued your interest. Hello, my name is Zara Shakbai. Today I will talk about a specialized modular architecture for collective intelligence and briefly about Hello, my name is Zara Shakbai. Today I will talk about Maple, a specialized modular architecture for collective intelligence and briefly about our in progress stream to explore world model architecture. Both project are inspired by active inference and aim to build um social and uncertainty operation that can better align with human values. This work is a collaboration between the institute for advanced consciousness studies mil and university of Montreal. Instead of chasing abstract AGI in the usual way, my work focuses on what I call social neuro AI and um brain inspired architecture. We take the brain as a modular pilot system and we use active inference idea to treat each module as an implicit uh model of the world and other agent. The big idea is that human compatible intelligence must be social from the start with architectural law prior that favor poor social behavior theory of mind and mutual modeling. Let me start with uh how complex social behavior emerge um and how has been studied today. Um in open AI hide and seek environment which was published in 2020. Selfplay create creates an autocurriculum of strategies like run chase build for use ram defend ram and so on. Um team learn um to use tools and division of labor but each agent is still uses a single monolithic policy network and the internal structure of these strategies is opaque and it's not interpretable. Roma take a a step toward a structure by explicit modeling rules. Each agent sample a role from its observation, condition its policy on the role and regularize the by that and push roles to be identifiable, compact and specialized. This allowed them to in to have more interpretable division of rap labor but rules are still imposed in a top-down way rather than emerge directly from a modular architecture. Our core idea is that big monolithic network entangle everything. So any change becomes risky and unpredictable. If one part breaks, the whole system can lose its ability. In contrast, true collective intelligence from ants to human relies on diverse experts. Decentralization and knowing who needs to track whom. We want systems that look more like a team of a specialized mechanism that share only a sparse information. So adaptation stay local and robust. To get this kind of specialization, we built on recurrent independent mechanism or rims. Instead of one big recurrent neural network, we use several smaller recurrent modules and each module has its own hidden state and its own parameter. At each time step, we run an input attention phase. Every module send a query. The input gives keys and um values and top case selection decide which modules actually receive and process the information. Only those active modules update their hidden state through their



recurrent cell. For example, a small GRU while the inactive one simply keep their previous memory. Then we run a communication attention phase where active module use query key and value over each other state. So they can exchange a small amount of information before going back to mostly independent dynamic. This gives us a sparse activation, a sparse communication and a natural way for u modules to be a specialized. Our first um maple architecture that we introduce here is the rim only model which is also our best performer. Each agent gets um an egocentric pixel observation for the policy and a global observation for the value which we pass through a normal CNN encoder. The encoder features then goes into a rim and the combined state is used to produce both policy and the value. Over time, different rim can specialize in tracking different aspect of the environment or other agent um giving us a modular social polic policy. We also use uh independent PPO as uh our multi- aent architecture base. The second architecture aim for a stronger interpretability and objectentric reasoning. Instead of a standard encoding, use a slot attention to factor the scene into a set of latent slot where each slot tend to represent an object or entity. We then feed each slot into its own recurrent module again a rim unit so that objects are tracked over time through dedicated memory pathway. This makes it easier to to say things like this module is tracking the opponent and this slot holds the share resource information. For this to work, the slot feature themselves must be high quality. We pre-train a slot encoder using reconstruction and a compositional consistency loss on data collected from the environment. So the same object is mapped to similar slot across different scene. The Tessani plot in the latent state of the slot here shows the initial result slots are starting to separate but clusters are messy and overlapping which push us to design a strandic a stronger inductive biases to sharpen these representation in the latent space and make them more cluster. We introduce our perceptual orthogonal projection loss or POPL. POPPL encourage different slot to be distinct. Push class center to be orthogonal and pull features towards their center. Different object type or rules should sit in orthogonal direction in feature space and each type should form a tight uh cluster. After adding POPL, the TES plus in the latent space should clean well separated cluster which gives our rim a much more disentangle objectentric input. So this avoid the interference between what um gets inside each of the rim module. This object centric is not tied to a single game across very different melting pot environment like alopatic harvest and territory room P O L still produce clear clustering and disentangle feature in each slot. This suggests that we are learning a shared object vocabulary that can transfer at across social games and can later be reused in benchmark like meta and common ground. Once a single encoder is pre-trained,

we copy it to all agent and allow only the high level representation to be fine-tuned. But we face a Classic stability plasticity dilemma. What does it mean? We want to adopt the high level features of a slot during the RL task, but we do not want to forget the useful object structure they learned during the unsupervised uh pre-training. To manage this we freeze low-level feature and finetune only high level selected layers which by adding elastic weight consolidation popl clipping and normalization. These tool protect important weights keep slot representation disentangle and still allow the policy to adapt to each social Here we compare our two maple structure side by side. The rim only model use a simple encoder plus rim and it's fast, lightweight and very strong in performance. It's good default when you care mainly about robustness, sample efficiency in complex game. This lottery model is more complex um because it requires pre-training and carefully fine-tuning but in return it give us interpretability objectentric slot and clear specialization of modules from end to end. So we think of it as a model for analyzing, auditing and reusing a skill especially when safety and alignment are priorities. We evaluate maple on melting pot which is designed to test generalization and collective intelligent. We focus on three environment that form a ladder of social complexity. First territory room as low complexity spatial competition with nine agents mostly working in isolation. Repeated prisoner dilemma as a mixed motivated game with eight agent and alopatic harvest as a high complexity coordination problem with 16 agent. If modular architecture really support social cognition, the benefit should be most visible in the harder This slide shows reward over training for each architecture. Our room only uh model is green. The slaughtering model is yellow and monolytic baseline like GRU and transformers which are used as um memory in the policy networks are in the other colors in both territorial and prisoner dilemma. The modular model learn faster and reach higher return than baseline. In alopatic harvest the the gap is the same. Modular agent achieve achieve a stable high reward while monolithic network mostly fail to coordinate in the complex environment. Now we are testing the scaling law and we ask whether performance is just about adding more parameter. We vary the number of rim unit and how many can be activated at once. We find that more modules helps only when activation stays sparse. When everything is activated all the time, performance drops slightly. So it's not just capacity but capacity plus a sparsity. Structure interactions matter much like having a bigger team of expert who do not all talk at once and they can communicate efficiently. to look inside the network we evaluate our model with using uh conditional mutual information as a metric. Uh CI ask how much information a given module K in agent I holds about the actions of agent J beyond what is explained by its own

actions. In other words, it quantifies how much a module is modeling another agent's behavior which is a proxy for theory of mind processing. As social complexity increases across these three games, we see CMI value rises and spread across more modules showing that the network allocates more capacity to more complex environments and more clustering and specialization between agents. We can visualize this as a heatmap where rows are modules and columns are other agents' actions. Some modules light up across many agents and look like coordination between agents, other track a specific partner or norm violation like monitors and sanctioners. Some are more sporadic like free rider. This connection back to Roma instead of adding explicit role parameters we see roles emerge from interaction of modular dynamics in the environment. So we move from blackbox emergent strategy to monolithic interpretable picture of collective intelligence inside our multi-agent system. So again putting this all together Maple gives us a small toolkit. If you want maximum performance and efficiency in complex multi-agent tasks, RIM architecture alone is usually the best choice. If you focus on interpretability, object-centric and reusable skills for alignment and auditing, a slot plus RIM is more attractive. I see them as complementary: simple RIM for everyday deployment, a slot-based RIM for probing intervention and safety research. These architectures are also well matched to upcoming benchmarks like Meta and Common Ground. Meta AI aims at open-ended multi-agent cooperation and alignment with agents embedded in the nested social circle. Common Ground emphasizes richer embodiment and multi-scale models of bodies and its rhythm dynamics. To succeed the agent needs both a strong internal world model and modular social machinery which leads us to the second part of our dream to explore aiming. Maple gives us agents a social brain at the policy level but they still need a deep theory like a generative model of the world itself. Amy, our dream to explore architecture is a single agent world model that learns to imagine possible counterfactual future trajectories and plans in this latent space that it learns through learning a world model. It follows active inference ideas. The world model is trained by minimizing free energy objective and control is learned through imagination rather than risky real world trial and error. With Amy, an agent can dream thousands of candidate future trajectories in its latent space, including bad and catastrophic ones, and learn to avoid them without even enacting them in reality. So Amy provides the explicit generative model and imagination engine needed for robust planning while Maple provides an implicit model for modular social machinery needed for social cognition. Let me briefly sketch Amy architecture instead of a simple Gaussian prior which is usually used in variational models and it suffers from mode collapse. We use a Dirichlet process Gaussian mixture model as the latent prior. We

also add a bottom up attention posterior from the current image and a top down attention prior for the latent state and context to mimic an attention schema uh process. During training we ask the prior and posterior attention to agree. So the model learn both what will happen and where it should focus next given current action. The nonparametric prior can grow new mixture component when the environment become more complex. So it can represent futures like either outcome A or outcome B instead of one blurry average and it gives better uncertainty on their distributional shift. Uh for dynamic we use variational recurrent models uh and we add perceiver inspired temporal context model. The perceiver reads full trajectory through to and use its bottleneck um as a context vector which condition both inference and learning the transition dynamic. During training we use a variational posterior and teacher context. uh and during imagination we sample from uh dishly process gia mixture prior and roll out future using a learn causal student context predictor. On top of this world model, we train adversarial guided active critic purely in latent imagination that produce with this generative world model and um it improves exploration and robustness compared to a standard dreamer style actor model. So to wrap up we have two main pieces. Maple give us modular brain inspired implicit world model architecture for multi- agent system with a clear evidence of emergent specialization and increasing social modeling as environments become more complex. Aiming gives each agent a rich prob probabilistic explicit world model that can imagine and plan over uncertainty of a future trajectory in the latent space instead of acting blinding in the real world. So many existential risky stories assume a single opaque monolithic super intelligence that we cannot interpret or control. Our work points to a different path. An ecosystem of modular interpretable agents with a strong inductive biases inspired by mammalian brain for cooperation and mutual mod. If we design these architectures carefully for from the start, alignment can become an uh emergent property of the whole society of Asia, not just the patch we add at the end. So I see these um modular world model based architecture as a concrete reason for hope a password AGI that is plural understandable and fundamentally built for cooperation. Thank you for your attention. Hello everyone. This is Sanjapi. Thank you for being here. It's a pleasure to share this talk with you. First of all, I would like to take this opportunity to express my gratitude to everyone um who has worked hard to organize and prepare the symposium. Thank you so much. So the title of my talk is modeling the emergence of perspective an active inference account of intentionality. But um before getting into the main part, I have to admit that I spent quite a bit of time thinking about what I should show and share with you today with while preparing this presentation. Since this

month, I've officially joined the active inference institute as a research fellow. So I wanted to share some of the research agendas I hope to pursue as a member of the institute but also wanted to make sure that this talk resonates within the spirit of the symposium and um which feels engaging for everyone participating. So um after much thought I decided to shape this talk not as a report of any finished results but rather as a reflection on why I have come to walk uh come to walk this path and what kind of inquiries I wish to continue exploring within this community. So rather than diving immediately into technical details, I'd like to focus on a few broader framing questions that include why should we study consciousness, why consciousness matters, and how might philosophical insights meet uh computational modeling under active inference framework and um why this convergence crucial for addressing the agent alignment, the social alignment problems including the AI alignment. ment and most importantly how can active inference serve as a framework in this process. It's hard to answer exactly what will emerge through the work and some of I think perhaps that uncertainty might be part of the process but my aim here is simply to be honest as I have always tried to be in my own research journey. So I often introduce myself as a consciousness researcher and I think it's worth now to share a bit about how I came to walk with this path and why I continue to stay on it. So personally I have had a kind of lifelong question on why can we never fully access another's inner world or why such subjective and personal experience are ineffable and unreachable. This question began from reflecting on the isolation of experience. That is, no matter how we deeply we empathize, we can never actually know what is it like to be someone else. Each of us is enclosed in our own private polia so to say. So I started to wonder how did the subjective living experience arise in the first place and is it truly something that can never be shared and why do such subjective living experience like suff like including suffering and joy uh seem to have such purely intrinsic and private urgency. So um those who live in flammness are clear to me that they cannot be simply reduced to behavior or m functionality. So there is something irreducibly first person about it. That is experiencing is something about having a perspective of mind. So to have subjectivity is to have a perspective and to have a perspective is to inhabit one's own universe. Each of us in that sense lives within a world model that is uniquely of our own. And then communication what we call understanding or alignment or even attunement is in its heart a meeting between universes I would say. So it is not about controlling or synchronization but it's about resonance or attunement between perspectives. So even though we can never become another person, we can still affirm their universe. We can recognize the limits of our own role models and respect the plurality of

values and perspectives that arise from this. In essence, alignment is all about perspective and resonance. So it is less about govern governance under law and order but about more about honoring the uniqueness of each way the world can be experienced and this I believe is where consciousness research and the alignment problem truly meet. So uh for instance, people who rescue refugees in the Mediterranean often say that um even though they know rationally speaking their work may be futile in many ways by many reasons but yet once they meet and resonates with the sheer urgency of the refugees by you know looking into their eyes and share that physical space together they say they that they simply can't stop doing what they do. But well, I'd like to say that this is not unique to the case of material refugees. So if we look across a wide range of so social conflicts from the struggles of underrepresented under represented minorities to the realities of war, we often find that what's missing is precisely this experiential resonance. So many conflicts arise when we fail to realize in a lived in a felt sense that the other is also a human being, someone just as real and alive as ourselves. So um here one could you know of course imagine developing a program of something like a phenomenology and practice quote unquote to cultivate such resonance. But for the purpose of this talk for now, I just want to emphasize that this is exactly where my own research uh research on consciousness began. And um it is also why I believe that understanding consciousness is indispensable for addressing the problem of social alignment. Well, pro-social alignment or what I sometimes call pro-social attunement. Again, understanding consciousness is ultimately what allows us to imagine aligned and coherent agents whether human or artificial. So when we talk about alignment whether in social context or in artificial intelligence the discussion often centers around reverse goals and accuracy but I believe alignment is at its core not just a technical problem of control or optimization is more of a problem of orientation and maturity. So um most of social discord and misalignment arise not because people lack information but because they lack perspective. So there is a kind of poverty of perspective which is essentially a failure to grasp how another another mind or another world model is structured and felt. So if we really want um aligned systems, we need models that encode perspective that reflect epistemic norms on how humans come to know and effective stance on how they care and feel. So here the science of consciousness grounded on phenomenology gives us a way to understand subjective experience in a principled and non reductive manner through agent-based modeling especially within the framework of active inference. We can simulate the agents that self-organize intrinsic value and perspective rather than being externally controlled by reverse or imposed objectives. So this would let us able to um model

how agents evolve and um evolve and adapt together not from not through reduction or one-sided control but through mutual constraint and resonance. So in the long run, my research vision I would say might be closely linked with integrating nominology and active inference in a computational manner to explore sentience about as something emergent and relational and situational and bound in context. And eventually for future visual systems this could mean modeling how perspective forms, how epistemic autonomy develops and how the um effective embedding that means the capacity to deal with um becomes an integral part of intelligent agent. Now let's take a look at what might be required to realize the computational phenomenology. In some of my previous work, I proposed a framework that attempts to unify four different major um major theories of consciousness based on phological analysis. That is main theories tend to describe only partial aspects of consciousness almost like the blind men touching different parts of an elephant. I won't dive in too deep here, but the paper's point is was that all the four different theories are in fact describing different features of conscious experience. That means none of them are entirely wrong in an ontological sense since each each captures a partial truth of conscious mental process. So building on this unified framework, what I now hope to do is to model the minimal conditions of mind or um of perspective perspective itself um as a step toward an active inference framework based computational Okay. So before I go any deeper um I'm sure many of you are already familiar with this concept. Um but let me just start from very briefly introducing what what is the phology. Um when when I speak the speak the word out this I argue phnology is important in doing the science of consciousness because consciousness essentially is an umbrella term. So it is important to discuss what exactly is the conscious um conscious fa that we are interested in and want to want to describe and um I argue conscious is not just the qualia or not just awareness but um every aspects of the entity of the experience. So um basically phenomenology is a philosophical view that begins with fa which means um the living experience as given. So taking the experience as given means that we should analyze the experience as itself instead of pre-implying anything before the existence of our experience. So the for cons for um conscious agent the experience is equal to the world and it is not something that should be described through some external framework. So this intuition is from the discovery that first person view always exists before the objective third person view that we can we cannot essentially escape from the first person view no matter what we do. So intentionally intentionality here means that every conscious experience is constructed through the first person attitude and situation. For example, if we are in a pessimistic attitude, uh

actually let me bring the frame figure here. Uh have filled the cup of water here would be seen as half of the water has gone. So um the is it blue blue colored uh part is part represent pessimistic attitude and the yellow colored part would represent optimistic attitude. Um so well pessimistic attitude half of the water has gone. Well under optimistic attitude that this half half filled glass um cup of a cup of water would be seen as half of the water is left and this difference results in totally different quality of experience. Um so our experience cannot really escape from the intentionality but typically phenomenology by itself explains the embodied nature of mind uh through by always starting from the experience itself. So um I would say this also plays huge significance to emphasize phenomenological discoveries when studying consciousness. uh and this is the framework that I proposed uh which is basically built upon the properties discovered through phenomenological research methods that I just introduced. So this diagram here illustrate a conscious agent finding themselves in a lab environment and spotting a half-filled glass of water which is their current focus of attention. um while on one hand perceiving it under a pessimistic point of view and the another on the other hand perceiving it under optimistic point of view. So this uh this different point of view is depicted as different intentional quality which give rise to different conscious experience left and right while having same objects at as the intentional matter. So here conscious experience A B and C is also different to one another depending how much this agent is deeply interacting with environment. So if the agent is closely closely interacting with um the cup of water like holding on to the glass and observing it that experience with the depict the cup of water in a much more vivid uh manner and as it's quite literally or graphically presented the vividness of the cup would mix in and decrease when the agent is not actively engaging in the cup of water. So here I point out four big properties of consciousness that necessarily should be included and investigated through any theories of that intends to explain conscious life. So the first is very reflective self acquaintance. That mean that means every conscious individual recognize their own uh conscious experience as belongs to their own whatever experience it could be that very experience always belong to our themselves or ourselves. Uh which means uh not to some other person than I. So um importantly this does not require any secondary or higher order reflection in realizing. So the name pre-reflect pre-reflective self-acquaintance. Second is contextualized and environmentally situatedness basically consciousness is a dynamic process of interacting interpreting the surrounding environments. So our experience of rather relative than being absolute. So this would naturally lead to the third aspects aspect um which is called the spatial temporal expandability. uh this points out that our conscious experience is



something that is continuous and expandable with regards of surrounding environment both spatially and temp temporarily. So the conscious experience could be depicted as the layered contour graph uh more specifically the contour surface of this diagram um and putting bodily self or the pre-reflective self recognition as the very core. The last part uh basically describes um A conscious conscious being those can apply reflective analysis to their own experience would inevitably possess the agency of shifting um attention and the ability to apply selective attention under their own volition. This could implo um involve multiple levels of shifting from simple perception to meta awareness. Now let's talk about active inference. In my opinion, active inference framework provides a powerful foundation for applying phological discoveries in an effective way because it offers a way to formalize systems that are self-organizing and values forming. that means systems that maintain themselves through intrinsic preference formation uh rather than being driven by purely external rewards. So within this framework um intentionality which is a basically key concept and phenomenology can be understood as a kind of hyper prior that shapes an agent's behavioral dynamics. Similarly, similarly pre-rele reflective self-acquaintance in phological terms naturally connects to the body schema or bodily um uh up to bodily sensation. So um active empress framework based on free energy principle specifically uh provides a computational handle on this uh suggesting how embodied states continually update the agent's generated model itself and as the models evolve an agent's world model naturally expands spatial temporally contextually and socially um reflective how perspective unfolds from immediate sensation to broader situational awareness well and including shared meaning which is essentially what it literally means by act of emperence. So um I kind of skipped over this part when I was talking about the theories of consciousness earlier but to be specific um active infer space predictive uh predictive processing framework doesn't directly explain the formation of process of qualia itself in full detail but what it can do is model the conditions for qualia and subjectivity. So basically modeling is a self-organizing dynamics that makes subjective experience possible in the f first place and um since what I'm trying to do as a first step is to model int intentionality or so-called perspective as a minimal form of subjectivity. I believe active inference framework offers significant uh theoretical advantage for that purpose also. Um this is a whole another branch of research that I didn't go through in this talk but um it's worth noting that this framework can coexist conditionally with integrated information theory which directly address the formal structure of qualia. So following active inverse framework doesn't mean rejecting or ignoring the exist existence of kia rather I see it providing a theoretical space where the

dynamics of subjectivity can complement these structural explanations offered by um IoT. Another benefit of modeling according to your active inference framework is um that it allows agent alignment to be handled in a more natural way within a simulation environment. So active inference allows us to model artificial agents without hard coding their behaviors. So instead of scripting what an agent should do, we can simulate the environment which includes the context in which the agents act, learn and adapt and let the agent well basically itself emerge agents itself emerge from the dynamics of self-organization. So um in such simulations agents don't simply follow the fixed of set uh fixed set of instructions. They adapt and grow within the context defined environments continuously updating their uh internal models in relation to what surrounds them. So in other words, their intelligence and value model is situated. And the same logic applies to the social dimension. When multiple agents interact, their goals and values begin to co-regulate over time through uh through repeated interaction. what we might call empathy or altruism can emerge slash evolve in a natural sequence as each agent learns to attune its internal model to others in its social environment. So in this way, active inference allows agent alignment to be handled much more naturally within a simulation environment as an immersion property of relational dynam dynamics not as a con uh constraint arbitrarily enforced from the outside. Now we move on to the step one perspective modeling. So when I would when I use the word perspective, I refer to something that deeply tied to what phalomnology calls intentional quality. I describe this as being, you know, optimistic versus pessimistic from this illustration. Um in computational terms we can think of this as how an agent assigns the informationational value to the things it encounters like different subjective experience arising from the same intentional matter such as the half-filled glass of water. This difference technically can be roughly captured by assigning task dependent loss functions. But that approach still require hard- coding the agent's preference structure in an arbitrary way. So instead if we decide to explain this regarding active inference grounded in the free principle these kind of preferences can be modeled as arising spontaneously through the agent's own inferential dynamics. In other words, instead of prescribing whether the agents should see the glasses half full or half empty, the active inference framework would would enable the agent to develop its own epistematic and epistemic and effective stance based on how it minimize the price in its unique environment so that we could computationally model the perspective as an emerging property of self-organized intentionality. So here I believe this diagram would oh sorry this diagram would better depict the sequence what I just explained. Uh so here we can model the experience of different

quality. So intentional quality under context like dislike indifferent just as like the context of being optimistic versus pessimistic. uh we can model each each experience being different uh being different uh sort of probability distributions. So here the whole idea is that we could represent each different um intentional quality as certain probability distributions. So here the subjective experience of same intentional intentional matter uh which would be well we um have flat filled a cup of water probably um over different intential quality is simply put as the conditional probability um we can essentially express this as different generated models under active inference framework. So um here each generate model should share the same hyper prior which would resemble the bodily sense bodily sense of self of this agent or pre-reflective self-acquaintance. Um also please note that prediction error minimization and basian update is not uh specified on this diagram. So if we ask how perspective can be modeled computationally perhaps the simplest way to describe it would be something like this. So if we draw an Asian network based on a generator model grounded in the four minimal um four properties of minimal consciousness, it could be a list roughly like this. I would say as you can see here both of the exterior dimension and the proprioceptive dimension operate through a chain of prediction minimization um where the loss function is formulated as a minimization of the prediction loss across both exterception and uh exterception would be the role model basically and proprioception which is the body schema. So these two dimension would jointly act as variables within the generative model determining the agent's policy which then leads to action and um of course that action in turn influence would influence the inputs of the uh the environmental input which would act as a full loop in a dynamic sequence. But the key point here is that I intentionally granted the generate model with a latent structure. This latent structure encodes information uh encodes information basically that's equivalent to possessing perspective in a veilance like form. So um in other words the worldview of the agent being either optimistic or pessimistic uh can be designed to stem out from these latent variables so that perspective itself becomes an emerging feature within the model. So this is still an ongoing project, but the goal for now is to implement this as a kind of wrapper function that could be applied within um a simulation environments like Python Gymnasium um which would make it available as a Python package um applicable for simulation. Now the next step two and three that I'm going to introduce are not just about realizing phological properties through active inference but about stretching this framework further towards the problem of alignment. Um after all as I mentioned earlier my own motivation for studying the science of consciousness in the first place has always been deeply connected to the question of self-

sufficient social alignment. after all. Um so this next strand of research is really about extending that inquiry um from modeling minimal consciousness to exploring agent alignment as this natural continuation. So let me put now um let me touch on a point that I find quite important something about how we often misunders um misunderstand what alignment really means. So to me equating training accuracy with maturity level is a kind of category error because acquiring specialized knowledge and developing the ability to truly take another's perspective are two entirely different functionalities. So a system can be extremely you know precise in prediction or classification and um but still yet remain completely blind to the life live the realities of others. So in human terms that's like being highly intelligent but emotionally or or both morally immature. So if you're serious about addressing AI alignment at a more fundamental level uh then perhaps the discussion shouldn't be about training accuracy at all. Instead it should be about maturity and specifically the capacity of perspective taking because it's at its core social alignment is all about respecting the unique conscious experience of the other. So that should be the basis of all true attunements whether between humans or between our visual agents. So when I say maturity I'm referring to processes like like perspective taking and um meaning making grounded in metacognition. So maturity requires a certain subject subjectivity at at the very least um which means the ability of an agent to reflect on its own standpoint. So here we can think of greater stages of ego development or Keegan's notion of the subject object shift where um growth is basically defined precisely by one's capacity to take um what one what once was me and hold it as object which means turning one's own worldview as just one among others. So what exactly do we mean by high maturity in development psychology? Uh this idea is closely tied to adult development theory um and has also been extensively explored in uh meditation research too. So in those context maturity is often evaluated through uh language patterns essentially. So that means that how people describes themselves and surroundings and how they reflect on their inner life. So it is less about what we know and more about how we relate to what we know. At a deeper level, higher maturity uh seems to correspond to the ability to affirm the plurality of conscious universe and to recognize that um multiple perspectives can coexist without collapsing into one single truth. By linking these findings from the phenomenology of meditation to the model introduced in step one, I would say we can begin to ask very concrete and testable questions including what kinds of environmental interactions would stimulate an Asia's growth and maturity and um within a simulation environment and how might we detect or quantify the markers of maturity as they emerge. And this becomes especially relevant

when we think about self-sufficient social alignment. So um if we want to cultivate alignment that doesn't rely on external control uh we need to understand what factors can route these type of effective stances like empathy, care and openness toward higher levels of maturity and perspective taking. So um insights from this approach could extend far beyond the artificial systems. So they could inform not only um AI human alignment but also human human alignment which means how understanding and attunement evolve naturally in social context. And importantly this idea is particularly applicable in active infra framework based multiple agent environment that will be described from the next step. Those settings would allow us to uh simulate the emergence of maturity as a property of relational So now let's take a look at how the concept of maturity can be applied within a multi-Aian environment. In development psychology, maturity is you know difficultly measured through language markers um as I mentioned before but that doesn't necessarily mean we have to apply human language directly within a simulation environment. So even without um even without language uh when the communicate communication is non- linguistic uh it's still functions as a form of information transmission and sharing. Um so whether agents use words, signals or actions, the essential process is that the information is being exchanged and interpreted and also transformed through such social interaction. So from this standpoint um it is able to define certain type of informational flow as proto proto linguistic maturity marker. Um in this view uh communication can be modeled as each agent reinterpreting a raw message through the lens of its own its own role model representation. So that re interpretation process would tell us a lot about the maturity status of the Asian's cognitive behavior. For for example, if you wanted to test a certain dimensions of cognitive maturity such as reflective judgment and interpretive complexity, uh reflective judgment RJ could be understood as the ability to interpret a raw message objectively. objectively with which means with less distort uh with less bias uh coming from one's own perspective. So say an agent with high RJ can recognize its own standpoint as just one possible lens among many. And while integrated complexity I see is the ability to consider and incorporate the world models of other agents when receiving or generating a message. um that would mark um agents with high IC's as um more likely to adapt its its interpretation in response to diverse perspectives. Now using these measures we can identify signatures of alignment failure modes. For instance, an agent showing high uh prestige bias that is overvaluing certain sources or hierarchies is likely to have low um IC interpretive complexity because it's not generally interpreting multiple uh viewpoints. And if an Asian seems trapped within its own world model um unable to update its interpretations

that would indicate low RJ. Now to actually test these ideas, the next um and for ultimate step is to set up a simulation environment that follows the active infra framework that um it would allow us to examine how maturity and alignment dynamics emerge through these um ascentic interaction. for a simple implementation using the RXMER package being developed from active empiric institute as a foundation would be um one possible method I would say uh especially utilizing the package within a relatively lightweight environment such as gymnasium or a pie game would be a good first step where these platforms are ideal for building early prototypes in terms of visualizing how Asians adapt and coordinate in real time. So good on visualizing uh for a more complex configuration. I eagerly look forward into multi-ion simulation spaces like Metag Grid or common grounds. These frameworks are particularly useful because they support contextsensitive social dynamics, meaning that agents would adapt their behavior based on changing social or environmental context. Um in particular metagrid is a project uh that is currently being developed within the active river institute. Um so if it turns out the perspective architecture can be applied within this u metag environment I believe it could uh strongly highlight the parts of the simulation that could benefit from active inference principles. And as for Common Grounds, which is basically a 2D Minecraft like gaming platform where humans can directly intervene in the civilization uh building sort of processes of agents, artificial agents. I find it especially interesting um because it allows us to design testable scenarios of pro-social alignment. And um by the way, from my understanding, there's a live chat scheduled about this project as part of the symposium. Um so I would recommend attending it if this line of work interest any of you. So in any case, testing self-organized pro-social behavior based on maturity dynamics within such platforms is something I'm planning to work on as part of my research activities at the institute. To close the talk, modeling consciousness leads us to understanding perspective. Understanding perspective in turn enables mature uh mature alignment. And while active inference is serving as a bridge by linking phological insights with computational realization, ultimately alignment begins where understanding becomes empathy. May we act and infer answer and share. Thank you for your attention. Um if you have any questions or would like to share your thoughts freely, please feel free to reach out using this email. Thank you. Greetings. Hello Maxwell and Patrick. It is November 2025 and this is the thermodynamic brains discussion for the applied active inference symposium. So let us begin with whichever order you'd like little introduction and then we'll jump into some questions and discussion. >> Awesome. Patrick, do you want to go first? Yeah, sure. Uh, thanks a lot Daniel for the invitation. Um, my name is Patrick. I'm a

physicist by training now working at Nomino uh as a researcher and we're trying to combine uh thermodynamic computing and active inference proistic inference um to advance let's say robotics and other applications. >> Awesome. I'm Maxwell Ramstead. I'm a co-founder at Numinal uh active inference researcher. I've worked closely with Carl Fristen. Uh very much interested in the intersection between um information theory and thermodynamics in particular non-equilibrium thermodynamics. Um so yeah, very excited to tell you a little bit about the thermmo brain and what we're up to at New Middle. >> Amazing. Yes. Let's get right into it. So what does it mean to say that a brain or reasoning in the brain is thermodynamic or that it's a thermodynamic process or computer? So I mean if you think of um and taking a few steps back maybe from um thermo itself if you think of the free energy principle uh what the free energy principle basically tells you is that um basically in the presence of uh in the presence of boundaries um what would ordinarily uh look like diffusion dynamics turns out to look like something a little bit different. Um I mean in effect what the free energy principle tells you is that uh the dynamics of systems that are separated but coupled uh by their maroval blankets will look as if uh they uh they implement some kind of parameter estimation where the systems that are coupled but separated through their markoff blankets end up I mean effectively um tracking each other's statistics or estimating each other's parameters. Um and um I mean this is in effect a thermodynamic process. Al although in the free energy principle literature we tend to um you know emphasize the dual information theoretic description this kind of coupling is ultimately a kind of thermodynamic equilibration. Um so what's what we what we think is striking really is that um almost all uh physical systems uh that learn learn through this kind of um coupling and gradient consumption where you know when systems kind of find a dynamic equilibrium or you know reach a non-equilibrium steady state they they end up becoming parameter estimators for each other. Um so uh I mean yeah the claim is essentially that all systems all physical systems that end up learning um the the structure of what they're coupled to learn through this kind of thermodynamic process with the exception um and I think it's a very important exception of the actual digital machines that we've built that should be you know the the main kind of learning machines so the these kind of machine learning stacks Um you know Hinton calls these immortal computers for a reason. Um but I guess the paradox is that um you know in um in architecting you know our digital computers such that they can uh you know uh execute uh instructions in you know wellgoed deterministic ways and you know hold on to pieces of information for a long time. in kind of architecting these systems, we've forced them to learn um in a way that's unlike the way that

anything else in the universe learns. So, I guess part of our ambition here is to allow our thinking machines to learn in the same way that everything else in the universe does. Uh Patrick, do you want to maybe like add on to that or expand or disagree? >> No, I think it's a very complete picture. Um, yeah, I I completely agree. I would totally second that. >> Okay. Where do you think reasoning fits into that? Is that process reasoning or is it another layer beyond that that would make something reasoning as opposed to doing something prior to or below reasoning? I I think an important thing to point out is that um you know what we're talking about here is kind of um implementing algorithms using the physics that naturally maps on to the algorithm. So this isn't necessarily about reasoning as opposed to perception as opposed to learning as opposed to policy selection. This is another kind of uh approach to um I mean codeesign in effect right so codeesign is this idea that algorithms uh so software right and then the devices that run those algorithms hardware should be designed together what thermodynamic computing allows you to do in effect is to replace um the simulation part uh in a lot of these uh sampling based algorithms like MCMC uh with the actual physics that you're interested in. Right? So MCMC Markoff chain Monte Carlo, right? The the Markoff chain part is the part where you specify the causal structure of transitions in the world using a Marovian model and then the Monte Carlo part um is about actually simulating the dynamics over time. And it turns out that um you know this kind of uh stochastic algorithm is extremely useful um in the context of machine learning. I mean you know if you've been on Twitter recently like everyone is now talking about diffusion models for everything and I think Elon Musk recently talked about like diffusion models for reasoning and this kind of thing. Um so the the important thing is um that all of these stochastic algorithms depend on simulating these kind of stochastic dynamics and this is where I think the fundamental contradiction of the current you know R&D landscape for AI like enters into the picture. So on the hardware side, you know, we spend a lot of time and energy to make sure that our digital computers are not stochastic, that they're fully deterministic, right? Like we drive the energies uh of these components to very high degrees to make sure that the the the uh bits are, you know, distinguishable in a reliable way that they don't bit flip and so on. So on the hardware side what we're doing is spending a huge amount of energy to drive the stoasticity away and then we end up you know from the algorithm side reintroducing it. Um so the idea is to design an al a hardware stack that is more apt to the kind of algorithms that we want to run on them. So ra you know design a chip basically such that rather than simulate all of these stochastic dynamics you can just have uh physics as computation. Um and that's really the key I think.



Uh Patrick do you want to add on to that? I've been talking a lot. I'm sorry. >> No no no I mean so I think also maybe to it's important to say that this idea is not that new. I mean uh a lot of people have heard of quantum computing. you know, you use a quantum system to do quantum simulations instead of running it on a CPU, which we know is very inefficient. But then there's also this idea of like analog computing in general. There's people, I don't know, using water pipes to simulate some systems. Um also even like actually putting a car in a wind channel is somehow a form of like doing physics based computation. Instead of doing this simulations in like a digital computer uh you use the actual physical object and wind and maybe some smoke to to do these simulations. No. Um so this these ideas are not like fundamentally new. Um but the evolution or the the technological advances of the last years in like especially CMOS design um allow us to actually build sophisticated devices that that actually do this physics as like that I think that's the the biggest change. Yeah, I mean maybe to address your question a little bit because I think what you're asking is like where does this fit into like current stacks like what you know you you mentioned reasoning but like why why would you want to do this? Um and I mean in effect um you know a lot of the so in the context of active inference since we're talking to you know the the founder of the active inference institute um I guess on one level the math of the free energy principle rests on these kind of stochastic uh thermodynamics non-equilibrium thermodynamics formulation. So if you if you open um Carl's monograph which is by the way going to be published with Springer I think either later this year or next year I'm sure you've heard of it um it's now called uh the free the free energy principle on the nature of things. Well so if you open the monograph the which is on archive in 2019 I think the original title was the free energy principle for a particular physics. Um so if you open that um monograph up literally the first equations that pop up are you know the longan equations the fauler plank equation this is fundamentally stochastic thermodynamics it's based on the this kind of stochastic approach to uh you know understanding uh dynamics. These are like fundamentally non-equilibrium formulations. Um so this math is important for the foundations of the free energy principle. Um and it's also very important I think to more applied work in the active inference space. Um so I mean as as I'm sure you know um by now uh it's it's turned out to be very difficult to scale active inference in part because it's so um purely basian it verges on the puritanic puritanical sometimes. So um in in some interesting work that has allowed us to scale active inference up, what we do is uh we create hybrid systems that employ active inference sure but also you know um MCMC family algorithms to perform for example policy selection. So, Zafas has this very cool uh paper or

rather set of papers where they uh ex where he and his team um explore um you know implementations of sophisticated inference uh the use MCMC to finesse the policy search aspect of that to to avoid kind of like having to deal with the computational explosion. And obviously the um the bottleneck there is you have to run the Monte Carlo steps um on you know a CPU or GPU or whatever you know is your favorite digital hardware stack. Um what this kind of technology the thermo technology uh promises is to you know make those uh steps in the hybrid algorithm like dramatically more efficient. So, uh, you know, the early numbers that are being reported here, you know, they're something on the order of like 10,000x less, uh, energy demanding. Um, so that's significant, right? It's it's significantly more than say like a 50% decrease. We're talking about like orders of magnitude efficiency increase. So, you know, I mean, to to to put it a bit more um simply, uh, what this allows you to do is cheap sampling. And uh you know for those of uh are those in the audience who are machine learners uh your ears have probably pricked up at this point because you know sampling is very expensive and it's arguably one of the main bottlenecks um in you know just general deployment of machine learning technologies at scale. >> Thank you Maxwell. Patrick, do you want to go into a little more detail about what's happening inside the algorithms, inputs, outputs, what languages or what similarities or differences with other kinds of toolkits that are available today? >> Yeah, this is like probably one of the the questions we get asked mostly is like, yeah, what do you actually program on these devices? No. Um I think the simple answer is that we all aim in this community of thermodynamic computing to abstract this away. So in the end if you're familiar with uh I don't know the active inference community that well but more like proistic machine learning let's say uh these frameworks like nonpyro standc if you're familiar with those it's going to be very similar to them if not even the same no because uh numpyro allows you to say hey I want to run this on the CPU I want to run this on the GPU Maybe in a few years we just can change the back end to a thermodynamic computer because fundamentally it does the same thing. Um it just does it in a very different way. Um same as if you change from a CPU to a GPU. Um you you don't mind just by using it in Python uh where it runs but you will feel it in the difference of like how fast it's going to be uh executed. Um and then yeah obviously what happens in the back end is is fundamentally different. Um so the sampling then really occurs through a physical thermal process instead of like some digital emulation of it. But to answer your question for the end user like most people in the field that do not care about writing their own compilers nothing will change. Maybe two lines of code just like send it to thermal or yeah, whatever we're going to call it in the end. Yes. Interesting.

So, I mean, what are those sorts of building blocks? How might you connect that to maybe Mars levels of analysis? Where do you where does the cognitive part come into play? Or what granularity do you do you think about a system or a problem >> and which pieces could you swap out model? Just to interject quickly, I'm not sure um Mars level of anal Mars level of analysis is an appropriate framework to think about this um because that framework supposes uh that there's a tight decoupling between you know the uh the different levels of analysis and in particular uh between you know the algorithmic and the physical implementation layers. What we're kind of saying is like you you you don't want to ignore those connections actually between the algorithm and the physics. The idea is to build a physical system whose uh kind of intrinsic physics if you will implement the algorithm that uh you want to run. So, um, yeah, I'm not sure that that kind of detachable or kind of separable concept uh is uh like as ill as as informative as it might be. I mean what what we're trying to say is like it's it's probably a mistake to separate the software from the hardware uh in such a dramatic way as see in today's stacks and things would be much better um if we went the other route which is to actually try to couple them more fundamentally. >> Yes, I I hear you on that. a little like an enshelved codependent marian. >> Yeah, precisely synthetic point. Um, from the point of view of a software developer, um, this just changes the hardware that is actually running your software. So, you know, why why you care is you're going to, you know, have a radical efficiency increase. Like things are just going to run faster um, at the end of the day. But uh I mean you'll still be um you know using the same kind of algorithms that are being deployed right now in software. It's just that the the actual physical routines that the system goes through to compute those algorithms is just going to be that much more efficient. Um and you know this also opens the door to um a kind of more extensive codeesign approach to algorithm development. So, you know, um, if you're developing for thermo, then knowing about the thermodynamics of a ther of a TSU, a thermodynamic sampling unit, and, you know, how that works is going to be pretty important. Um, so yeah, I mean, I think your knowledge of how the hardware works is is now going to possibly contribute to how you're thinking about software development. Um, but in terms of model building, as Patrick was saying, uh, it it's pretty much plugandplay. I mean you might as you were saying you know route route your algorithm through another kind of chip >> but ultimately um you know this is this is more going to be a gamecher for from the point of view of performance and you know long-term code design than um just like in in an immediate day-to-day kind of thing for developers. And I mean just like adding to that it's like there's a lot of people uh on this planet that that care about CUDA kernels and

optimizing them but a lot of machine learning practitioners have never touched pure CUDA which is still a relatively high level language if you think of like actual transistors and the GPU know so but even that level of abstraction which is already pretty high most practitioners never touch that's the job of of like really specialized folks and this still will exist. This kind of profession will continue existing and people will have to think of like hey we have this particular sub routine that people really care about in let's say active insurance like this this specific way of sampling something um let's try to make this more efficient and then there's also really like the the hardware level where people think of like actual single transistors and how to communicate them together and so on. These kind of professions exist also now in SMOS. It's just like no one really talks about them in in machine learning or there really this are abstracted away fortunately otherwise you'll not be able to do any meaningful machine learning if every machine learning engineer has to think of bits and transistors. >> Yes. So to fuse a few questions together, let's say you were setting up a research and development lab over the coming few years time scale. What does that lab look like in terms of skills and hardware? And what kinds of problems? What sectors is it tackling and why? Hello, I'm Michael London and I am a member of the science advisory board at the active inference institute. I live in Washington DC. Have an academic affiliation with a couple of local universities and then by training a Hello, I'm Michael Lennon and I am a in Washington DC. I have an academic universities and then by training a an economist and a technologist and today I'm going to share some reflections that um through the mostly lens of an economic technologist you could say um about um the impact of AI. Uh and so uh about a year ago or so I co-wrote a book with um six others seven others about the possibility in the future of AI. It was um mostly optimistic. Um I was one of the skeptics. You can see the chapter I wrote there dancing with a fire. Um but it's been um an area of interest for a long time the kind of tensions that exist with uh uh artificial intelligence and over the past year some of these have simply continued to sharpen. um one of the ways certainly within um not just economics but from the mindset of of economist um two mindsets about the future impact of of AI on on economies and on humanity and one is that um it will take over inefficient humans that it'll shrink the bad in the world but uh but really it's going to be a kind of this someone would say this a technoutopian vision of hyper efficiency um a different technoutopian vision is that it will help grow the good in humans and amplify human capacity for meaning and contribution and um and this debate is going on in in um in many circles. Um one of the ways in which I found um talking about impact with folks is u uh can be illuminating is using this this particular diagram.

I've u I developed it a while ago for comparing investments in different uh uh technology projects around the world and comparing where they were uh targeting. the institution that I was doing the work for was um had a global portfolio and um one that wasn't just economic but included um social and other kinds of impacts and um and yet all the reporting was done through an econ an economic lens and by asking people to talk about a whether they were trying to address finance or experiential life or or the larger system such as >> Hello, I'm Michael Lennon and I am a member of the um was one way of uh fairly easily contrasting uh these these different projects but often these projects also had two very different kinds of objectives. Uh one was often a shrinking of a problem um and the other was often growing a new capability and just like fixing a broken arm relieves your misery and uh and uh helps create a sense of being better than when you were miserable. It does not make you an Olympic athlete. And so often these uh by by distinguishing between what are what are interventions that shrink harm versus growing good, one really gets much greater clarity about what is the invisibleized impact of of of the project that one is undertaking. And so um often in one class of political mindset, there's the the growing the good is often uh solidarity type economics, folks that uh that are trying to grow the solidarity, the well-being and uh and such. Whereas often for um efficiency and um uh minded folks um the idea of shrinking waste uh uh is at the heart of uh of what there is to do and um and where for example um neoliberalism often finds and people who self-describe as neoliberals uh often find themselves setting up the the the focus of the work that they're doing. And so one of the things that both these narratives are um useful and and they represent um uh perspectives on realities. One of the things that's been interesting about the emerging science of predicting economic futures has been that it turns out that one of the most influential aspects of predicting uh a country's future economic performance is um its networks. what kind of networks in which kind of it's learning networks frankly um uh learning cities learning relationships between the cities all of these um and the complexity and sophistication of these all are much more accurately predictive than many of the uh what have been traditional uh indicators used to to look at at what the future will hold and um and that these particular um uh networks um operated at different levels that there's both uh networks of people individuals networks of groups networks of firms and that Each of these are like um layers of a living system that each are learning at in their own way at their own level and and all are contributing in uh in um in their own ways. And so really beginning to understand how the economy is a is a often could be looked at as a kind of learning system that is learning how to do things better in the place where the economy is and add value uh

based on what is there in the economy. that life does that often with the energy and nutrients that receive from the living ecosystems in which it's in. And so a takeaway could be that uh you know a priority for for AI in the future is to boost the learning capacity of networks that we do this for ourselves that we do this for the groups that we're in and that um and that when we're having conversations with others that it's like do we understand what are the networks and how can we boost the the learning and interarning and it's really important to sharpen our understanding of what these different um learning units are because we have vocabulary that is mostly individual and it turns out that as a as a group or collective learns it it happens in a different place and in different ways. So, um I learn within my brain, if you will, uh to oversimplify things, and a a hive of bees learns by by interacting with other bees, signaling where is the honey, where are the flowers, and then responding as a group. And so, we can be talking about learning and um and be uh referring to what is happening within an individual or we can be talking about learning and and be talking about what's happening across a group of individuals that are a self-organized group or it can be between groups of groups and and and so on. So uh the learning edges whether they are within whether they are between or whether they are beyond the group all of these are um different learning edges that um that can be adding value and um can be step points. Uh so back to the original prompt for this particular presentation. Um there was last year in Buffalona uh a s a group of creatives and uh technologists and activists and civil servants a variety of different communities all in this vibrant city that were hearing that uh that generative AI was going to change the future and they weren't sure whether it was all rainbows and lollipops. Some of them were skeptical and so they were kind of uh how do we get smarter about this and how do we begin to grow the civic conversation about co-parenting a vision a shared vision forward that uh that is grounded in in um the information that we each have our own individual perspectives and woven together into a larger narrative. So, so each of these communities um each brought their puzzle pieces and they tried to assemble into a simplified map that became then a discussion board for moving forward and um it did help uh demystify and kind of build a greater pool of knowledge as well as surface some of the tensions and controversies that that exists. And uh among some of the outputs if you will was to identify 21 topics uh that um that are that they identified that were affected by this emerging phenomenon of generative AI. um they had under each of these topics whole bodies of additional content. But one could summarize these and they did in the following way that that this phenomenon of generative AI was um was magical was the word they used perhaps oversold um and that there were there were

fascinating dazzling things about it and um hidden harms and that some of the pattern of accelerating harms that they saw were that extractive practices of data of resources of humans, exploitation particularly uh in foreign places, the invisibleized um uh the the amount of humans in in in far away places that support the cleanup of the data and the delivery of AI is just starting to come to the surface. And they they were beginning to recognize the hidden human factor in in AI. Environmental destruction as well as the concentration of economic power. All of these things were being intensified by this by this process. And so um as a group it started to kind of feed a kind of maybe we need to update our our myths about um what is possible that it's not just all about being smarter leaner than what we were but also recognizing um how is this affecting the the groups and the intergroups in in interesting ways and uh and uh one of the things that I uh would have served here some additional limitations of of LLMs that are known to have problems. uh for me the one that I want to build upon in this conversation is the the architectural failure of chat bots to uh to support nested learning uh of networks. This kind of the different parts of of the being and the different layers of the being. Still um uh the work was very very um fruitful and um and it was well um built upon. Uh the EU um uh incorporated whole aspects of it into its vision for the AI Euro stack. um in China uh several um organizations, learning festivals and such have have built upon it um in different ways. It's been translated I think in over seven languages and and is being um elaborated evolved forward um in in many many ways. And so I had as a as an aspiration uh because our community is a learning community how to bring if you will the not just the engineering technosavvy perspective but also um other non- US perspectives on on on the macro trends and how might that um serve or feed our uh our own active inference community and um I think one way of summarizing that would be to say that uh that people are also infrastructure and so it becomes an invisibleized aspect of um of really um optimizing the place in space if one sticks to to the to the tools and the automation without um without really accounting for how the humans are part of that living infrastructure. And so um but then um as I opened up conversations I ran into some problems. I I was u I was engaging in bio regionalism supporting uh this environmental movement called bi-ism uh that looks at kind of what are living areas that uh it might be a watershed it might be a forest that are self-contained um uh unique living systems and often adjoining each other and that this might be another way in which humans can be in relationship to the places that they are that they're not just in a in a for example uh the state of a political state but that for example I who live in Washington live on the Chesapeake watershed and I'm in a in an ecotone that is three different uh three

different uh bio regions uh the South Atlantic, the North Atlantic and the Appalachian uh mountains uh come into a kind of confluence and create a biodiversity hotspot that is not how most people talk about Washington DC. So this movement about um orienting to life and living systems um has been u a growing movement worldwide. Um and um and then one of the questions was how can AI be talked about in those spaces and um and some of those conversations turned out to be uh uh well that would trigger the oh AI is going to kill everybody or AI is you know the the the usual. And so um after a couple of attempts um I have um landed on these three topics as ways to both talk uh across different uh different communities. And this is part of what I'm testing here in this in this delivery here today. Um that uh the following three themes are ways to think not exhaustively but usefully about um about human and AI interactions and and learning. So about data, data is a well understood uh phenomenon by many um and yet um most of us um are the data of our own lives for example is uh managed by others in other places away from us. And so I think of um in this picture you see uh a picture of my Gmail inbox and my communication patterns. It turns out that relational dynamics are predictive of your happiness, of sales effectiveness, of a whole variety of things that people uh tend to uh explain and and figure out through other lenses. And if you're able to take your communication patterns and render them uh in a social graph, you're able to see um dynamics about your relational patterns that um that enable you to choose differently than you might when you're using linear and rectangular presentations of your own data. And this is a really important notion to get because it's this whole systemic uh relational data uh is something that happens in layer after layer after layer and sometimes we have to see it at the level of me in order to then feel more skillful or more confident about talking about and how does this play itself out in other spaces. So this uh level of seeing our systemic data and taking ownership, having support for easier ownership and integration of our data so that it's not fragmented across Google and Yahoo and and 20 other thousand places, but that we have tools that enable us to to mine our own insights about ourselves instead of it being corporations that are selling things to us is um is one of the um uh ways in which a civic platform can help um make our lives easier and see our own data in in more novel and useful ways to ourselves. That pattern repeats at the level of groups. This representation that you see here is uh seven different data coding teams that all used the same architecture, the same platform on GitHub to um to code a solution to a particular problem. And um and what you see u represented there is their data calls, the components that they used, the pattern, how they solved the problem, even if they were using the same tool sets in a sense. And you begin to see that there are all



additional ways to see how the the living sense making is happening whether it's at the level of individual at the level of teams etc etc and so a third dimension of data that uh can be uh of interest is what I'm calling uh intersubjectivity now we all have um a sense of our own our bodies naturally give us sensory information uh that we then translate into our own actions uh as groups we also use language to get synchronized and um and orient towards common goals u one of the things that um that science is helping um sharpen is u well let me reframe it this way um a Nobel Prize was won by Elanor Ostrong who who recognized that um that when groups organized they that econ economists used to describe it as either as hierarchies institutions as marketplaces and she recognized that there was a different kind of human grouping um she called it a commons a group that self-organized and practiced shared care it could be like fishermen around the fishery and that their rules were not um hierarchical or marketplace place but but were others and that these kind of self-organizing groups these natural groups um were everywhere and that they basically operated uh according to eight principles if they were sustainable and functional and these eight um principles uh can be used to to gauge perceptions of how functional are we being what is our level of symbiotic cooperative fitness and that that becomes a way to to support the evolution of of groups uh forming into uh from from from baby groups groups into mature adult symbiotic groups and that this is a a class of of data that pays attention to interiority um uh in ways that also builds relational spaces between humans. So switching now to the next topic away from data and on to compute here I'm just going to talk about what for me are some of the interesting uh for me have been some of the interesting insights from active inference it is far from complete but uh when I tried to talk to people outside of the community some of the what's in it for me that might be different from a computational perspective and um and perhaps one of the hardest and both intuit but most useful might be the idea that AI that everybody gets is kind of like uh pre-trained it's It's it's uh it's like canned beans. They were harvested. They were prepared. And now you can eat them as prepared, but they're not going to change or evolve in any way. They it is pre-trained. And so, uh these chatbots are things that can ingest the entire web. And if you want them to be current the next day or 3 days later, you have to reingest the entire web. They are not really they're capable of doing the ingestion, but they're not continuously adapting and learning. So the architecture of active inference is one that both these types of of computation have learning methods. But in a sense, the FGBT, and this isn't an oversimplification that many in this group will probably debate, is that not only is it um related to to only the past in a sense, not the present or making predictions about the future, but um

but it completes its training loop and then it works for whoever the user is of that particular canister. uh that cognitive capability. Uh active imprints uh can behave at the clustering level and so it crowdsources um what are the predictions of the group and what are they predicting and then based on errors is readapting or reevolving forward um what has been um accurate or inaccurate about the group's crowdsourced prediction. And what that enables is it means that as the group faces stable conditions, they stay on course whatever course that they were on. But if new conditions change or new circumstances arrive, then the updating of expectations and the co-synchronizing of behavior all happens in real time, which is how living systems kind of stay attuned and adapting forward continuously. And so it's a kind of distributed learning that is not present in these other forms of uh what I call pre-trained uh GBTs. Um a second is that that dynamic capability allows for new degrees of freedom is how I describe it. But it allows for new capabilities to emerge um a new creativity which we see over and over and over in the ingenuity of living systems. And in part the logic of this is that in order to there's in system thinking there is a uh there is a uh rule so let's say a law that says that in order to control that system let's say it has 10 dimensions to it then you must have control mechanisms that either match the number like 10 or exceed it. If you have fewer then you're only able to influence but you cannot control the complexity of that system. And so matching the complexity of the system that you're facing is an important aspect in developing greater and greater mastery over things in the world. And one of the things that language allows for groups to do together that they cannot do alone is that as groups new language can be developed to describe what is being seen collectively. And then that terminology can be used to then co-behave differently, co-adapt. So it allows for groups to also perceive together and then synchronize uh and do as groups what they cannot do alone. And so this linguistic development is um is the mimetic or the linguistic the development of the vocabulary is what enables the creation of new forms of competence. And um and that active inference has algorithms that when we were looking at the previous diagram on the previous screen that the ability to evolve its assessment of what the error was and then how does it weigh uh correcting forward means that as a group the co-correction can become a new shared um capability uh that allows for the group to do at the level of group what it can't do alone. And so this this is part of the miracle of adaptive uh learning. you could say it allows for the continuous adaptation the way that life does. Last thing I'll talk about um computation is that there are new classes of learning metrics that become available. Um what is learned by which action? what is the information gain of

any particular learning activities and um and in the environmental space for example there have been new ways of talking about the thresholds of planetary carrying capacity versus individual and how does the relationship between the individual for example consumption of water and um the capacity of the watershed how are those two related and those uh enable for new quotients new ways of measuring new ways of um monitoring the the sustainability of let's say a watershed the use of a watershed, water footprint or carbon footprint that become these vocabularies that we now use to understand the flow of water, the flow of carbon through a system. Similarly, um this idea of agentic learning and the field learning um can have um baselines and thresholds that are related to each other so that these become benchmarks that one can begin to make new kinds of comparisons, new kinds of metrics for for the learning of groups and groups and groups um and their overall the value and totality of their of their learning. So jumping down to the third topic, the final topic and I know this is running a little longer than expected is um and all of these u different data types, different informational sense making types all create new ways of learning and new ways of weaving new learning to add value to the new economy. And whether that is to make faster cars or leaner cars or cheaper cars or whether it is to interact with other um social systems and ecological systems in entire new ways, there are a whole new set of job opportunities um arising from these new competencies. and um and so positioning to provide some of these um advisory services, these knowledge collection services, these knowledge curation services is all part of the reason why this idea of a net zero from AI is entirely uh plausible. Um I'll just share an anecdote um uh before moving to the next screen which is there's a variety of countries that I think of the richest the richest region of Italy Emilia Romagna is um is composed of almost one over one-third of his cooperatives and as they uh went through digitization and automation over the last few decades um they did not lose any jobs um they uh continued to lead in productivity and performance but it was a commitment of the community and of the co-ops to keep people employed. Germany has a similar kind of approach to labor development. They outsource the the um if you will the the simplest work and they keep the most knowledge intensive work locally in the design and the engineering of the work. And so they stay at the learning edge and they stay wealthy as a result of moving the needle and staying on the edges of uh the knowledge economy retaining their people and their their uh human networks as having a component of that adaptive learning as much as um as as the codified knowledge in systems. And so um this kind of learning um is happening in level after level after level. examples here um from developing economies is some fishermen who had

some reporting to do in um on the fishery uh in um I believe this was Latin America. Um it was a bit u for them and they started pulling uh resources and activity to both uh track and uh and monitor the the data together and they started both becoming more um effective at their reporting as well as developing visibility into the kind of things that help them manage their own fishery better. And this kind of self-organizing at at the microcosmic level is thing that happens over and over. It's not just the globalized systems that can benefit from these AI systems. But each of us can be tailoring at our local place. And so whether it is health data co-ops, um patient owned platforms, um other forms of co-ops, these are are coming up left and right. And there's as much a revolution going on in the development of machine learning capabilities, whether it's translating linguistically in different ways or tailoring these capabilities to the characteristics of the place that they're in. So um there's a whole range of opportunities for prototyping ventures for adapting the governance of these and for bringing things like geo infer to the um Q infer which is something Daniel prototyped as a synthesis of active inference and GIS data sources to to bring the capability of active inference to GIS databases. And so these are just three um of the methods. Um I was going to show a list of allies but had not consulted. Uh there's of course um I already talked about the service so I'll move along. Uh active inference can help with sense making through with human networks through um through traceability of AI which is a real problem for LLMs. Um recasting digital footprints which we talked about this idea of having methods for how metalarining and metalarining happens as well as these additional tools. Finally, I'll throw out that um a consortium of social impact investors in the US have uh pulled together half a billion dollars recently in the last few weeks to drive three levers that they hope will influence the present and the future of AI, expanding the groups of people that participate in AI design and development, strengthening the organizations that play these coordination roles, and supporting the creation of public goods. So, we are wonderfully positioned as a community to help with these. And this is one of the questions that I like to throw out about what comes to mind for people as opportunities for action. Um so um I'll leave as closing thought you know how to better enrich the how can AI better enrich the learning networks in which we are all in. Um and um and I'll stop there if uh you have questions or would like to reach out to me individually. Uh [lenin.ci.org](mailto:lenin.ci.org) is my email. Um and um thank you very All right, this is going to be the final pre-recorded video. It is active inference for the social sciences state-of-the-art with AEL. Here it goes. It is a lengthy 1 hour and 43 minutes. Enjoy. So hi uh I am Kush and this is the round table about active inference for the real science for the active inference

symposium as I so we have uh well myself raising it of course we have zitski we have pashia we have we have malarasa and what we will discuss tonight is uh basically go over what that inference does for these true senses if it does anything which you know we may discuss at some point. uh we will uh each try to give a very broad impression of what our work is about and each our work is relevant to this question hopefully and then we will uh discuss uh you know what do we do with act in France what can be uh what can it be used for etc such questions so uh first of all I want to go over a very short very um accessible hopefully presentation of fat inference which I guess people in the symposium will not need uh very strongly but uh we're trying to make this an invitation for people who are not necessarily you know cognitive sense people to collaborate and to uh sing together uh what I in France brings uh to the table and so I believe there is a large proportion of the audience that will necessitate primary inference if they are to follow anything that we say so uh for those people welcome and thank And I will now go over uh the state-of-the-arts. So uh first of all I will uh define social sciences very broadly as the study of societies. I know it come as a shock. >> You have uh many kinds of social science with uh different methodologies and different topic. For example, by convention, due to historical contingency, anthropology is mostly interested in stateless societies while sociology, political sense will be more interested with societies with states, large scale organization, parliament, selection, etc., etc. And there are other disciplines that interested stuff in the past to stuff in the past that has material culture, respective history, and archaeology. There is a very broad variety of social senses. This is why we have a plural to the term. I will not try to define specifically what they do. But I would say in broadstrokes uh social senses are interested in social phenomena and they're interested in finding patterns in social phenomena and they tend to hold that those patterns are actually meaningful and explanatory and this is where the notion of social structure uh comes into play. Most social scientists I would say uh would agree that there is such a thing structure and that are uh from the very basic Wikipedia definition I found it is uh the aggregate of pattern social arrangement society that are both emergent from and determinant of the actions of individual. It's something that explains individual behavior in the context of society and that is not you know really inside individuals. So this notion brings a lot of questions actually foundational ones. uh first if there is such a thing as social structure how is it implemented in human minds? We are not born with a sense of class or with uh the specific arrangement of French. This is not something that is implemented at inner brain at birth. So at some point we integrate those notion and there is specific process by which and this you know process necessarily will be

limited in some way. So the question of how specifically structures implemented matters quite a deal. Relatedly there is a question of uh what is the relation between social structure and agency. So if structure that is something that is not not really inside our heads can determine our behavior. It does not entail that we are just mindless robots. It does not seem we are. So there is some level we can decide what we do and some level we are determined by society. This is non trivial. The articulation between the two is non trivial and it is a widely studied topic in aology specifically. And third uh is there is such a thing as flow structure. uh it seems to change over time. We are not in the same societies as uh our ancestors 20,000 years ago and the specific rules that apply to social interactions engagement they seem to change between context. You do not talk to your boss like you do your dating buddy. Hopefully for you and uh these questions uh I believe you will see this coming. I think uh sorry can be approached through the lenses of active inference which is the product we are selling to you right now. Sorry about this. So let's try to uh take a look more specifically to what is this active inference thing which I did not define at any point so far. uh let's get back to the basics which is uh the concept of ping predictive processing and the friendship principle. So in 2006 uh British new scientist called Carl Tristram proposed a very basic principle for commission that is basically the idea that we do bayesian inference. I will not get into it but bayesian inference is the optimal way uh that we can mathematically determine what kind of uh stimuli what kind of object out there would cause sensation. Uh here uh h correspond to hypothesis and e corresponds to evidence and the hypothesis is you know the concept of what's out there affecting us and the evidence is how specifically it affects us. Imagine if I hear mowing maybe there is a cat but maybe there is a audio box that meows all the time and I can use the probability that a cat would indeed meow and the probability that audio box meowing would be out there uh and determine the specific probability that there is indeed a cat given that there is meowing that's the core idea so and the slides have frozen again. Sorry about this. We have a reoccurring technical issue with uh Google Meets. Hopefully, this will uh arrange itself. So, uh you don't need to get the math. I just want you to know there is mathematics. The fact is when you consider complex problems with a lot of parameters, uh just calculating bayesian probability of something is not possible. But what we can do is compute what we call variational free energy which is uh much more simple much simpler to compute under realistic circumstances and that has the nice property of banning surprise which means if you minimize free energy this is a hard upper limit on how surprising the world can be to you. So this is as good as it gets really in terms of practical inference and the idea of calf freestanding is that uh neurons do that they

minimize version 3 energy uh over models of what is going on out there at the scale of the brain and within the brain at the level of lower neurons on the neuronal hierarchy. So uh assuming that there is indeed nested layers of terinal activity uh that's organized as to be able to infer sensations and predict what's next uh then yeah there is a strong case that the brain could indeed be able to understand the world and navigate it in the way we do. So uh this is very abstract at least for people that are not uh already into neuroscience. I will ask you to do a very small exercise uh to get better grip on the idea. There is a text on the screen that says please do not look away from the screen. I will ask you to look at this text and not to look uh away from the screen until inserted other ways. And while very carefully looking at your screen and not elsewhere, I will ask you to pick out something in your peripheral vision that is outside of the screen, something around you and uh something that you believe you know what it is and then while still looking very intently at the screen, try to uh retrieve specific details uh from it. For example, I can see the box to my mic. I don't lost my mic. Uh but what I can see now is very much nothing. I can see it's very square. I can see white. I can see there is dark in the white. But practically the sensor information I can retrieve is much more limited than what I need to infer it is the box of my mic. The only reason I know it is is because I knew it is before I looked. uh because I know I put it there and because there is only so many things out there that could be white and placed there. So you can now look away from the screen. Hopefully most of you will experience the same thing I did. Uh pick out a thing, realize that you just do not have the sensory information necessary to make this inference. Uh at some point you had necessarily some prior model of what was going on that allowed you to recognize uh what it was. And before I look the exercise, it seems seamless. You just knew what it is and that is at core uh what predictive processing uh says. So I forgot a bit of a slide. This is a phenomena that is well known to philosophy of perception and to neuroscience and it has been for a while. So this for example is a rabbit duck illustration from pagander blatter I do not know how to pronounce this that was made very famous by the philosopher vinstein using it to show the um let's say duality flittiness of human perception if you are like most people uh you can see a bird a duck specifically or you can see a rabbit but you cannot see both and the point I believe vidgenstein was making and the point I'm making now is that it kind of goes to show that perception is not purely information processing. It is uh judgment. It is decisions. You can apply the concept of a rabbit to this and you can apply the concept of a duck to this and you will just see a rabbit or just see a duck and this is something that will be constructive of how you engage with a thing at a very very basic level prior to your conscious

experience. So uh let us get in a bit more detail about active inference. active inference is that it is the idea of predictive processing as I presented to you. But because of mathematics I will not get into uh uh freestone and most of us here believe hold that not only agents engage by predicting uh what causes their sensation predicting u what is out there and cause you to see this white blob on the wall or whichever it is that you picked. We also select actions uh the same way. The idea would be that we not only predict sensations but also actions that we uh maximize the basian coherence or minimize version of energy over both and that therefore both uh perception and action are guided by near optimal prediction of ongoing sensory motor flop. uh this idea I believe is very powerful and this idea I believe solves lots of very foundational problems in the cognitive sense. I could go on about this for hours which we do not have so I will not do that but I will tell you one thing as an engineer and former engineer physicist technically physicist I'm interested in models I'm interested in mathematics and um something that is very important in most fields of engineering at least signal engineering uh automatics this kind of thing is inverse problem if you have a forward model you know exactly how hypothesis would shape the Uh maybe you can run uh this through and you can know exactly how a given object would um sorry how would your given object would look like to your perceiving robot or something. But uh knowing what object is out there given the array of vectors that your sing robot would see requires the inversion of your forward model and for reasons I will absolutely not uh get into. It's not practically possible for the wireless category of models. Uh let us take a look at this vehicle quickly and this unlin again. Sorry about this. This is a painting. I believe you can see a shape that is orange under a dark background. I do not believe you could practically recover what the painting is unless you know what it is uh from this picture. And actually I know you can't. The painting is this and okay linked again. Sorry about this. The painting is this William Blake ancient of days and just because of the nature of the process which is a very simple blurring process. Well, you can know precisely how this picture will be affected by blurring but you cannot recover the picture from blurring. And this kind of problem is the kind of problem that we have to resolve every single day. Every time we try to pick up perceive something to decide on what our plan and there is no way that we brute force it uh we have to work around the problem somehow and this is what active okay this is what active inference uh proposes that we do this is at the most basic level why I believe that inference is a correct model of the world maybe not like you know specifically precise in the most metaphysical way, but it seems to capture something very important, which is how we use prior information and we use selfization in the brain to



circumvent the inverse problem by anticipating what it is that we will do and what it is that we will. So let's hope that it did not link. It did not link. It is incredible. So what this means for our experience of life as active influence agents I believe you have a you know experience of life as active influence agent because I believe that we are that but we do sense we need to put words in things and I would say the most basic level if active inference is verified then we experience the world as a landscape of agencies affordances being a technical ecological psychology term that says actual or perceived opportunity fraction. So for example, this is a very nice picture of of a path in Galysia that's towards a archeological site we will not get to see. Sorry about this. And if I removed it uh briefly you would remember mostly there is a path and there is green because the way it is structured uh it is the possibilities for action this image and tail that you know jump to eyes and so you see the path and you don't see that much else. And in addition to that, if active inference is verified, uh there is not only the prediction of uh states outside you, which is called exteception, but also of the states inside you, which is called interception. And because the two are predicted in a dual manner, just like because sensory states, active states are created in a dry manner, there is a very efficient use of context, there is a very efficient way in which our bodily perception will shape what we look for there. If it's thirsty, we look for water, etc., etc. And uh all of this again runs at a very basic level uh far prior to our conscious experience. And this entails I would argue the dissolution of agency as we know it. The idea that there is a little thing inside of us that has opinions on what we should do and that you know neutrally analyze world out there and takes decisions for us. It is not the case. Uh the world as we experienced it is loaded with value. It is loaded with affordances and a lot of the processes that shape our activity exist outside our conscious grasp. We flow in uh practical terms along our predictions of what the world is our world bundle and our prediction of what it is that we do our self model. And this is a very fundamental insight that has many application to visual sciences which I hope we'll discuss later. Here I will only uh take a very quick look at what was canonically done uh to relate active inference to the social senses. Uh the first paper that I believe qualifies as social scientific is this paper by Brunberg and Vidvelt that analyzes that says basically what I just told you. If the acting story is verified then we experience the world as a field of currencies a field where we can do stuff and due to versionization we strive to maintain an optimal agreement and it happens to be structured by social and cultural resources. So this is the basically everything that active inference sociology people did relies on that framework skill skilled intentionality framework. Uh the first major paper I'm aware of is the Ram title cultural affordances that uh tries to get

into more detail about how we can specifically integrate and perceive such uh culturally and shaped differences. And their argument is that there is a thing called regimes of attention which is way of paying attention. Wave of cues outside out there matter and what sense to give to them and that uh the action of those regime is basically forced by selectivity. So when you perspective you follow along regime of attention that thereafter you can mobilize to uh pick out cultural affordances. Uh from there uh there was the thinking through the minds framework that basically agrees with the uh prior ranch paper and that's uh further opines that the transmission of culture as in a sense of human specific culture. There's a lot to be said about animal culture. This isn't what we're talking about. Is enabled not only by participation in social activity that are embedded with regimes of attention, but also and perhaps most importantly by the ability to think through other minds by the attitude of when we are confronted to a situation to a problem to a thing that calls action. uh wonder what would someone else do instead of us which entails a coupling mechanism between ourselves and other people that's mediated by expressive capacities and that essentially entails a runaway evolution of uh culturally embedded cognition and then there is the paper by Mu and me automated narrativity which is interested more in the role of material culture in uh shaping humanity and our argument is that because of the normative nature of perception and direct inference because of this affordances thing. Metal landscape are a direct participant in cognition by the fact that they constrain activities towards certain perceptions and actions which is uh the essence of you know cognition. if the version of attention picture is correct and which is the basis of behavioral regulation and also there is other uh more out there stuff I did where I argued that such constraints over regimes of attention collectively constitute forms of forms of life strictly speaking biological things biological if not biochemical that are embedded in patterns of human activity I don't know if we will get into it a lot because this is perhaps a bit abstract but I thought this was a important thing to say because if it's true we have two level autonomy ours as biological individuals and collective autonomy of this form of life that we collectively enact. So uh let us run over core message core ideas that are relevant to social science very quickly. Uh first uh active inference if it is correct validates some ideas about uh within s science. It basically provides a framework that proves that if our nested recognition is correct then this is true. It also grown society in the sense that it provides a way to do very specific models and very specific predictions according to those. So this is in my opinion a pretty interesting thing if not relation of direct constraints. So the first thing that active inference I would argue validates and grounds is daily that social structure. It is not

something that you know exists out there. It is embedded in systems of cultural meaning in way that we have to attribute well meaning to stuff out there and those systems exist prior or within our basic phenomenology or experience of the world and this idea is a very central idea to anthropology to at least second half of the 20th century anthropology because it was popularized by Claude Lévi-Strauss within his anthropology structure. The second idea I think inference, validates and grounds is that and I need to be on the slides is that social interaction is embedded in such systems of cultural meanings in the sense that those system are constrain are constraining but are also expressed within social interaction and social interaction is not only you know determined by such system of meaning but it also participates in making them evolve and this is I would say uh very broadly speaking the point of symbolic interactionism which is another of the major mid 20th century sociology school. Well, this one is a both sociology and topology actually. And third, uh the same applies to matter landscapes. The way in which our physical reality is organized, it signifies stuff. It constrains our attention. It participates in uh you know constructing systems of cultural meaning that may or may not be uh strictly speaking life forms. So uh this is it once again we want uh this session to be dedicated to inviting people to collaborate on the project of uh building active inference for visual science. And so I will leave you with while with these two QR codes that are respectively a link to the discord channel of our project and a link to the course that some of us me and myself organized I think three years back to give a much more exciting perspective on what we can do with France than we'll possibly do during this round table. So I thank you for attention and now we'll uh go towards uh this presentation of the different participants of the round table. Uh starting with Tarius and I hope you got the QR code because I will stop sharing now. Okay. Allow to share. No, no, that's not right. >> Excuse me. >> Excuse me. Sorry. Uh, I got to I think I hit the wrong thing. Uh, wait. Okay, hang on a sec. Um, share screen. Allowed to share screen. Where is it? Allow to share window. Why is this not working? I'm having some trouble here. Sorry about Oh, wait. Show all windows. Here's There we go. All right. Where is it? There it is. Okay. Can you all see this? Hello. >> We cannot see anything. >> Oh, thank you. >> Um Oh, share this window. Hang on. How about this? Can you see this? >> Yes. Yeah. >> Okay. All right. I'm going to get started. Um, is everyone ready? >> Yes. >> Okay. So I love uh Avil's presentation and I have a you know I've have a lot of work that bears a close affinity to what he was talking about but today I'm going to talk about a new kind of new direction in which I'm taking active inference in my thinking uh and it's uh about um I think it provides the resources to uh critically interrogate some uh very uh old and classic assumptions about

human thought and action. embedded in the European philosophical tradition and offer an alternative to them. So intuitively there is this distinction we make between patterns we find in nature. For example, laws of nature that science finds and patterns we make um uh cultural phenomena mainly music, architecture, dance, technology, uh language and so forth. And this distinction has a very old provenance in European philosophy. Aristotle made a distinction between *theoria* which uh is a uh capacity to discover patterns in the world and praises and poises which is about making patterns. Um and he even he even hypothesized two different cognitive processes. One theoretical the theoretical syllogism one the practical syllogism underlying these two capacities. This has been taken up in cont uh re more recent European philosophy in terms of the uh distinctions in direction of fit. So some cognitive acts and speech acts have a world-to-word direction of fit. What does that mean? That means if there's a discrepancy between what you say and the world, you alter the world in some cases. But in other cases, there's a word to world direction of fit where you alter the word to uh respect the world. So uh Elizabeth Anscombe has a very memorable illustration of this with the grocery list. Suppose you have a list um and you're doing your groceries. Um there the the word is d is regulating the world. So if you don't have what you want on the grocery list, if there's a discrepancy, you change the world in order to match the list. But you can imagine a private investigator following the person around and making an identical list. Um but there the direction of fit is in the opposite direction. The the list is modified in light of what the person puts in their shopping cart. Surl uh uh uh marked this distinction as the distinction between declarative and imperative speech acts. Some speech acts uh declaratives aim to track the world and if they don't track the world it's the speech act that has to be changed. Other uh speech acts like commands imperative speech act aim to change the world and if there's a discrepancy it's the world that has to change. Um this has been applied to models uh classical model cognitive models of the mind where there's a distinction between belief modules and desire modules. Uh Hume is very explicit about this distinction. Uh Peter Kurthers is a contemporary example where belief modules aim to track the world and desire modules produce states that aim to alter the world primarily uh the body. Now the interesting thing in uh active inference is um that the proposal is the brain doesn't work that way. Here's Andy Clark's uh very uh interesting recent article where he writes desires and motivations are fluently accommodated within the unifying predictive processing schema where they emerge as webs of prior beliefs that sculpt probabilistic predictions some of which become positions so as to bring about actions. Importantly, a single construct here plays the role of belief and

desire. So, it's a departure from this classical dichotomy between uh states that have a word to world direction of fit and states that have a world-to-word direction of fit. Um, now what I find interesting about this is it can make sense of certain um uh um phenomena, linguistic and cognitive phenomena that don't fall neatly into either category. And one uh such example is the concept of ideology which has gotten expression in uh any number of thinkers including Marx, Fukco, Nichze and Freud. But basically what I want to suggest is a a kind of quick uh fast def definition of ideology is that it's pattern making and masquerading as pattern finding. So we we uh think that we're discovering a a pattern in the world but actually in believing in that pattern we are making the pattern where it it becomes a sort of self-fulfilling prophecy. The pattern wasn't there but by believing that it's there it becomes true. So in ideology, the basic idea is we the you promote the interests of the powerful by concealing true patterns with masking patterns that victims unwittingly work to reproduce at least partially. So you might think sort of the Christian um belief that the meek shall inherit the earth or God will redeem our suffering. these uh help make patterns um rather than track patterns that are there independently of these beliefs or uh example from Fuko it's only natural for men to desire women again it's pitched as a description of a pattern but it functions more to make a pattern true okay so I think active inference is a really nice way of making sense of a phenomena like ideology uh and that is because as a pointed out. Um what active inference says is that all that matters is that the discrepancy between a pred prediction or an expectation and an observation is minimized. But there are actually two ways that one can minimize uh these discrepancies. One can adjust the prediction so it better fits the observation or could one can adjust the world through action so that it better fit uh fits a prior prediction. Now that uh raises the following question um why don't we just adjust expectations always adjust our expectations in response to differences um discrepancies between extterceptive feedback and expectations. Um well the idea is that some expectations for example those related to fundamental requirements on survival like uh the your body temperature or levels of blood sugar are sticky and the only way to minimize the discrepancies with relevant feedback is to act. You can't adjust the level that your blood sugar should be at or the level that your body temperature should be at. The only way of minimizing the discrepancy is through action. So as Clark puts it, bringing about action requires attenuating, that is assigning low precision to the sensory information currently indexing the actual disposition of the body so as to enable precise propriceptive predictions to prevail. Intentional action thus depends upon a delicate balance that combines precise propriceptive predictions with attenuated information concerning current bodily states. An

interesting consequence is that action thus depends on a kind of systematic misrepresentation of how our body is currently arrayed in space. I emphasize that because it's very similar to the idea of ideology. Ideology is a systematic misrepresentation of the way of the social world but it leads to a kind of alteration of the social world to more approximate it. So one of the interesting consequences of this picture is that we can replace this classic pattern finding and pattern making uh disjunction that we saw in SURL and Anskam, we could replace it um with a more nuanced um continuum. So here rather than two categories into which we could place thoughts or or judgments or sentences, we have a continuum that and anything you think or say can be placed on different places in this continuum depending on how sticky it is. That is how likely are we are to hold on to the belief no matter what and the prediction error that you get. So diet doesn't impact health. Presumably that's pretty we're willing to revise that and if the prediction error is high as it is in this case but something like life has meaning even if there's a lot of prediction error um we we can't help but hold on to it because it's very sticky and but it can vary for individual to individual or occasion to occasion and it's it's um a continuous variation not a discrete one like in the classical understanding of these things falling into one of two categories. A final area that I I'm really interested in applying this to is an another ancient tradition, an alternative to the European philosophical tradition, the Buddhist tradition. This is Amber Carpenter channeling Christopher Gowens's uh talking about the belief in the self. So she writes the false belief in a substance self that there is a stable entity distinct from phenomena with which to identify leads to the sort of mental activity of grasping and aversion that creates generates and constitutes the continuum that is what we in fact are process cells. So falsely taking myself to be substantial generates the kind of thoughts that keep keep my process self my continuous self going and more broadly u I'm not going to read this uh citation out of interest of time but the I the very Buddhist idea that desires help construct our our reality is um I think very compatible with the active inference framework. The idea is that we divide the reality up into the chunks that we see based on what we want based on our preferences and uh if we didn't have those desires we might divide reality up in a different way and I think that fits very nicely with the active inference framework. So anyhow that's the direction I'm taking it uh these days and I hope you found this interesting. >> Yes. Uh thanks a lot. Uh that is uh so now yes let us move on to M and then we'll move to Reish. >> Thank you. So I think the talk I'm about to give is somewhat congress in the sense that uh you both discussed what it is that active inference allows us to do and and how it allows us to frame how we carve reality. But I think I'm about to talk about the

mechanism at play when we talk about the ways in which our perception overlays with group dynamics and then um shapes that reality itself. Um and I'm going to go rather relatively quickly to give Andrew and Remy a chance to um talk as well. The core idea here is that every adaptive system biological or social operates by minimizing uncertainty. And so active inference starts with the individual. So the self model maintains coherence between internal states of the world, internal states and the world. Uh but because no one can model the entire world alone, cognition extends beyond just the individual. And so uh to reduce uncertainty we extend our cognitive boundaries by coupling our inferences with those of others. And the shared modeling of reality depends on uh shared prior. So overlapping assumptions about what it is likely uh meaningful or true. And so um trust here arises when we assume that others self-evidence as we do that they also act to maintain coherence between their predictions and the world. So through this mutual self-evidencing, individuals become part of each other's generative models, they enable social cognition, uh the the the ability to think together, to coordinate and to construct shared worlds. And so these distributed inference loops become the foundation of social identity and group belonging. And so self-esteem here functions as a precision regulator, a dynamic estimate of how well our internal model aligns with social feedback. So when prediction errors are low and feedback is affirming, confidence increases. And when errors rise and feedback turns negative, self-esteem decreases, prompting potentially self-correction. Um overconfidence and shame emerge when the system miscalibrates precision. So over confidence inflates certainty discounting error signals and preventing learning and shame amplifies precision on negative feedback overweighing error and suppressing action. So in this sense self-esteem is a sociometric gauge of prediction error keeping our self models coherent with the social world. And so through this continuous calibration, individuals align with collective expectations and they form more stable social groups. And so social groups can form around shared expectations or beliefs that collectively minimize uncertainty. And so over time, confirmation bias reinforces these shared priors. individuals selectively attend uh to evidence that supports their existing models and ignore what contradicts them. It's it's uh it links to what Tadu was saying about motivated reasoning. And so this feedback signalizes social identity but also traps the group in local minima um of of belief space the the states that feel coherent internally but are resistant to change. So within these echo chambers, prediction errors are reduced not by updating beliefs but by filtering experience itself and such stability preserves belonging at the cost of flexibility. So the system maintains low surprise through mutual reinforcement rather than genuine adaptation. And so

once we have these shared beliefs, once they are stabilized, um power and influence can determine whose priors guide the collective models. Power here emerges when one agent or an ideology uh or ultimately a group can modulate the priors that guide collective sensemaking rather than controlling actions directly. Influence operates through the precision weighing of beliefs by determining what counts as credible, relevant or true. Leaders, narratives, institutions steer collective attention. They um shape how groups interpret evidence and update their models of the world. Under this view, power is the capacity to steer collective belief updating. It is to define the boundaries of what can be imagined or questioned. And so the group's generative model thus becomes a shared cognitive architecture governed by those who set its priors. And this hierarchical modulation scales from individuals and small groups to entire societies. But finally um we can express cultural, political and interpersonal processes in terms of shared inference. And we can formalize how meaning, trust, and coordination emerge without producing their complexity or richness. The goal here is not to necessarily mechanize society, but to make its hidden computations visible to describe how collective sense making evolves, adapts, and sometimes fails. And so we can use this to design systems that learn and align together. We can apply it to artificial intelligence but we can also apply it to policy large and that's the direction I hope to take here. >> Thank you. It was fascinating. Remy, are you ready to share? >> Yes, I'm going to share my screen or attempt to do it. Okay. Um, can everybody see the presentation? So, um, yeah, I'm just make it full screen. Okay. So, um, yeah, I'm going to talk about, uh, things that are really similar to what Tad presented a bit earlier. Um, so, uh, yeah, I'm just, it seems that it froze. Sorry. Okay, I'm just gonna Okay, so can everybody see the presentation? >> Yeah. Okay. So uh the what has drawn me to active inference is really something that's similar to what Tad was talking about which is uh this kind of um questioning of traditional distinctions uh which have really ancient origins in accidental philosophy and have been represented also in more con um more contemporary philosophy of mind. So this distinction that Tad discussed between what is sometimes called epistemic attitudes which have the function to accurately represent the world. Uh so things like beliefs and perceptions and uh on one hand and on the other hand prottitudes which uh like intentions and desires which have the function to change or transform the world. Um and as Tad explained there's no such distinction in active inference. there's only predictions which can be more or less like beliefs more or less like desires depending on precision modulation as Tad explained and so I think this has a potential to explain to um eliminate a wide variety of social phenomena as Tad explained I think ideology is one of them but what I want to focus on here



is the phenomena phenomenon of social norms so what are social norms norms. Social norms are simply behavioral regularities that are socially maintained in a group. So simply these are patterns of behavior which are um socially maintained which means that uh members of a group will behave so as to incentivize reward behavior that follows the norm and disincentivize or punish behavior that infringe upon the norm. And so uh social norms are just like patterns of behavior which are enforced through uh patterns of social sanctions. And um how do these social sanctions uh come about? How do they emerge? Um I think that uh the active inference framework has a lot of uh things to to say about that. Um the idea as I said earlier is that in the active active inference framework there's no strict distinction between beliefs and desires. So uh when an agent predicts something about its world, it can either alter the the prediction so that it fits how the world is or it can also alter the world itself so that it fits what it model is predicting predicting. And so in a social group uh there will be various kinds of social regularities, ways that people tend to behave in in this group. And so uh we can expect the members of this group to learn these social regularities. So these the models of these agents will tend to integrate the these social regular regularities they will tend to form social predictions that predict these regularities. And um when uh uh uh these predictions are fixed in the the model uh you can expect uh something like social sanctions to uh to to uh appear because uh as we said earlier uh when these social predictions are well uh well confirmed well placed in the in a model um the the agents will not simply uh um update their um their social predictions. concerning the environment. They will also try to minimize social prediction error by intervening in their environment so as to confirm these social predictions. And so we can expect members of a group that have learned uh social uh regularities of their group to enforce these social regularities. that is we can expect them to uh start behaving so as to produce in their social group the behavior that follow the norm that follows the norm and uh to produce behavior that punishes uh behavior that uh that violates the norm. And so uh I think that this uh erasing of the distinction between belief and desire to have just one notion of prediction can help us explain how social norms appear in social groups and how they stabilize in social groups because members of such groups will attempt to minimize social prediction error and not only by revising their own model of the world but also by actively enforcing these social predictions on their social environment. I think so that we have now better tools to understand how social regularities in social groups can become social norms. So that's what I had for today. Thank you for listening to me and uh yeah I think it's close to what dad was saying about ideology but it's kind of a spin on the notion of social norm. >> Okay.

Uh thank you Reie. Uh I will do the next one if I can share my screen. Okay. So this is uh you know I think it was originally a 45 presentation. I will go through it very quickly. So I the thing I want to talk about is uh my contention that we can actually make things up and it matters which I will explain uh very shortly. So uh first uh the way that we engage with the world displays uh very clear signs of what we call opening evolution. We can make new things up that do not seem to follow very clearly from published law principle. For example, we can build a city and then there is an entire new level of organization that follows its own rules that are constrained by but not strictly by prior law of physics. We can make signs that mean things and because we treat them as meaning things, they do indeed seem to mean things. uh that at least they work to signify stuff and they work to constrain mission and so there is a I would say very very very very strong case and a quite trivial one that that we can build uses of the world that causally matter and uh there is an entire sub field of physics that yes >> well I'm sorry I think your presentation is um >> frozen thank it's back >> and so I will not get to detail but uh it's hard to explain physically because in physics we tend to have specific list of states that define a system and you know we can't just tell to the math to invent new math in the middle of doing math that does not work. So most of it is what I showed you about inference. What I want to get at is the following uh contextuality in life and cognition. There is a very basic concept in quantum physics that is I think it's simple but uh it's the reason why everyone is so confused about it and it sounds very weird. That's called contextuality. So at a very basic level I don't really want you to read that but I want you to know that the math exists. What's we the kind of system we study in quantum physics there are vectors and there are vectors that are kind of like probability distributions up to uh squaring but whatever and when we do an observation what we observe is not a vector but uh on value um multiplicatively stable value of the vector uh given the operator. So if I look at energy, I can find any of the level of energy that are defined by the operator with which I look at energy. And what this means is that um the physical act of observation in teran gladly it uh I don't have much longer to talk. So the physical act of observation uh in a contextual theory cannot be understood to reveal appreciates even reveal up to noise. It brings them about and there is a quite simple uh way to show it from a thermodynamic perspective that you would know about if I could make this slideshow work. uh which is that if you want to observe an object, you're reducing the entropy and the entropy has to go somewhere because we live in a bonded universe. And so every time you observe some information, you destroy some information somewhere else. And this is something you cannot escape if the universe is bound by the which I will very much assure it

is. And my contention here is that uh no is that uh human perception is in fact uh contextual maybe not quantum. I do not have the contention that our brain is a quantum system but at the scale of functional activity where which is relevant to explain and understand our behavior. It is contextual and the duck rabbit is a case in point. You can observe it as a rabbit or as a dog. You cannot whichever would be a neutral process of observation that would naturally constrain uh action down the line. You either perceive it as a rabbits or you don't. And uh in the same okay in the same way uh when we observe uh material landscape here uh pathway in the forest uh we can observe it as a pathway we are actively projecting context uh that are defined by the priors with which we look at situation and the priors in the rabbit duck case is that there is such a thing as a duck and such as a rabbit and they are different things and the prior here is the fact that we are physical bodies which walk upright on feet and can walk in a pathway but not through a bush. Uh mice would likely not see the picture as we do if you could get the mice to look at the picture which is an entire other story. And this this is nerve-wracking. And this becomes very important uh when we consider uh the constructive nature of uh cognition and of well uh the communication system that we build. We can just decide to that something is a letter and then we can use it to communicate. We can just decide that something is a word and then we can use it to communicate. I can just decide that this gesture means okay and this gesture also means okay and another one I will not show means I don't like you. We can decide this and then we think in those terms and then it becomes actual determinance of social organization and the entanglement of this if I am uh correct in the broader scheme of things is that uh there is a process uh by which we can observe uh reality observe social reality and material reality in a way uh that basically tells us what the constraints over what can perceive are uh what are the letters what are the kind of path you can go down and the kind of path that no you cannot I assure you you cannot do that and uh because this process is not uh you know purely patterning transmission of pre-existing ideals because it is something that requires trial and error and misunderstanding uh it's enables the possibility for open-ended evolution and this is this uh you know tension between what we want the world to be and what the world is uh that enables that and moreover I would claim that if this thing is related we can understand sociality at least the kind of norms that was talking about but more more broadly the constraints that material and social uh context averally described as constraints and they can be formally defined as a closed system of constraints where every constraint brings about other and is brought about by the other and that's what uh at least some corners of the elective literature called biological autonomy. So if

this is correct then there are circuit life forms and there are several layers of that are embedded uh in humanity from the level of basically down to the level of ourselves and up to the level of judges parliaments uh discussion about whether social organization and this is I would say the core idea I'm complaining for. So, uh, thanks for your attention. I hope it was a short talk. Uh, it's it's a your turn, Andrew. Okay. Can you guys see and hear everything? Okay. >> Mhm. >> Great. Um yeah, thanks. So um wasn't quite sure what to expect from the round table. So I did not give any kind of like redux of active inference. Um I've done that a lot uh over the past couple years. I'm also going to set a five minute timer for myself. Um but um so this is effectively just like uh kind of a a snapshot montage of some of the things that I've personally been working on. Um, and it starts with uh hopefully this maybe fills in some gaps regarding something like affordance sharing for anyone who watches this. Um, so there's three slides total. I always overpack them. Uh this first one is essentially a summation of sort of what I did last year at University of Pennsylvania, which was uh give a three-hour tutorial somewhat Ryan Smith style for anyone familiar with with his work uh on uh creating agent-based models uh using the active inference framework, but now in more of a uh social science focused multi- agent setting. Um I think from my perspective of the field there's a lot more work to be done uh on multi- agent agent-based modeling especially with applying active inference um specifically or even just any kind of uh sort of cognitive or autonomous uh agent model in general. Um during that uh 3-hour tutorial, I kind of make this this narrative, this this argument um that seems to be happening in computational social science where um those who do engage in uh agent-based modeling are repeatedly coming back to ideas of what is cognition and uh what what does it mean for an individual to to have beliefs? And so whenever it comes to doing agent-based modeling where traditionally due to low computational resources and power, we've largely relied upon very simple conditional rules that agents follow uh whenever we actually simulate their behavior. Uh which you know there sort of the justification for that is well one we don't have the computational power. to uh create an agent uh who's who's necessarily um uh able to emulate sort of real world behaviors that we find in other areas of science. Um but then also well if the model is sort of real enough if it exhibits enough sort of ver similitude then um then that's enough of a justification and then whenever you start to imagine trying to simulate say hundreds of agents um you're you're you're definitely reaching the limit whenever it comes to concerns about computational power. So I'll I'll attempt to speed through the rest of this, but this was simply uh anyone who's interested in that tutorial I gave. There's roughly 70 to 80 slides of material uh as well as all

the code for this particular simulation which I taught everyone sort of how to recreate. Um this is a classic uh computational social science uh simulation by Laszlo and Friedman uh at um at um Northeastern and Harvard respectively. And this is a way of looking at agents in a way that um they're composed in networks. And so suddenly network science becomes relevant looking at agents who maybe interact with particular agents but not others. Uh we often think of you know societies or collectives at varying levels. And so basically this this almost provides a kind of sandbox to allow for testing agents in different kinds of networks with one another with different network network uh configurations and have them uh do what we could phrase as parallel problem solving. And so this this kind of boils down to how do agents collectively solve problems which is something that I'm personally very interested in. Uh I think that culture in some ways could be read that way. I think that sociology in some ways could be read that way. I think that if we're able to have different kinds of proxies for I don't know say we we need to have food uh to live and so whenever we organize different kinds of agricultural and distributional systems around food and those become sort of a you know a basis for being able to have a society otherwise it dissipates. uh that you know having those kinds of perspectives of these are agents who are collectively you know trying to maintain their homeostasis or or however we want to phrase that. Um but that's sort of what those are some of the kinds of dynamics that this simulation gets to and then by being able to use active inference agents. You can see how the sim simulation is reconfirmed using the old agent-based model as well. But here we have agents who who who learn who who adapt to the particular network configuration that they're within the kinds of ideas that are being shared. We do find semblances of a lot of the kind of phenomena that uh that I think ma was referencing earlier. Uh here there are agents who they they share ideas with one another trying to find the better solution. But whenever agents are too heavily connected uh with one another, sort of like our our modern world a little bit and thinking about interactivity on the internet, um attention is so broadly distributed that what happens is that agents tend towards uh actually converging towards whoever seems to have the best idea. And then whenever that happens, the diversity of ideas and solutions to a problem starts to dissipate and everyone's kind of limited by the leader at the top. And so if that leader at the top sort of doesn't quite have the right idea or it's not very well reflective of reality then uh what you find are a lot of agents kind of converging towards um you know ideas that might be very much maladaptive or be more focused on the social dynamics between the agents rather than actually solving the problem. So that's a bit abstract but um it's just I what I find very interesting in the framework is uh the rel of active

inference is relating it to neur biological plausibility of getting down to sort of first principles. Um I've done a variety of different kinds of projects. And so tomorrow, which has kind of been my main focus this week in preparing for a presentation was uh this um model that we're actually making in more of a computational psychiatry setting. And it uh we're so I'm actually working with a clinical therapist who's done a clinical study with seven individuals who have PTSD. And this is sort of that how do we bridge the qualitative to quantitative gap in the kinds of data and information we can collect and and then kind of compose and and analyze or interpret in the formation of a study to yield some kind of insight. And so um we're effectively making a model that somewhat um kind of replicates a lot of the behaviors synthetically produced from these seven different accounts of individuals with with PTSD as well as of course a lot of other sort of empirically inclined literature. Um there needs to be in in active inference there needs to be a lot more kind of work on validation of different kinds of models. Um there's been a lot of great work uh already done with like Aumura uh Takuya who's who's done a lot on on validating this sort of at the at the cell culture or or neuronal level. Um and so I'm very excited to see where the social sciences uh can kind of take this next and recreating like broadly social behavior. So with this particular example, this is an agent who um you know analogously exhibits the symptomatology of PTSD, but they're in a social task. And so we're looking at something like epistemic trust. We're looking at uh what does it mean to pursue your own individual goals while simultaneously needing to potentially rely upon others? How do you establish that kind of trust? Uh what happens when trust is lost? What happens actually in the body? Um so so there are aspects of like modeling the autonomic nervous system here and relations between um affect and emotion. Um yet another kind of reference back to some of the things that uh I think Ma brought up earlier with with uh sort of having a sort of emotional sensor or looking at um precision modulation changes in confidence. All those dynamics are here. Um there will be more on that tomorrow. And then finally, this is a little much, but this is just sort of a quick like uh overview of some different kinds of things I'm I'm looking at right now with agent models. Uh I'm deeply involved with the active inference institute where I teach um the textbook uh with uh Daniel Freriedman. I'm involved in a variety of projects. Uh, I look frequently at different computational resources that are being like developed that people could use, figuring out how I can use them and then how I can teach others to to use them in in potentially new or creative ways. And so um you know this is I'm just interested to see like what we can do not just with optimizing agents and systems for accomplishing real world tasks in the way we want them to do but actually

looking at how do agents on their own so to speak kind of interact amongst one another. So um the the upper left this is just um I'm interested in mix models. you know, a lot of people are using uh POMDP models to emulate uh decision-m, but I'm more interested as as someone who my my main career is as a data analyst. I'm looking at different kinds of data all the time. And so I'm interested in like how do we include continuous data? Of course, there are many ways to do that in active inference, but it's just what are the ways that sort of neurobiologically plausibly align with what what we can make, what we can emulate and then what can actually scale, right? Such that we can reach those kind of dreamed uh uh 100,000 million agent level uh agent-based model simulations and what can be fit to real empirical data otherwise. So, uh I'll I'll stop there. I went much went much went much went much went much went much went much went much went much went much went over on time but thank you for your attention on all that >> thank you Andrew uh so we did uh we went a little bit uh of time uh we spent 70 minutes on the uh presentations instead of 45 this is okay uh so to my knowledge we do not have uh any hard time limits now is there a limit on the Google call thank So we don't have any term limit. I don't have any term limits. So like for me it is uh very much okay. Oh what I would like to open for the rest of the session however uh lensing gets is uh for us to discuss uh what active inference brings to a table for social science. Uh it is uh I would like it to be a very open conversation. We can talk about anything that we like like collectively individually. uh just as to be somewhat scholarly. Uh also I'd like us to talk roughly equivalent amount of times each of us. I don't think we have a timer but I guess we can be mindful of that. And uh I would >> excuse me do you mind if I take a quick break before we start because I >> Okay, >> I'll be right back. Sorry. Uh and uh last thing we can uh like if we want to show things we can show things we don't have to just talk there is a interesting graphic we think of we can share it. So uh just to break the ice I'd like to ask one question. Um I would say we are beating upon the contention that there is indeed some value that active inference has for this science. I would say there is a pretty strong case to believe there is not social sense existed for a long while. They have existed with very information about cognition and they they functioned uh this way. You don't need to have a very specific model of the hypocalus to know what symbolic threat is. And uh yeah so what I would like to discuss is your opinion what it is that active inference brings to the table for the social science uh how valable it is what are the terms in which it works. Do we reduce social science to cognitive sense or is this something more integrative? And if we are convinced that there is indeed value to act inferred, how do we uh tell this to social sense people that maybe are not very interested in learning a lot of

mathematics for a pretty questionable value from their perspective? >> Oh, I'm hard balling. Yeah, we don't have a Please go ahead now. Yeah. Um, well, I mean that's that's what most of my thesis motivation section was about. Basically saying why would we want to apply active inference to social science and what's in it for them. Um, and my answer was predict predictive power. That's one of the things that social science really doesn't very much. We have a lot of descriptions. We understand we think we understand things. we can discuss about phenomena and critique it, but when it comes to actually predicting things, we're very bad. Um, and that that's partly because social systems are extremely complex and they span over multiple scales and a big chunk of it is subjective, i.e. not accessible through real objective metrics. Um, so it's it's it's really hard for us to to get a real insight. And that's part of what active inference also gives us. It gives us what we could consider to be a mechanistic map of the processes that we can then leverage to make predictions based on measurements we've taken for years and years and years. Sociology, psychology, anthropology have made a lot of measurements, if not a lot of very good predictions. Uh and now what we can do is okay well now we've got this mechanism we can potentially fit to all those measurements we've taken what does it allow us to say about you know the future at x time scale >> yeah uh so we don't have to raise the ends every time I guess if it gets a bit competitive we can do that but just to for context uh what I asked question which was a bit of a hard ball which was why do we believe that active inference has anything to bring to social science? >> Oh okay. Um, can I should I go now or I just wanted to riff off of what um, Ma Mah said, which is um, just the only thing I'd add is that the other a nice thing about the active inference framework is it it doesn't just give theorists a predictive purchase on this complex domain, but it explains how humans non the non- theoretically can adapt to this complex domain without without having an understanding of it. Right? that that's what I find very interesting is that um because we uh through the mechanisms that I was describing regarding ideology but also me about norms it shows that you could have sort of relatively cognitively cheap ways of um uh of creating uh uh social consensus and enabling coordination making people more predictable to each other without them having to understand how it's done, how this very complex domain uh functions. So, uh I think that's a really nice kind of um uh corollary maybe of of of the active inference framework. Yeah, just um quick comment that uh I think because I'm regularly interfacing with people who are very interested in active inference and trying to figure out how to apply it in their respective field. It's something I notice people frequently kind of flock to is the notion of the marov blanket. whether they're very familiar with the



notion of like you know what it means to have like a conditional independence or not. um the idea of being able to separate out what is a thing uh which becomes very relevant in in social science where we say like what is a society what delimits that society what is inside of it and what is outside of it and I think that uh sort of you know I I enjoy I've enjoyed talking about the notion of the Markoff blanket with others whenever they try to conceptually model out like map like what what is a society what are these different institutions and then if you could reach some point where you could delimit what are sort of the inputs and the outputs or the states and the blanket state uh active states uh in relation to those entities then you become much closer able to actually specifying a system where you're not getting so confused whenever you change different scales right because it's social science is like we'll break things out into a micro scale and a meso scale and a macro scale or others will look at things at different time scales the analog school with with history and Fernandelli that sort of thing uh and I I just think it's Great. But I think that active inference supplies a very strong framework for looking at many many phenomena as opposed to kind of you know side siding with your own like little sub sub discipline silo in how you look at things and then kind of either casting out or getting in a debate with others but not making much progress beyond that without making too much assumptions about other how others operate. Yeah, I mean another thing that beyond the interdisciplinarity which I agree is because it acts as a lingua franca due to its precision and due to the fact that Carl and most of the people who have developed it come from to some extent a degree of the social sciences at the very least cognitive science. So the the language that they use has been nicely couched in this sort of veneer of accessibility um for social scientists, cognitive scientists. But I find that another benefit which perhaps not everybody will think is a benefit but coming from queer studies, gender studies, um is its amenability to postmodernism. Yeah, >> you mentioned it, Tadius, with the constructionism, but I I feel like it gives you a concrete foundation to make those claims beyond just the observation that it really does seem like the concepts have different meaning for different people and they seem to lead to different kinds of conclusions for different people as well. But under active inputs, you can literally couch that under. Yeah, your mapping is different because you've carved reality at its joints differently. And therefore um now that you've carved reality at its joints differently, you're going to sample reality and shape it through your sampling also differently because of uh this this uh joint distribution and therefore the ways in which your boundary coexists with the world and and and forces it into a certain way. Um and and that's profoundly postmodern. And I think it also is some of the the issues we found with

with postmodernism which is okay but now what if you accept that things are relative and and hard to measure that measurement itself is a bit um not facious but uh begging the question to some extent what do we do what do we measure what like what's the future I think active inference sort of answers that question like okay yeah sure we we can map all these different ways of couching reality and it is amanable to a dynamism of perception itself and to cultural um influences. But also we can map things to each other. Like we can understand how perhaps through this anchoring with common observations you have seen something slightly different given your vector. And we're now coming to the point where we can apply things like perhaps uh category the theory, coology, uh and all kinds of mathematical tools to understand how these different things map to each other without having to resort to well nothing's real, nothing's true, we can't do anything. >> Yeah. So I uh strongly agree with what M just said up to postmodernism. might say I'm not a fan of uh postmodern theory for forat reasons. I can stand some fuko but uh that's pretty much it. What what I would say is uh there is a very strong record of social science that try to work with positivistic social sciences uh that try to measure behavior in isolation and then use it for prediction and this record is absolutely terrible. There is no probability for anything. uh most of the time there isn't even a coherent theory behind predictions and at some point uh I will share a thing because such is my power uh at some point uh there is a very strong contention that agency matters uh and agency is if the active influence picture is uh concrete uh more constrained by how we happen to perceive a given situation than by um Craig's mechanism or social structure in a very strong Marxistish sense and then what we are left with is what Gopan calls frame analysis. It is semiotics. It is specific fields of visual science that have studied how we signify stuff and how we understand stuff. And um to me it's a very strong result that if factory inference is correct then those fields are in their basictology correct. the object they posit exists. They are valid construct and we can work with them because such is not true for say social psychology. We cannot just well we can't just measure effects and then expect them to replicate uh outside of context but it did not work. It did not work very well and so active inference seems to be correct from an empirical perspective because again this is my basic contention. we do not have another theory of how we can circumvent the um inverse problem and maybe it's uh formulation is not wide enough it's not it's too restrictive I do believe that the formulation of the fy principle is restrictive it's my one of my horse battle but the basic structure of it seems to work there is no competitor to that and seems to flow from very basic physical principles so I think it's a very strong indication for some tradition of study which do include qualities which include symbolic

interactionism which include the most constructivist interpretation of structuralism and to me it's a very strong result because we can not only say that again those fields have construct that work but also now we have a way to model it that is not based in speculation that is based in very specific uh computational hypothesis that translates the basic physical principle of in cognition this is a I cannot say that this will make and breaks is ridiculous. This is another field with its own theory. This is I would believe a force that is very important to validate and verify specific theories within specific fields of the social science and this is where uh M was talking of lingua frana I think this does that I think inference is a lingua franka it's a way to talk and representational phenomena and so on a bit more negative note I think also we must be aware to not think it is a selfstanding theory of sociology that will uh replace uh this or that approachology. This is a cognitive science thing and it works and it can be made useful for syology but there is also a distinction in methodology and in ontology and in what we are trying to do at the most basic level between the two disciplines. Uh so Remy you did not say anything yet. Do you do you want to wait in on this issue >> specific? Uh yeah, no I do not have that much to add but thank you very much for for giving me the opportunity to to intervene. Um I I uh I I wanted to come back like I'm a mere philosopher so I don't know much about empirical applications of vector inference but I was wondering um so so ma you said earlier that uh there's a potential for better predictions uh in social sciences with active inference. >> Oh yeah. Okay. Sorry the Okay. So yeah. So I do you know if there's like work that are that is it just like we're we're thinking that there might be better prediction or do you know if there's there there's current work in social sciences that use accurate and that yield better predictability uh do you do you have any I would be very very curious to see if we have this additional benefit of active inference in in the social sciences. Um yeah, anything that relates to social systems I know but it's generally used in industry right um so for example the company versus which I'm part of uses it to do market predictions um I know that Connor had done a lot of simulations for social groups one of our simulations was about predicting opinion dynamics within um within social media um we're being approached by multiple companies to do say um marketing trends etc. So it's it's just that um I think what's missing a little bit um is heavy work on empirical validation. Part of the benefit of active inference is it can be leveraged by one person. So you can get into your computer make a little simulation and it has some degree of validity because you can reproduce an existing phenomenon. But gathering a lot of data is really expensive. So we're still missing large scale studies of the kind. But Ryan Smith is doing such studies right now. Um and he is getting predictabil predictability. So it's

not that it's not happening. It's that these are the kinds of coalitions that take a lot of funding, a lot of coordinated efforts and ultimately the things which social sciences are good at which is taking measurements of given social groups. >> Thank you very much. >> Might just um briefly add to that too that uh I agree with a lot of mouse points. there does need to be a lot more uh empirical data collection. And so what I like just coming from the kind of computational social science side uh I see the strong division between those who would use something like agent-based modeling and recreating dynamics uh which are some of the things we've kind of talked about here today and then on the other hand there are those who are only looking at very large like kind of big data level volumes of data um textual corpus uh looking at a lot of um downloaded X or tweet data or otherwise. And so you get this weird division between like natural language processing over here versus like kind of recreating essentially dynamics that come out of physics um numerically. And so I I would love to see more of an integration between those two kinds of perspectives. I can't say precisely how that should happen, but I only say all of this because I think that there are a lot of social scientific theories that maybe haven't been able to see their day per se because someone hasn't been able to simulate something or test it on data or otherwise. So, I I think that active inference here at least gives us the framework to start doing that. Um, again, we just we need to really build out the the toolkit, uh, so to speak, and and get more people involved and feeling capable that they can sit down at their computer, give something a shot, and publish their results so we can keep this going. >> Uh, yeah. So, I would, uh, want to get back on what you just said about based modeling versus, uh, big data analysis. I would say that act in France gives us a very strong ground to believe that uh sorry me could you cut the audio I can hear people behind you I think >> thank you uh so I would say it gives us a very strong ground that cognition is contextual in a practical sense that it depends on context and uh therefore I would place much more faith into detailed agent based analysis of specific phenomena to reconstruct I don't know dynamics of influence ritual of power in network etc etc. Uh then we could have in a very large physical analysis because your stats will be based on a very great variety of context none of which understands and for which you have no appur reason whatsoever to believe that it generalizes. Uh so it's a problem with experimental methodology. We never test for erodicity. ever test that the uh effect that we measure is invariant between populations and I would contend that most of the time it is in fact not and so I think we I think we need to move toward much more detailed qualitative analysis of uh how people think things and I think we need to move to models approaches to computational methodology that reflect that and that's

more in the agent based modeling brand than it is in the data analysis uh strictly speaking you're still muted. >> Can't hear you. >> This is a different topic. I'm going off a bit. Um but in in a number of the presentations people talked about how um active inference can explain why um non-adaptive or maladaptive belief systems are maintained I think especially in Mao's talk but I'm wondering what the definition of adaptive and maladaptive is in that context because of course for the individual it's adaptive because you need to um respect the norms of your community or the the prevalent ideologies in order to succeed in that community. So are you in you know is it um is the idea that the group becomes the the group becomes the um unit of analysis here. So that uh it's it's a maladaptive group because um in the long run say the ideology gets the group to deplete natural resources at a at a rate that threatens its sustainability or something like that or is there some kind of normative um background that's being presumed in claims that these beliefs are maladaptive where you're just saying they're just wrong. it's wrong to uh to treat women differently from men or something like that. So, uh yeah, I'm just wondering what the thoughts your thoughts are on that those sorts of questions. >> Yeah. So, I've given this actually quite a bit of thought. Um I don't have a formal answer, but I have a thread of research that aims in that direction. Um and I think you have to think of it in terms of paths and scale of paths. uh and something we're developing with David Highland and Dalton Shakivel and Alex Kefir uh and other people from uh the empowerment literature is and and Ben Girtzil as well there are paths on a given time scale which are irreversible. So, for example, uh if you jump off a cliff, even if you're in the middle of jumping off the cliff and you're not dead yet, effectively you can't come back from that. That's that's an irreversible irreversible path. And so whilst not everything is as um obvious as that, you can sort of couch your paths a little bit like um like a like fineman uh paths where you can see there's a probability distribution over certain paths and then uh the the probability distribution means that somewhat they're they're most likely connected, right? There's a there's a probability of going from one path to the other. closer they are and the closer they are the more likely they are you know to be um uh under the same distribution. Uh and so before knowing what a good path is if we apply some normative aspect to it there still are some paths you know cannot be returned from and some paths you know have a a high ability to be um flexible so you can go from one to the other. So assume you don't know what a good path is. You still know that if you go in that direction, you're most likely to be able to switch. If you if you were wrong, then if you go in that direction where you will not be able to switch and if you're wrong, you're you're you're a bit And so that's that's the difference between like a rigid

path and a flexible path. So we can sort of take this as a sort of metric that doesn't have any veilance to it. Now considering scales of interest let's consider that on my life alone I can make a judgment of suffering say based on suffering well okay well my life will probably suffer less if I max out on one thing but if I look at a larger scale and say my children or my children's children or whatever and that's usually the uh the kind of uh thought process that um long- termists take um yes I could max about my benefit through like convenience uh reality that we take right now but I'm depleting all the resources for everybody else and so again this notion of path yes my path is still somewhat flexible but on a larger scale um that path also entails that their paths are now extremely rigid they have no other possibilities afterwards because their resources are depleted so you have to consider scale of concern as well and then the last thing is functionality like it is true that um some beliefs can be maintained without losing a degree of functionality if you are not actively acting on those beliefs. So there are scales of beliefs that are differently related to your day-to-day life. And I'm going to use an example and it's that it's potentially um um sorry controversial, but if you believe angels exist and that if you jump off a bridge an angel will catch you, you will die. No angel will catch you. We can test this like it's it's testable, but if you hold that belief with high precision and you take some actions based on it, you will die. But there are people in the world today that hold a belief that is somewhat related to that if not that. The thing is they sort of shelter themselves from areas where they actually have to act on those beliefs. So if you believe the earth if you believe the earth is flat, you probably shouldn't be an aerospace engineer. >> And it so it turned out that most of them aren't aerospace engineers. So they can maintain such beliefs in a form in a sort of normative way without them being dysfunctional. But that means they are limited to that sphere of reality. So all of these three things together I think lead us to beliefs which can be more or less maladaptive regardless of the cultural context which would qualify certain beliefs as adapt as adaptive. Then you can talk about just existing within a social context. Relative to a social context, walking around naked is maladaptive. Now, it holds no value other than you will probably incur social um social consequences along the lines of what Remy was saying. And if you live in Canada, you'll probably like me have issues if you walk outside naked. But if you live in, I don't know, somewhere really warm and nice, then the only the only issue is social. So perhaps you're only maladaptive relative to the current um context of the group. >> Could I just quickly respond to that uh before people go? Yeah. So I I really love that and I especially like the first idea that uh it's uh without committing to any particular normative uh preferences it's better to have open paths than than

committed paths is kind of fallibleism. It's a very minimal sort of um and plausible uh constraint on on uh what is adaptive certainly. But uh my worry is that there might be situations where just false beliefs basically are extremely adaptive because our most important decision- making is social. And so we uh it's not just a matter of you know you might get some sanctioning in the way or um Remy suggests but literally like um life and death decisions decisions affecting our biological fitness, who to trust and who not to trust, being able to to uh find uh mating partners. All of these things rely essentially on having beliefs which are complimentary to uh those of members in your society and and that could be anything right I mean if in order to get a partner a life partner you have to believe that men's and women's roles men and women are sort of naturally suited for specific kinds of roles and that is it doesn't really matter if that belief is um uh um false or anything like that. All that matters is that it matches the beliefs that it complements the beliefs of of the available pool, right, of people you have to interact with. And that seems to be a case where uh you can have very false beliefs equivalent to a flat earth, but they're not quarantined from action. they're actually they're actually involved in action because all that matters is that other people have equally absurd beliefs. Yeah. For for those sorts of social actions. And I guess that's the that's the case that I'm most interested in. Um all right. >> Andrew, >> you all just covered a lot of ground, so I'll try to keep it short, but this is a very interesting piece on the maladaptive versus adaptive. And I do think that both of them seem to entail the dimension of time. And so we're talking about something changing and it can become very abstract where like okay well anything that changes based upon like some kind of mutual interaction and relationship with things around it influences must be adapting whatever that means. And so yeah, I do think that to say something is adaptive expressly as as opposed to being maladaptive is to apply some kind of criteria for how you can judge like how well something adapted or something. Um part of what brought me to active inference aside from a few dozen other reasons was uh an interest in in mental health and clinical care. That's the computational psychiatry project there uh that I showed earlier. And uh one so one take on the idea of adaptive versus smile adaptive is if an individual themself expresses a personal disturbance with how their life has been and they notice it's coming from certain kinds of things that don't appear to be immediately external to them. That is they are advocating for themself I am having a problem then it suddenly becomes a little bit more potentially like ethical or reasonable to apply the word maladaptive. Oh, there's something maladaptive that their life is being led in a direction or I guess we could say a path that they themselves don't like that that that is harmful to them in

some way, shape or form. Right? So, so that's one take on that kind of divide. Um and and it's so difficult as social scientists whenever we want to make these kinds of divisions, right? because we're immediately hit with like well aren't you just agreeing with one particular set of norms when something is adaptive rather than my adaptive right and why are your norms better than my norms or what the norms I think should exist right so it's just we're we're scientists within the phenomena that we are trying to study and it's so tricky so I don't know if I gave a lot of useful material there but I hope it is helpful in some way >> so we'll need to uh close this out sadly for logistical reasons. Uh I just want to say one thing. Um I think when we talk about adaptive and maladaptive what we tend to imagine is some kind of utility theory, rational choice theory of decision because this is what uh I don't know about the rest but at least the western cannon has theorized for thousands of years and we think that we infer belief somewhat rationally and then from this belief we infer uh desired actions somewhat rationally and this is not how people uh work. So Tadas uh described in my opinion very uh relevantly the social meaning of beliefs. There is a social value to being believed as a to being perceived as a flatters person. And so maybe there is interest to talk about flatters and this is something that is that has been studied by these school senses for decades. Actually I think the most revealing thing I've uh come across well I'm doing hyperbole now but uh something very interesting is the study of authoritarian personality by Adorno uh Frankfurt school which is by all right someone I should be sworn to kill because I'm not into Frankfurt but um which so the point I'm getting at is that you have uh authoritarian personalities are by the studies uh people from very authoritarian families and there is always the double discourse of oh yeah my father was great it was the best of fathers well sometimes the ex spoiled and he told me it was my fault and then he beat me in the belt and then he locked me in a room that was dark and then I starved but you know he's a father he's the best there is a way in which uh violence is not only naturalized but presented as a sign of goodness and this is because within a society and a small society of family that is uh this violence you you'd rather agree with that and so you integrate it as the normal way of doing things and as uh mostly already pointed out norms are not there is no clear distinction between what is normative and what is uh descriptive. So the norms that you integrate, what you believe the world is is what you believe the world must be. At least this is uh what modes of behavior your attention uh will be tuned towards. And I want to very quickly point out one political philosopher, my favorite contemporary political philosopher um which is uh yes I want to do that which is Washington >> and uh he pointed out in my opinion very very very accurately that uh in authoritarian systems that I'm interested in as anyone that



has open news since 2016 understands why Um in this context uh the role of propaganda is not to affect beliefs because you cannot just yell at people to believe the earth is flat. They will maybe stop answering you but they will not change beliefs. What's the goal of propaganda is to change what uh public discourse is accepted. What you think you can say without getting bad things happen to you. And with four reasons that are uh very much related to inference, what you kind of behavior you expect yourself publically will cut very closely to what you believe to be the case. And I think that's that little tiny thread that comes from the solution of the directive inference. Uh it is something that is very strong. It's something that's very robust. This is something that is uh basically what we can bring to social science as a cognitivist that have a strong interest in active inference and that understand how this kind of thing work because not only we have some kind of conceptual thread that has a very high level of validity. Uh but also we have a way to model it and to uh translate it into specific computational hypothesis and specific frameworks to interpret uh evidence. So this is uh yes I did not want to articulate closing thoughts but we are closing at the time articulate my thoughts. So I guess this will work as a closing thoughts uh I'd like to thank you all for the opportunity to talk today. I'd like to thank in particular Mao for uh helping me with recording well for recording the session and hereby helping me organizing it. It was a pleasure talking to you and I hope we have some opportunity to uh get things get a little deeper at some point in the near future. We will uh stay in touch. So thank you and thanks to our audience for bearing with our rants about mathematics and committive